Topic 1: Introduction to HCI (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

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01 OVERVIEW OF HCI

02 EVOLUTION OF HCI

03 HISTORY & FUTURE OF HCI

2

OVERVIEW OF HCI

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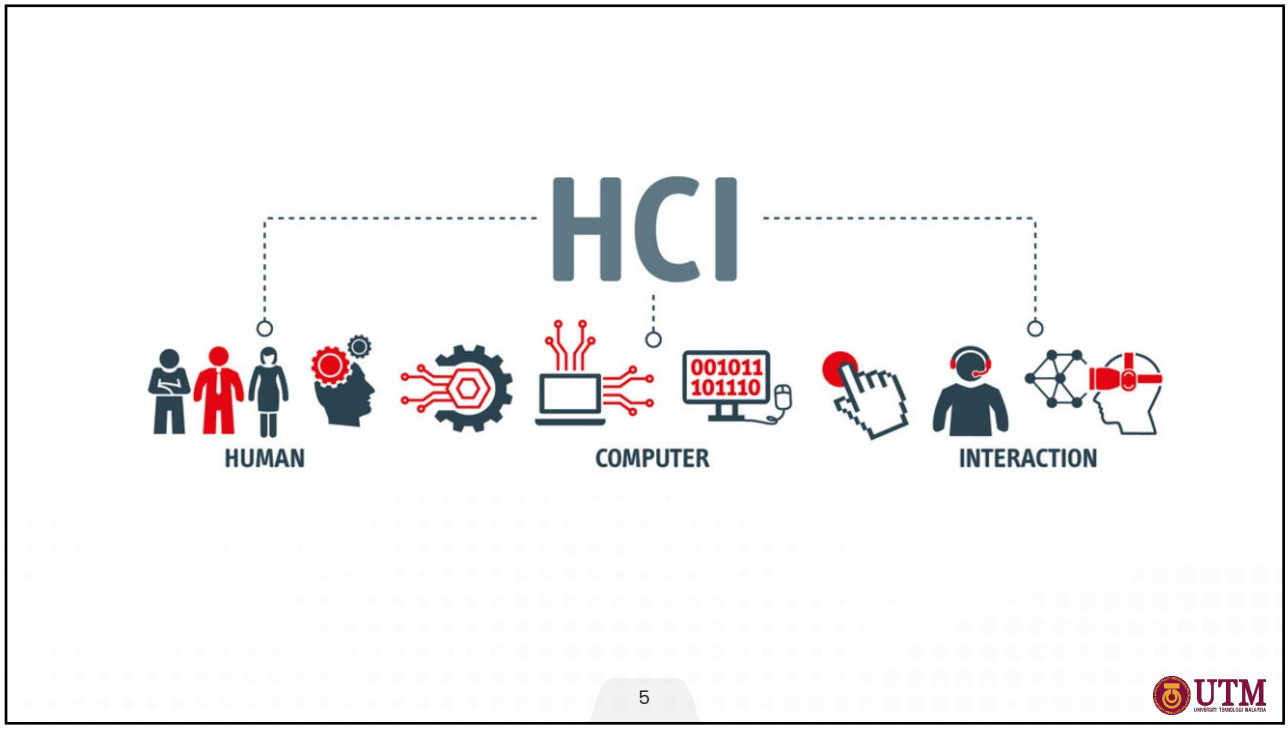
What is HCI?

https://youtu.be/yNzLBI0wsGU?si=0g16ZUp8SZPYvHRX


HCI
001011
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HUMAN
COMPUTER
INTERACTION
5
UTM

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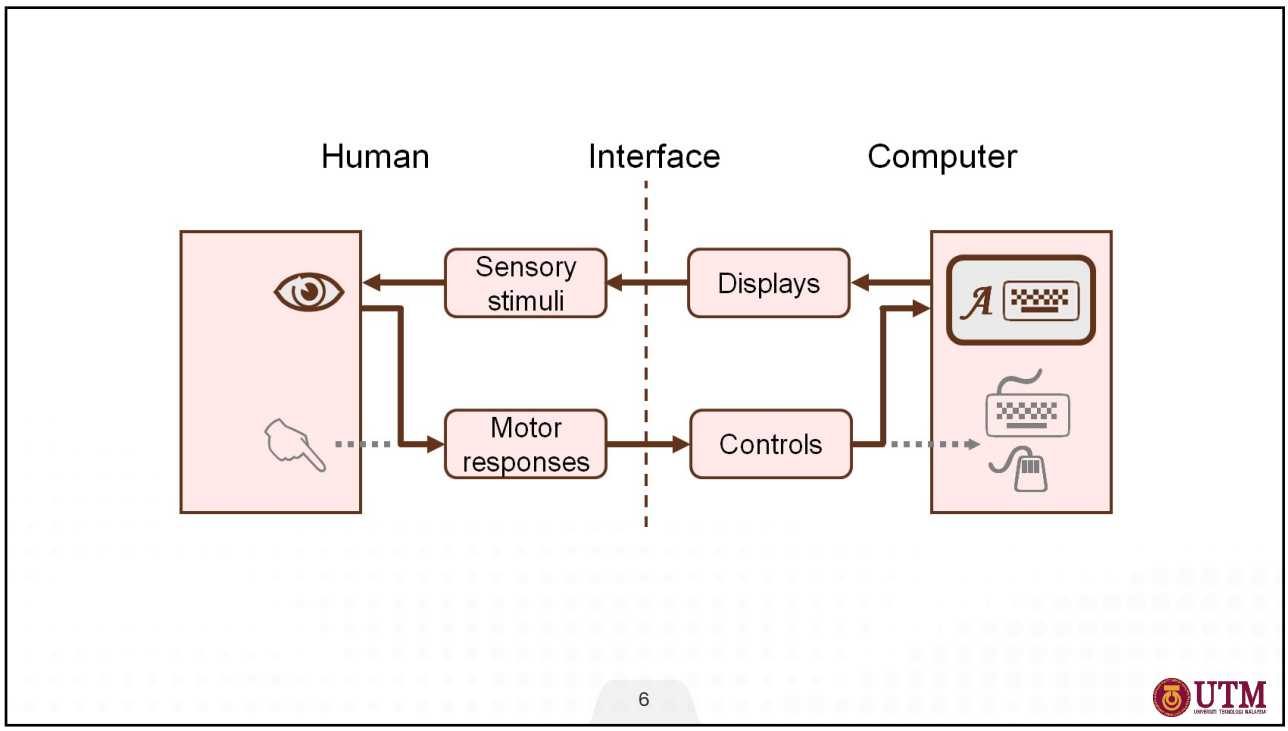
Human
Interface
Computer
Sensory stimuli
Displays
Motor responses
Controls
6
UTM

6

What is HCI?

HUMAN	COMPUTER	INTERACTION
• Single user • Groups • I/O channels • Memory • Reasoning • Problem solving • Error • Psychology	• Desktop • Embedded system • Data entry devices • Output devices • Memory • Processing	• Direct/Indirect communication • Models • Frameworks • Styles • Ergonomics

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Why study HCI?

- To employ people more productively and effectively
  - people costs now far outweigh hardware and software costs
- People now expect "easy to use" systems
  - generally they are not tolerant of poorly designed systems
  - if a product is hard to use, they will seek other products

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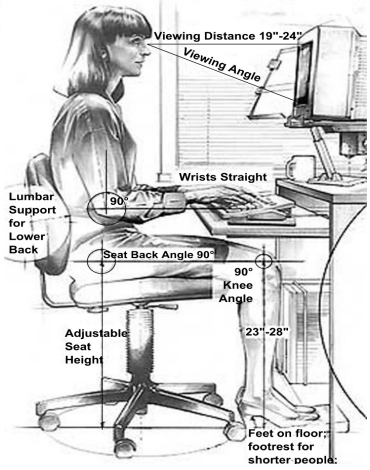
Why study HCI?

HUMAN FACTORS VIEW



- Human have limitations
- Errors are costly in terms of
  - loss of time & money
  - loss of lives in critical systems
  - loss of morale

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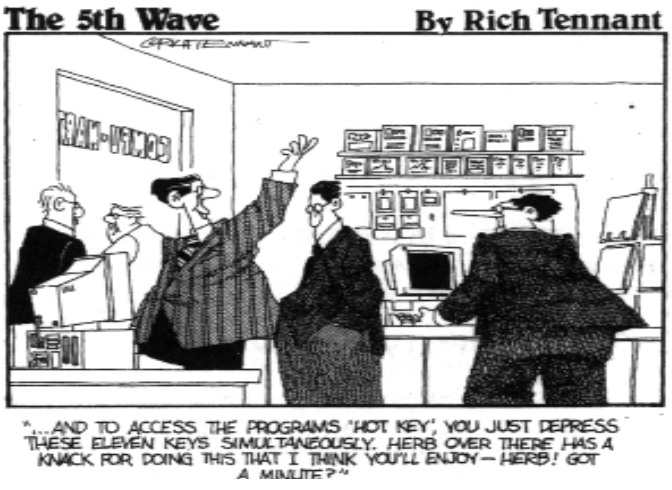
How do you feel?



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Why HCI is Important?

- Fail to design well, it can affect
  - Effectiveness
  - Productivity
  - Morale
  - Safety

- Example: a car with poor HCI

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Activity

- Take 5 minutes for everyone to write down one common device with substantial HCI design choices and discuss with the neighbor the pros and cons.

- How does it affect you or other users?

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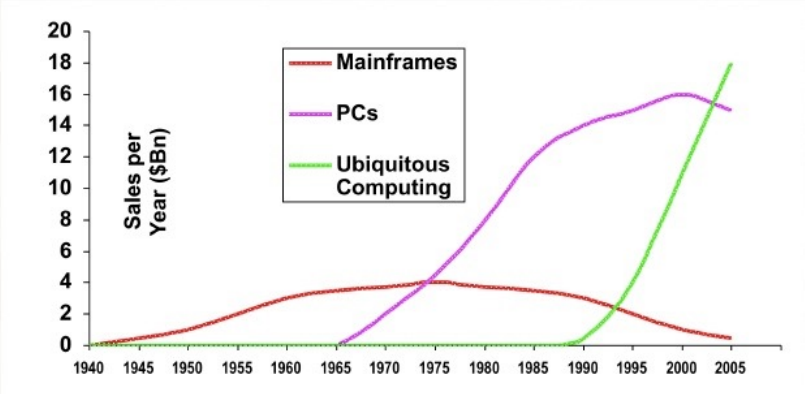
Weiser's Waves of Computing



The Three Waves of Computing, Mark Weiser, 1990

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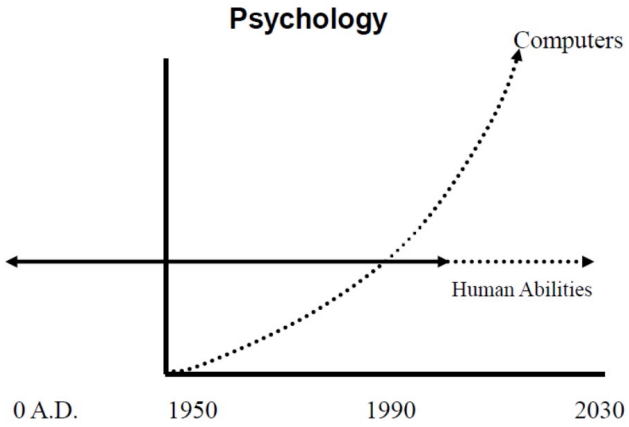
Human vs. Computer



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Definition of HCI (1)

"Human-computer interaction is a discipline concerned with the **design, evaluation** and **implementation** of **interactive computing system** for **human use** and with the study of major phenomena surrounding them."

[ACM SIGCHI Curricula Human-Computer Interaction]

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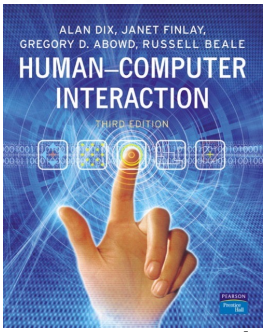
Definition of HCI (2)

"The study of interaction between people and computer-based systems; concern with the physical, psychological and theoretical aspects of this process."



[Alan Dix et al. 2003]

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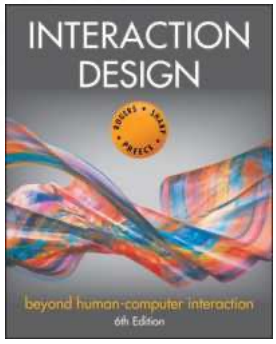
Definition of HCI (3)

""Human-computer interaction (HCI) is about designing computer system that support people so that they can carry out their activities productively and safely."



[6th edn Jenny Preece et al. 2023/2024]

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Definition of HCI (cont.)

- HCI has a **role** in the design and development of all kinds of systems, ranging from those like:
  - air traffic control and nuclear processing where **safety** is extremely important,
  - to office systems, where **productivity** and job satisfaction are paramount,
  - to computer games, which **must excite and engage users**.

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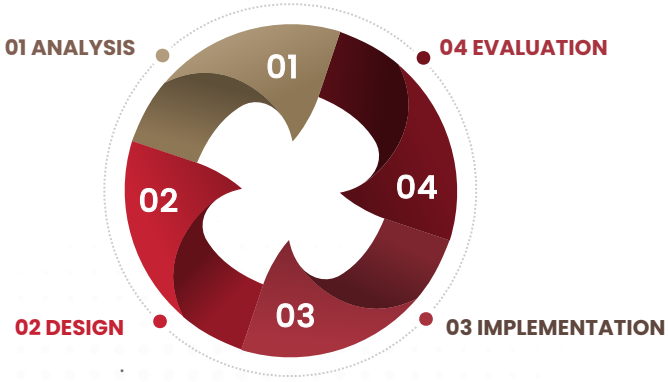
Components of HCI



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Goal of HCI

To develop or improve the safety, utility and effectiveness of systems that include computers, often through improving usability

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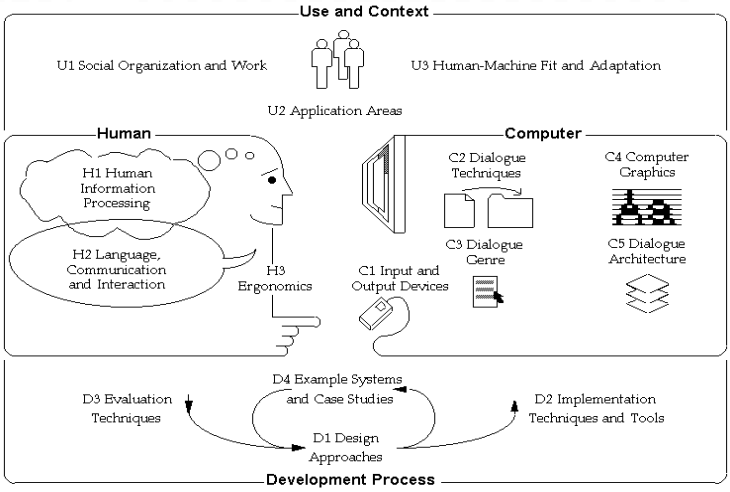
Field of HCI

- Psychology
- Sociology
- Cognitive science
- Ergonomics / Human factors
- Computer science
- Visualization
- Design
- Interactive art
- Information science
- Information security
- Speech-language pathology
- Personal information management
- Phenomenology
- …

Meta Models of HCI



The Discipline of HCI [Source: Book "**Human Computer Interaction" (1994,p.16)**]

Why HCI is important?

"Human-computer interaction is the kind of discipline which is neither the study of humans, nor the study of technology, but rather the bridging between those two. So you always have to have one eye open to the question: what can the technology do? How can you build it? What are the possibilities? And one eye open to the question: what are people doing and how would this fit in? What would they do with it? If you lose sight of either of those, you fail to design well."

(Terry Winograd, Professor of computing, Stanford University)

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EVOLUTION OF HCI

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Mac Evolution

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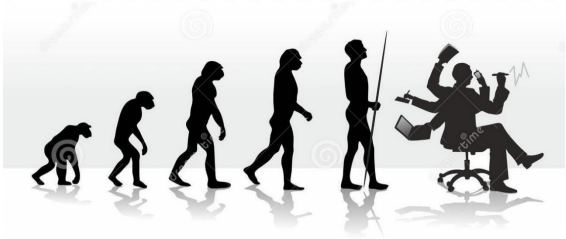
Human Evolution (?)



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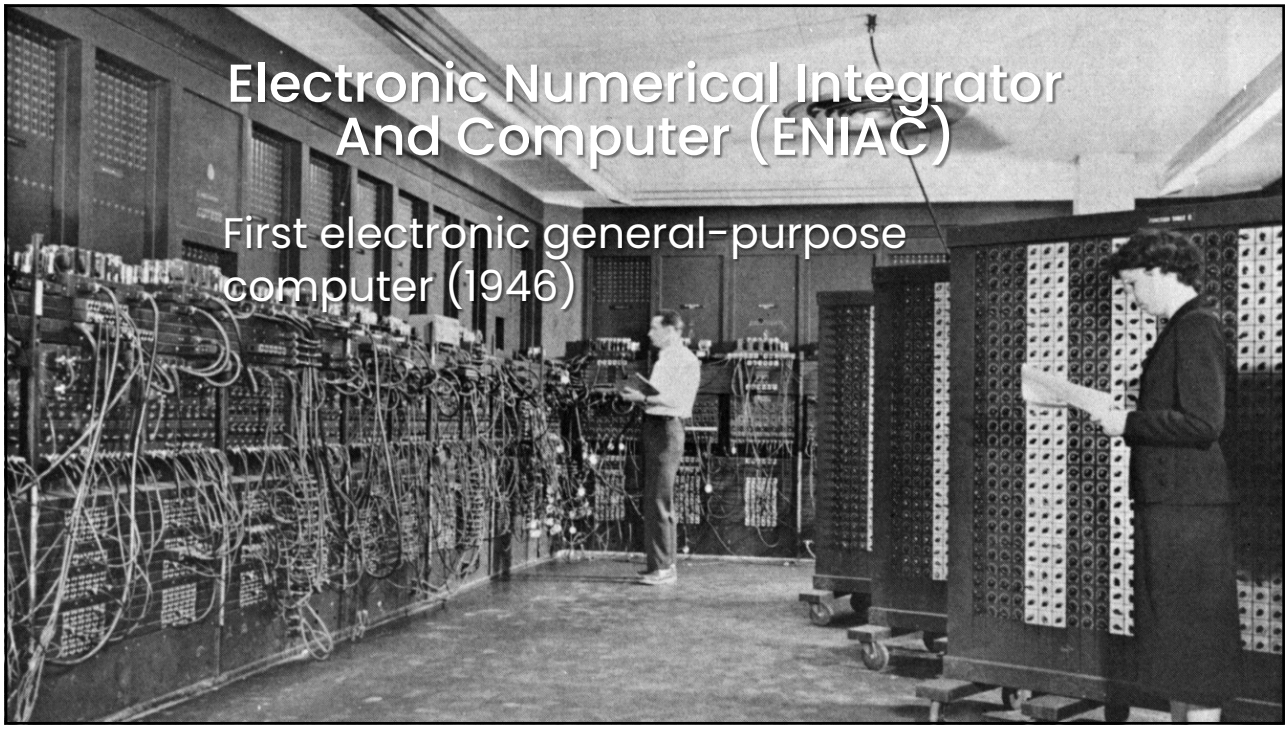
Electronic Numerical Integrator And Computer (ENIAC)
First electronic general-purpose computer (1946)

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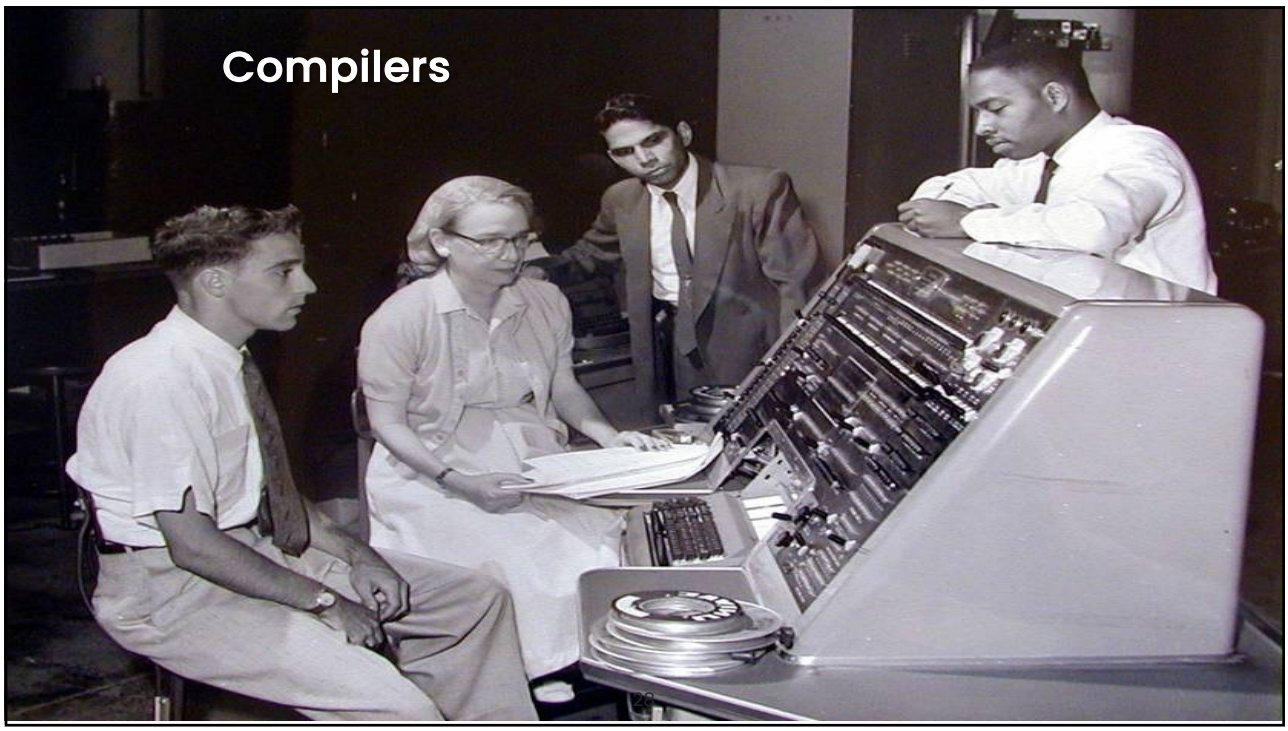
Compilers

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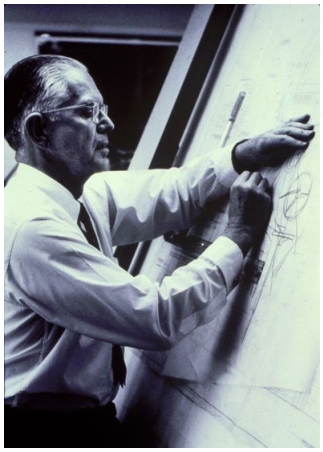
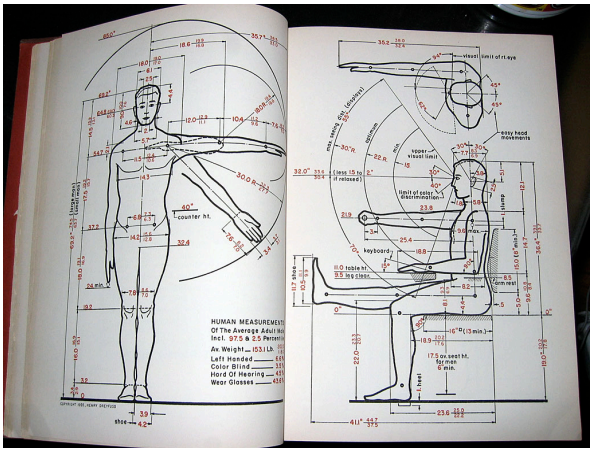
Designing for People





Henry Dreyfuss designed specific measurements & proportions of the human body for specific work & environment. [Source: Book "**Designing for People" (1955)**]

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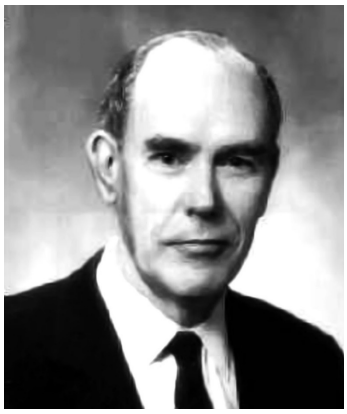
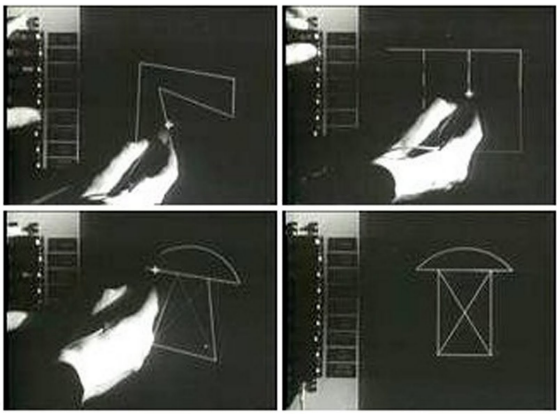
Graphical User Interface (GUI)





Ivan Sutherland's Sketchpad (1963)

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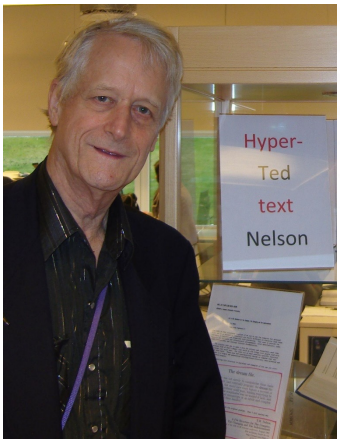
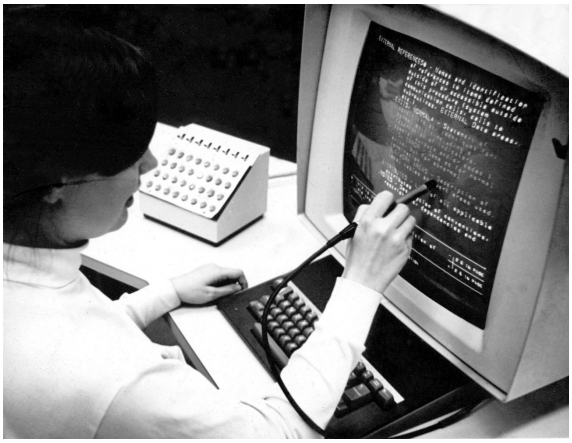
Hypertext



Ted Nelson introduced the terms "hypertext" & "hypermedia" in 1963



Hypertext Editing System (HES) was an early hypertext research project conducted at Brown University in 1967

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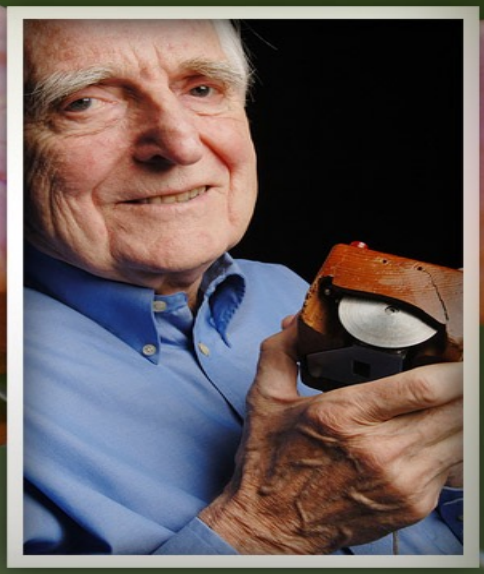
Mouse

- Douglas Engelbart was an inventor of the computer mouse
- Invented in 1960s



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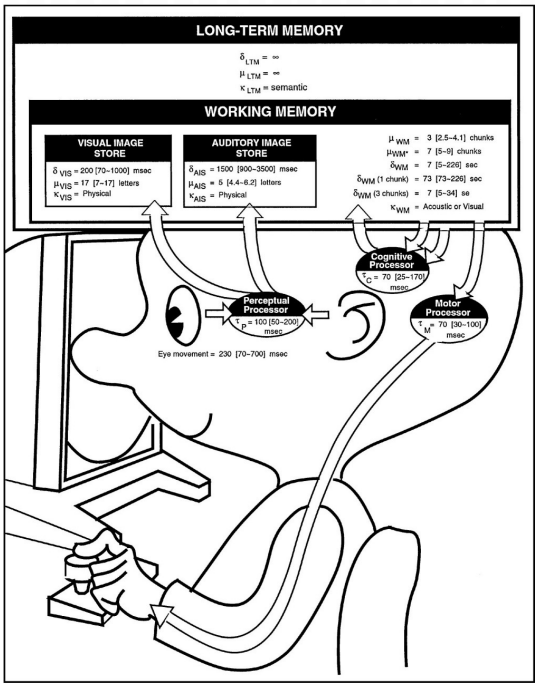
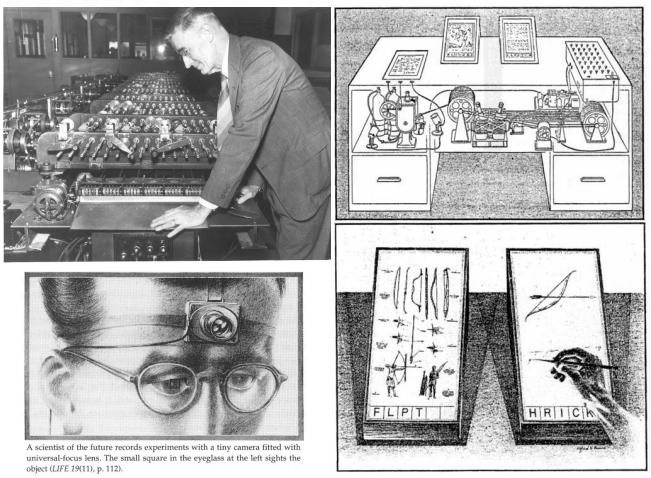


Model Human Processor

- Introduced by Card, Moran & Newell in 1983
- To calculate how long it takes to perform a certain task using cognitive modeling method

Memory Extender (MemEx)



A scientist of the future records experiments with a tiny camera fitted with universal-focus lens. The small square in the eyeglass at the left sights the object (*LIFE* 19(11), p. 112).

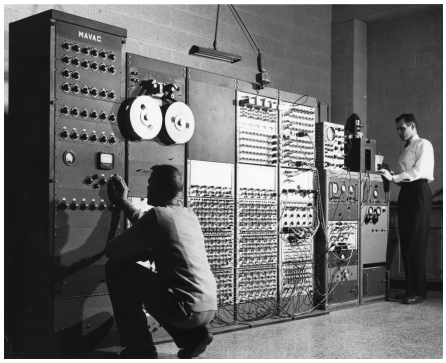
- A device to store all books, records and communication
- Mechanised to be consulted with exceeding speed and flexibility
- Support for full-text search, text & audio annotations and hyperlinks

["As We May Think", Vannevar Bush, 1945]

Man-Computer Symbiosis

- Tightly coupled human brain & machine, speech recognition, time sharing and character recognition

[J.C.R. Licklider, 1960]



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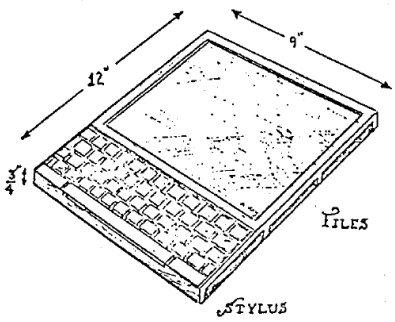
DynaBook

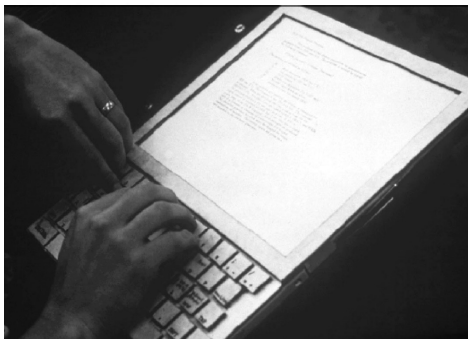
The DynaBook

"I wish to God these calculations were executed by steam!"
Charles Babbage (age 19)
ca. 1803

"The Analytical Engine weaves algebraic patterns, just as the Jacquard Loom weaves patterns in silk."
-Ada Augusta
Countess of Lovelace



[Alan Kay, 1968]



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Xerox Machine

XEROX Alto (1973)

XEROX Star (1981)

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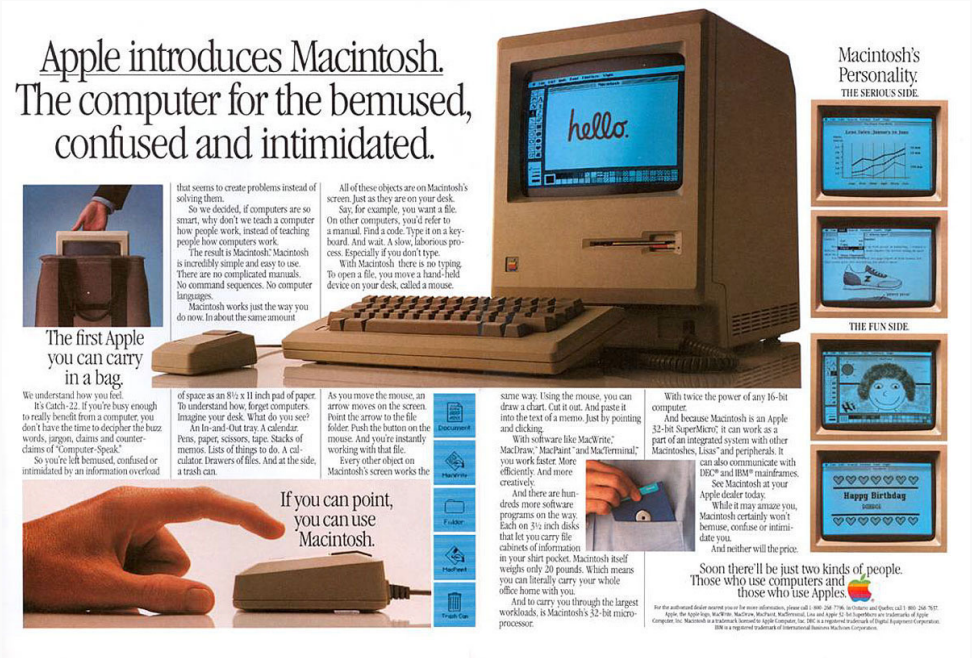

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Challenges to HCI Designers

- Two important challenges to HCI designers:
  - How to keep know what's going on within the development of technology (changes in technology)
  - How to ensure that their designs offer good HCI and take advantage of the potential functionality of the new technology

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HISTORY & FUTURE OF HCI

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Where we are today

- Large displays
- Small displays
- Peripheral displays
- Alternative I/O
- Ubiquitous computing
- Virtual environments
- Implants
- Speech recognition
- Multimedia
- Video conferencing
- Artificial intelligence
- Software agents
- Recommender systems
- etc ...

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The Reactable

- An electronic musical instrument with a tabletop/mobile device
- A multi-touch interface for playing music
- Performers can simultaneously interact by moving and rotating physical objects on its surface
- Use Interfaces Gestures to interact

Source: https://reactable.com/

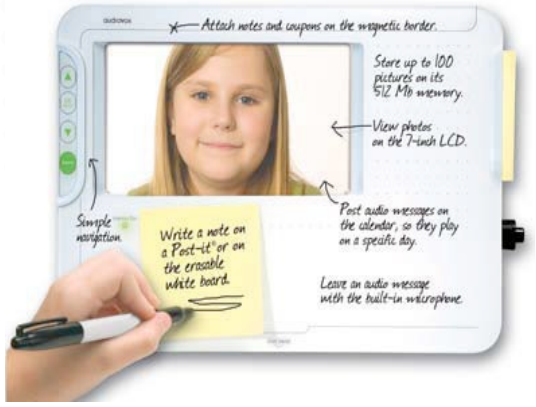
MIT's Transform

- Dynamic and Adaptive furniture
- It holds and moves objects like fruits, game tokens, office supplies and tablets; creates dividers on demand; and generates interactive sculptures to convey messages and audio.

Source: https://tangible.media.mit.edu/project/transform/

Audiovox's Digital Message Center

- Digital Message Center is designed to be attached to the refrigerator.
- Letting families scribble digital notes and leave audio and video messages for each other.



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Duality

- Duality ART+COM's artistic installation called Duality, located at the exit of a metro station in Tokyo
- Passers-by provoke virtual ripple effects with their footsteps, as if walking across a pond

Source: http://angesleva.iki.fi/art/duality/

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Mobile Phones Evolution (50 years)

From the first mobile phone 'brick' to the latest Apple iPhone: as the size reduces, the potential expands.

Source: https://www.sellcell.com/blog/first-mobile-phone-call-50th-anniversary-history-of-the-mobile-phone/

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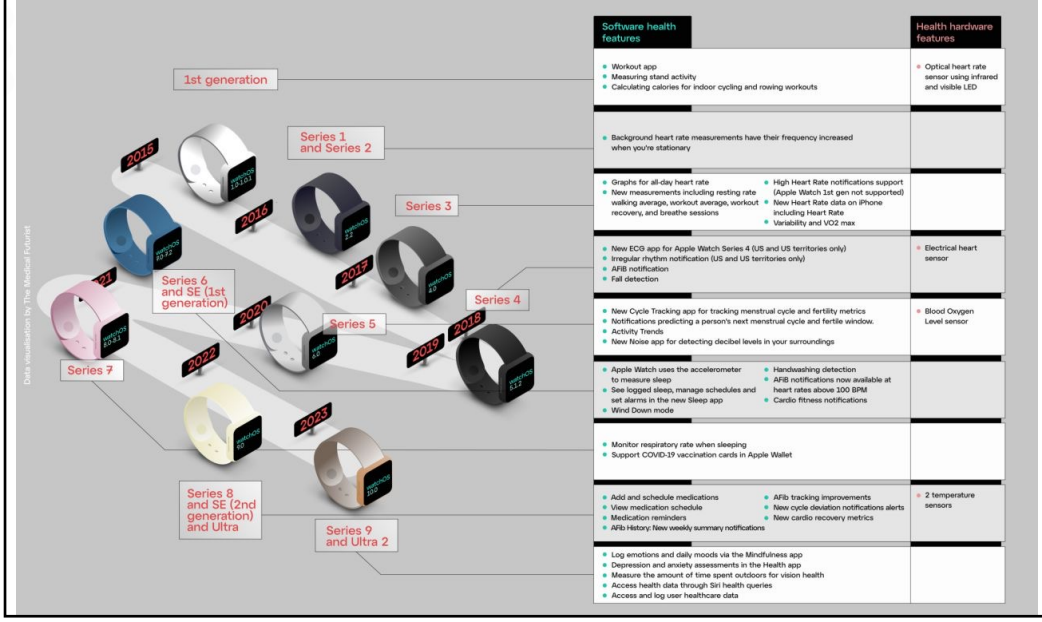
The Evolution of Apple Watch Health Features



Source: https://cdn.medicalfuturist.com/wp-content/uploads/2023/10/Apple-watch-evolution.jpg

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Smart Living

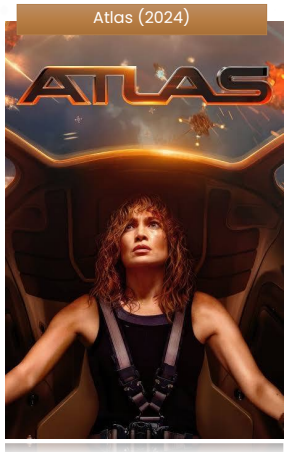
Source: https://www.tianjian.com.sg/smart-home-solutions

Film Productions

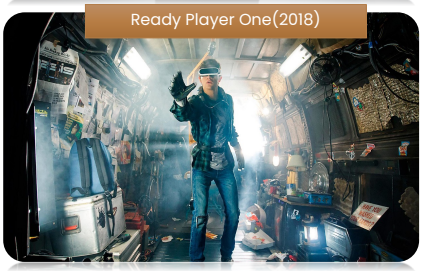
Minority Report (2002)

Iron Man (2008)

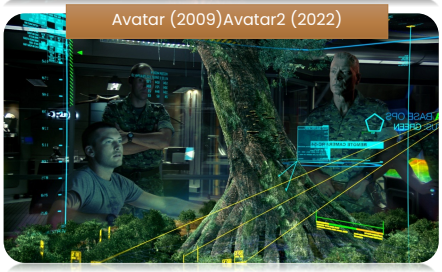
Atlas (2024)



Ready Player One(2018)



Avatar (2009)Avatar2 (2022)



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Topic 1: What is Interaction Design (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 GOOD AND BAD DESIGN

02 WHAT IS INTERACTION DESIGN

03 UNDERSTANDING PEOPLE

04 USABILITY & USER EXPERIENCE GOALS

05 DESIGN PRINCIPLES

2

GOOD & BAD DESIGN

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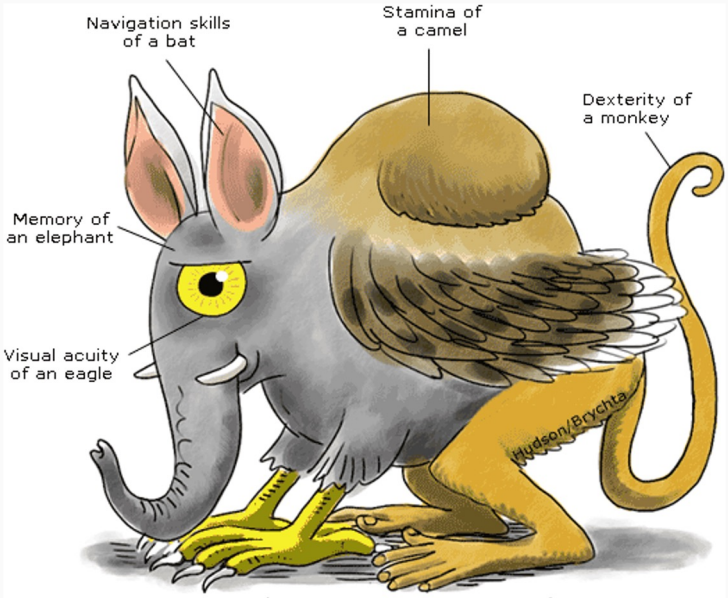

"The Perfect User"

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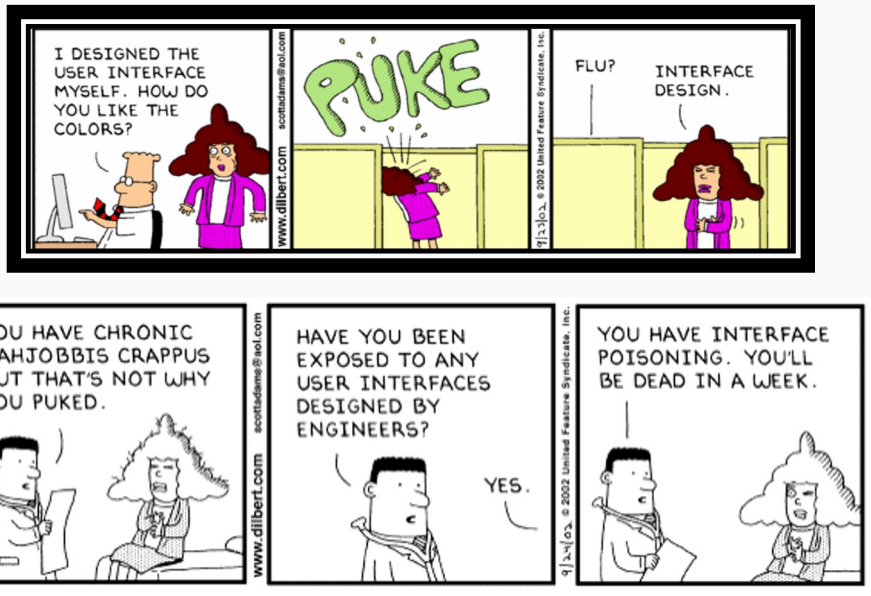

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Bad Design: Lift Control

- Lift controls and labels on the bottom row all look the same, so it is easy to push a label by mistake instead of a control button

Source: www.baddesigns.com

- People do not make same mistake for the labels and buttons on the top row. Why not?

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Bad Design: Why is this vending machine so bad?

Source: www.baddesigns.com

- Need to **push button first** to activate reader
- Normally **insert bill** first before making selection
- Contravenes well known convention

Good Design

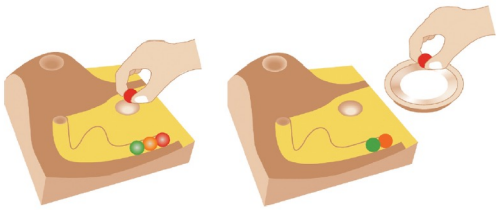
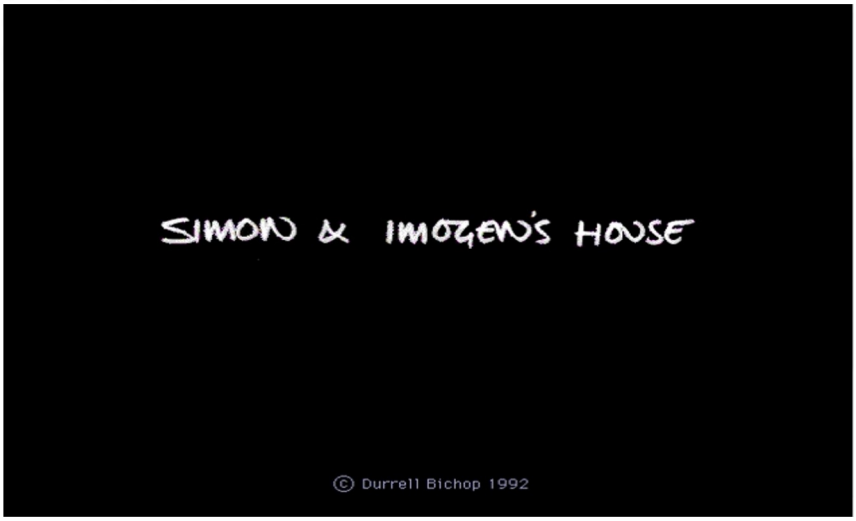


Figure 1.1 The marble answering machine
Source: Adapted from Gillian Crampton Smith: "The Hand that Rocks the Cradle" *ID Magazine*, May/June 1995, pp. 60–65.

- Marble answering machine (Bishop, 1995)
- Based on how everyday objects behave
- Easy, intuitive and a pleasure to use
- Only requires one-step actions to perform core tasks



How does the "marble" answering machine differ from the voice mail system?

Why is it a Good Design?

- Use familiar physical object "marble pinball" that visually represent how many incoming message been left.
- Play recorder message
  - dropping the marble into a slot in the machine
- Dial the caller who left the message
  - Dropping the same marble pinball into another slot on phone
- It is aesthetically pleasing and enjoyable to use
- It only required one-step actions to perform core tasks
- It is a simple but elegant design
- It offers less functionality and allow anyone to listen to any the message

Good and Bad Design

- Why is the TiVo remote much better designed than standard remote controls?
  - Peanut shaped to fit in hand
  - Logical layout and color-coded, distinctive buttons
  - Easy-to-locate buttons

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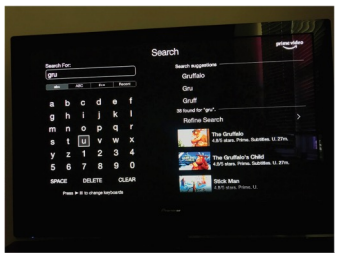
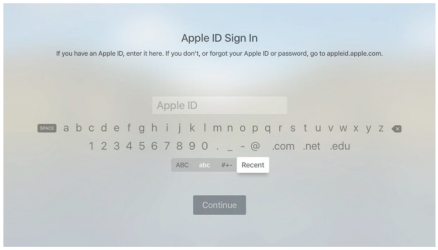
Dilemma

- Which is the best way to interact with a smart TV? Why?
- Pecking using a grid keyboard via a remote control
- Swiping across two alphanumeric rows using a touchpad on a remote control
- Voice control using remote or smart speaker





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Switching to Digital

- Many activities that used to be done with a physical artifact have gone digital
  - e.g. ticket machines, parking meters
- Benefit
  - Makes many tasks easier, quicker and more convenient
  - Our details can also be stored ready for next time
  - Don't have to wait in line before when buying a physical ticket
  - Easy to swipe QR code to gain entrance through a ticket barrier
- Disadvantages
  - Not everyone has a modern smartphone
  - Some people still prefer to use older phones which the apps won't work on
  - Some people prefer not to divulge personal information online
  - Some people prefer to talk to real people when making a purchase

What to Design

- Need to take into account:
  - Who the users are
  - What activities are being carried out
  - Where the interaction is taking place
- Need to optimise the interactions users have with a product:
  - So that they match the users' activities and needs

Understanding User's Needs

- Take into account what people are good and bad at
- Consider what might help people in the way they currently do things
- Think through what might provide quality user experiences
- Listen to what people want and get them involved
- Use tried and tested user-centered methods

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WHAT IS INTERACTION DESIGN

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What is Interaction Design?

- "Designing interactive products to support the way people communicate and interact in their everyday and working lives."

Preece, Sharp and Rogers (2015)

- "The design of spaces for human communication and interaction."

Winograd (1997)

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Goals of Interaction Design?

- Develop usable products
  - Usability means easy to learn, effective to use and provide an enjoyable experience
- Involve users in the design process

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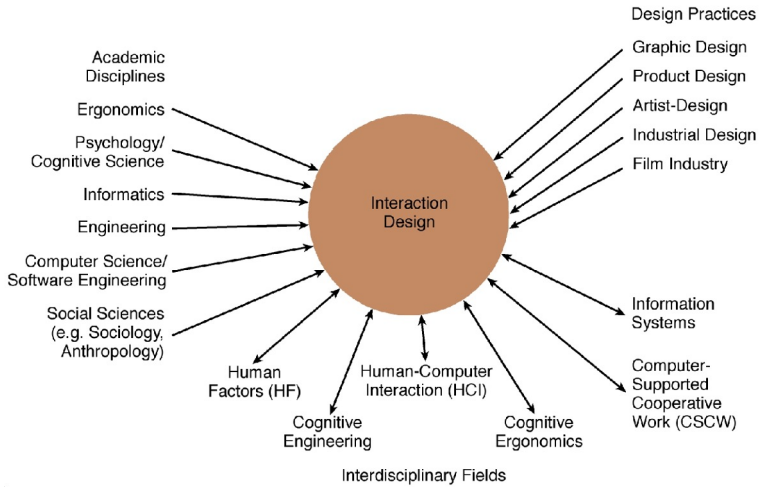
Which kind of Design?

- Number of other terms used emphasising what is being designed, e.g.
  - user interface design, software design, user-centered design, product design, web design, experience design (UX)

- Interaction design is the umbrella term covering all of these aspects
  - fundamental to all disciplines, fields, and approaches concerned with researching and designing computer-based systems for people

HCI and Interaction Design



Relationship between ID, HCI & Other Fields

Academic disciplines contributing to ID:

- Psychology
- Social Sciences
- Computing Sciences
- Engineering
- Ergonomics
- Informatics

Design practices contributing to ID:

- Graphic design
- Product design
- Artist-design
- Industrial design
- Film industry

Relationship between ID, HCI & Other Fields (cont.)

Interdisciplinary fields that 'do' interaction design:

- HCI
- Ubiquitous Computing
- Human Factors
- Cognitive Engineering
- Cognitive Ergonomics
- Computer Supported Co-operative Work
- Information Systems

Working in Multidisciplinary Teams

- Many people from different backgrounds involved
- Different perspectives and ways of seeing and talking about things

- Benefits
  - more ideas and designs generated
- Disadvantages
  - difficult to communicate and progress forward the designs being create

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Interaction Design in Business

- Increasing number of ID consultancies, examples of well-known ones include:
  - Nielsen Norman Group: "help companies enter the age of the consumer, designing human-centered products and services"
  - Cooper: "From research and product to goal-related design"
  - Swim: "provides a wide range of design services, in each case targeted to address the product development needs at hand"
  - IDEO: "creates products, services and environments for companies pioneering new ways to provide value to their customers"
  - *Stampede: "Transforming online experiences for business and organisations with incredible, innovative solution since 2006"*

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UNDERSTANDING
PEOPLE

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People-centered Design

- Involves understanding how people feel about a product and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it.
- Their overall impression of how good it is to use
- The quality of the experience
- *"It is not enough that we build products that function, that are understandable and usable, we also need to build joy and excitement, pleasure and fun, and yes, beauty to people's lives."*
  - Don Norman (2004)

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The User Experience (UX)

- How a product behaves and is used by people in the real world
  - the way people feel about it and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it
  - "every product that is used by someone has a user experience: newspapers, ketchup bottles, reclining armchairs, cardigan sweaters." (Garrett, 2010)
  - "all aspects of the end-user's interaction with the company, its services, and its products. (Nielsen and Norman, 2014)

- Cannot design a user experience, only design ***for*** a user experience

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Defining UX

- How users perceive a product, such as whether a smartwatch is seen as sleek or chunky, and their emotional reaction to it, such as whether people have a positive experience when using it.

  (Hornbæk and Hertzum, 2017)

- Hassenzahl's (2010) model of the user experience
  - Pragmatic: how simple, practical, and obvious it is for the user to achieve their goals
  - Hedonic: how evocative and stimulating the interaction is to users

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Why was the iPod UX such a success?

Figure 1.6 The iPod Nano Touch
Source: ©Press Association, reproduced with permission.

- Quality user experience from the start
- Simple, elegant, distinct brand, pleasurable, must have fashion item, catchy names, cool...

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29

and today..

Spotify

RUNCLOUD

NETFLIX

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Core Characteristics of Interaction Design

- Users should be involved through the development of the project
- Specific usability and user experience goals need to be identified, clearly documented and agreed at the beginning of the project
- Iteration is needed through the core activities

Why?

- Help designers:
  - understand how to design interactive products that fit with what people want, need and may desire
  - appreciate that one size does not fit all
    - e.g., teenagers are very different to grown-ups
  - identify any incorrect assumptions they may have about particular user groups
    - e.g., not all old people want or need big fonts
  - be aware of both people's sensitivities and their capabilities

Accessibility and Inclusiveness

- ***Accessibility***: the extent to which an interactive product is accessible by as many people as possible
  - Focus is on people with disabilities; for instance, those using Android OS or apple voiceover

- ***Inclusiveness***: making products and services that accommodate the widest possible number of people
  - For example, smartphones designed for all and made available to everyone regardless of their disability, education, age, or income

Disabilities

- Whether someone is disabled changes over time with age, or recovery from an accident

- The severity and impact of an impairment can vary over the course of a day or in different environmental conditions

- Disabilities can result because technologies are designed to necessitate a certain type of interaction that is impossible for someone with an impairment

Understanding Disabilities

- Disabilities can be classified as:
  - Sensory impairment (such as loss of vision or hearing)
  - Physical impairment (having loss of functions to one or more parts of the body after a stroke or spinal cord injury)
  - Cognitive (including learning impairment or loss of memory/cognitive function due to old age)
- Each type can be further defined in terms of capability:
  - For example, someone might have only peripheral vision, be color blind, or have no light perception
- Impairment can be categorised:
  - Permanent (for instance, long-term wheelchair user)
  - Temporary (that is, after an accident or illness)
  - Situational (for example, a noisy environment means that a person can't hear)

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Being *Cool* about Disabilities

- Prosthetics can be designed to move beyond being functional (and often ugly) to being desirable and fashionable
- People now refer to "wearing their wheels," rather than "using a wheelchair"

Fashionable leg cover designed by Alleles Design Studio

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USABILITY & USER EXPERIENCE GOALS

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Usability Goals

- Effective to use
- Efficient to use
- Safe to use
- Have good utility
- Easy to learn
- Easy to remember how to use

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User Experience Goals

Desirable aspects

satisfying	helpful	fun
enjoyable	motivating	provocative
engaging	challenging	surprising
pleasurable	enhancing sociability	rewarding
exciting	supporting creativity	emotionally fulfilling
entertaining	cognitively stimulating	

Undesirable aspects

boring	unpleasant
frustrating	patronising
making one feel guilty	making one feel stupid
annoying	cutesy
childish	gimmicky

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Usability and UX Goals

- Selecting terms to convey a person's feelings, emotions, etc., can help designers understand the multifaceted nature of the user experience
- How do usability goals differ from user experience goals?
- Are there trade-offs between the two kinds of goals?
  - e.g. can a product be both fun and safe?
- How easy is it to measure usability versus user experience goals?

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DESIGN PRINCIPLES

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Design Principles

- Generalisable abstractions for thinking about different aspects of design
- The do's and don'ts of interaction design
- What to provide and what not to provide at the interface
- Derived from a mix of theory-based knowledge, experience and common-sense

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Visibility

www.baddesigns.com

- This is a control panel for a lift
- How does it work?
- Push a button for the floor you want?
- Nothing happens. Push any other button? Still nothing. What do you need to do?

- It is not visible as to what to do!

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UTM

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Visibility (cont.)

www.baddesigns.com

…with this lift, you need to insert your room card in the slot by the buttons to get the lift to work!

How would you make this action more visible?

- make the card reader more obvious
- provide an auditory message, that says what to do (which language?)
- provide a big label next to the card reader that flashes when someone enters
- make relevant parts visible
- make what has to be done obvious

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UTM

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Visibility (cont.)

What do I do if I am wearing black?

Invisible automatic controls can make it more difficult to use

Feedback

- Sending information back to the user about what has been done
- Includes sound, highlighting, animation and combinations of these
  - e.g. when screen button clicked on provides sound or red highlight feedback:

Previous → "ccclichhk"

Previous → Previous

Constraints

- Restricting the possible actions that can be performed
- Helps prevent user from selecting incorrect options
- Physical objects can be designed to constrain things
  - e.g. only one way you can insert a key into a lock

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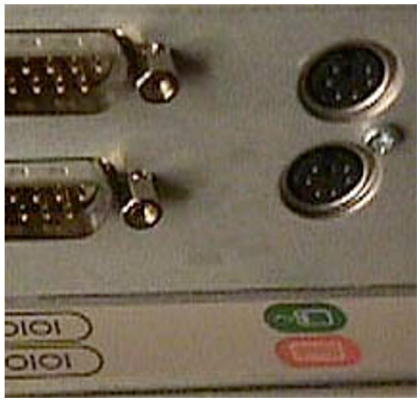
Constraints (cont.)



www.baddesigns.com

Logical or ambiguous design?

- Where do you plug the mouse?
- Where do you plug the keyboard?
- top or bottom connector?
- Do the colour-coded icons help?

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Constraints (cont.)

www.baddesigns.com

How to design them more logically?

(i) A provides direct adjacent mapping between icon and connector

(ii) B provides colour coding to associate the connectors with the labels

www.baddesigns.com

Consistency

- Design interfaces to have similar operations and use similar elements for similar tasks
- For example:
  - always use ctrl key plus first initial of the command for an operation – ctrl+C, ctrl+S, ctrl+O
- Main benefit is consistent interfaces are easier to learn and use

Consistency (cont.)

When consistency breaks down

- What happens if there is more than one command starting with the same letter?
  - e.g. save, spelling, select, style
- You have to find other initials or combinations of keys, thereby breaking the consistency rule
  - e.g. ctrl+S, ctrl+Sp, ctrl+shift+L
- Increases learning burden on user, making them more prone to errors

Consistency (cont.)

Internal and external consistency

- Internal consistency refers to designing operations to behave the same within an application
  - Difficult to achieve with complex interfaces
- External consistency refers to designing operations, interfaces, etc., to be the same across applications and devices
  - Very rarely the case, based on different designer's preference

Consistency (cont.)

Keypad numbers layout

- A case of external inconsistency

(a) phones, remote controls

1	2	3
4	5	6
7	8	9
	0	

(b) calculators, computer keypads

7	8	9
4	5	6
1	2	3
0		

Affordance: to give clue

- Refers to an attribute of an object that allows people to know how to use it
  - e.g. a mouse button invites pushing, a door handle affords pulling
- Norman (1988) used the term to discuss the design of everyday objects
- Has since popularised in interaction design to discuss how to design interface objects
  - e.g. scrollbars to afford moving up and down, icons to afford clicking on

Affordance (cont.)

What does 'affordance' have to offer interaction design?

- Interfaces are virtual and do not have affordances like physical objects
- Norman argues it does not make sense to talk about interfaces in terms of 'real' affordances
- Instead interfaces are better conceptualised as 'perceived' affordances
  - Learned conventions of arbitrary mappings between action and effect at the interface
  - Some mappings are better than others

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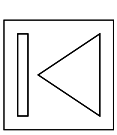
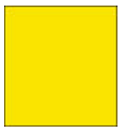
Activity

Virtual affordances

- How do the following screen objects afford?
- What if you were a novice user?
- Would you know what to do with them?





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Shneiderman's Eight Golden Rules

The principles	Questions to consider
1. Strive for consistency	Is the style of this element maintained across your site/app? Is this content placed in the correct location according to the site hierarchy? Does this follow the conventions for your chosen platform? How can you make your designs more consistent?
2. Enable frequent users to use shortcuts	Are there shortcuts available for your more experienced users? Who is this product designed for? Will there be a need to consider experienced users? How can you make it easier and quicker for experienced users?
3. Offer informative feedback	Does the user know where they are at in the process? Does the user know what they have done after performing this action? How are you communicating this feedback to your user?
4. Design dialogue to yield closure	Does the user have to do any guessing here? Is it clear and obvious enough for your intended audience? Are there any next steps for the user? How are you communicating the system status with the user?
5. Offer simple error handling	Have you done everything imaginable to prevent this error from happening on your end? Is this error avoidable in the first place? If the user does make an error, how easy is it for them to fix it?
6. Permit easy reversal of actions	How many steps does the user have to take to reverse their actions? Will the user quickly realize they need to reverse the action in the first place? How can you make your users detect the possibility of reversal?
7. Support internal locus of control	Will the user feel in control at this specific touch point in your app? Will they be surprised in an unpleasant manner? Does the site feel easily navigable? Does the user feel safe and in control? How can you make the user feel more safe and in control??
8. Reduce short-term memory load	Are there enough visual cues here for the user to find the functionality or item? Do they have to remember things to understand what's going on? How can you help the user recall?

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Key Points

- Interaction design is concerned with designing interactive products to support how people communicate and interact in their everyday and working lives
- It is concerned with how to create quality user experiences for services, devices, and interactive products
- It is multidisciplinary, involving many inputs from wide-reaching disciplines and fields
- Optimising the interaction between users and interactive products requires consideration of a number of interdependent factors, including context of use, types of activity, UX goals, accessibility, cultural differences, and user groups.
- Design principles, such as feedback and simplicity, are useful heuristics for informing, analysing, and evaluating aspects of an interactive product.

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Topic 2: Interaction Design Process and Practice

SECV2113 Human-Computer Interaction

Faculty of Computing
Universiti Teknologi Malaysia

1

01 WHAT IS INVOLVED IN INTERACTION DESIGN

02 SOME PRACTICAL ISSUES

03 AGILEUX

04 DESIGN PATTERNS

05 OPEN SOURCE RESOURCES & TOOLS FOR INTERACTION DESIGN

2

WHAT IS INVOLVED IN INTERACTION DESIGN

3

What is Involved in ID?

- It is a process:
  - Focused on discovering requirements, designing to fulfil requirements, producing prototypes and evaluating them
  - Focused on users and their goals
  - Involves trade-offs to balance conflicting requirements
- Generating alternatives and choosing between them is key
- Four approaches: user-centered design, activity-centered design, systems design, and genius design

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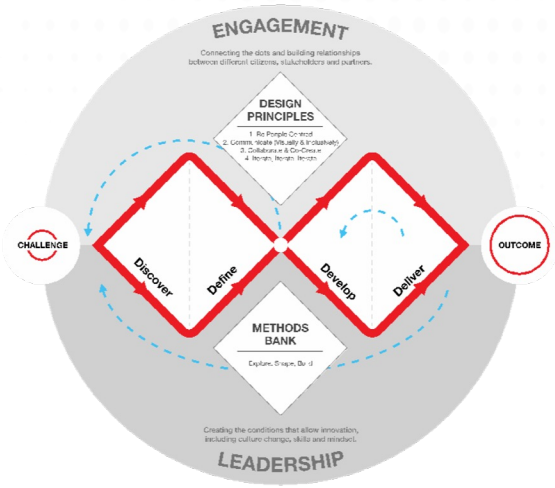
The Double Diamond of Design



Focus:
General design thinking

Source: Adapted from www.designcouncil.org.uk/news-opinion/what-framework-innovation-design-councils-evolved-double-diamond

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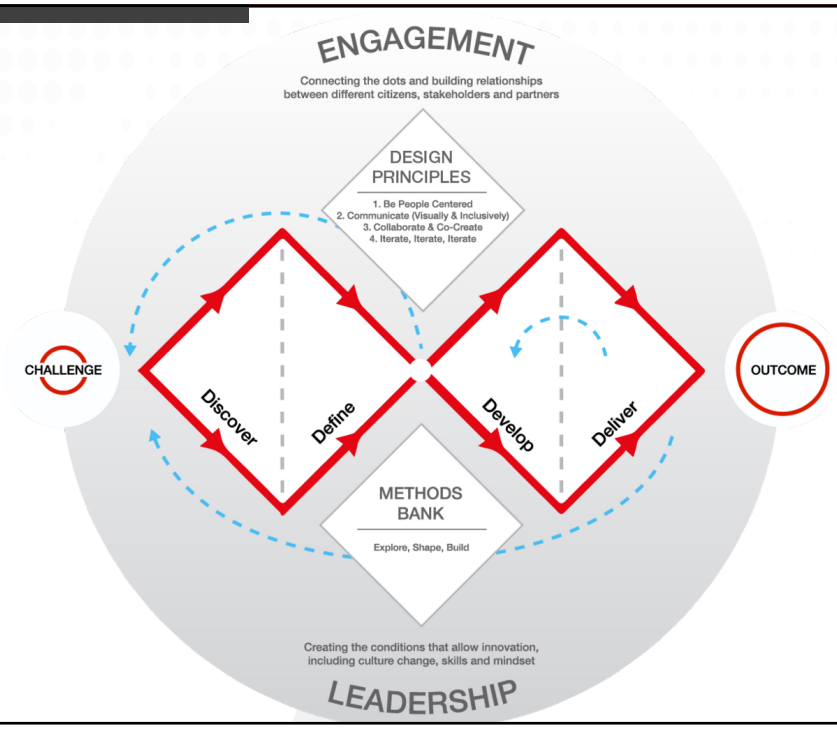

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The process: using the Double Diamond

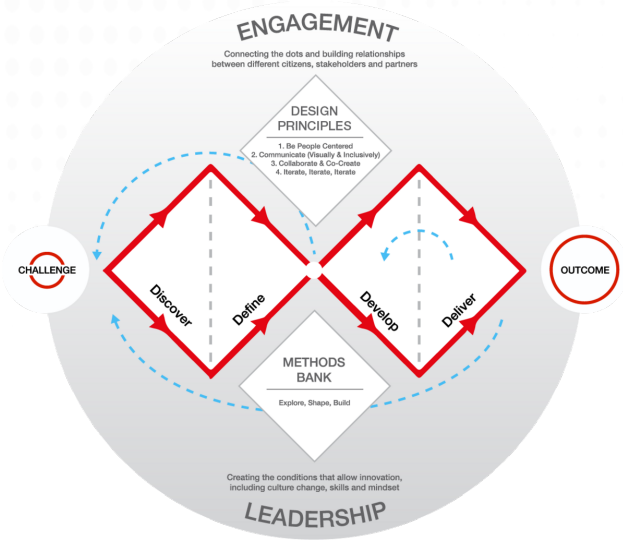
Design Council's Double Diamond clearly conveys a design process to designers and non-designers alike. The two diamonds represent a process of exploring an issue more widely or deeply (divergent thinking) and then taking focused action (convergent thinking).

•**Discover.** The first diamond helps people understand, rather than simply assume, what the problem is. It involves speaking to and spending time with people who are affected by the issues.

•**Define.** The insight gathered from the discovery phase can help you to define the challenge in a different way.

•**Develop.** The second diamond encourages people to give different answers to the clearly defined problem, seeking inspiration from elsewhere and co-designing with a range of different people.

•**Deliver.** Delivery involves testing out different solutions at small-scale, rejecting those that will not work and improving the ones that will.



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Who to Involve in ID?

- Identify stakeholders for the product
  - Individuals or groups who influence or are influenced by the success or failure of the project
  - Users are not the only stakeholders
  - Direct and indirect users, developers, executives, legislators
- Involving stakeholders is not always easy
  - May not need to involve all groups
  - Knowing the stakeholder groups is only the first step!

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Degrees of User Involvement

- Member of the design team
- Small group or individual activities
- Online contributions from thousands of users
  - Online Feedback Exchange (OFE) systems
  - Crowdsourcing design ideas
  - Citizen engagement
- Participatory design
- User involvement after product release
  - A/B testing
  - Customer reviews

What is People-centered Approach?

- **Three principles** for "useful & easy-to-use" products
  - Early focus on users and tasks
  - Empirical measurement
  - Iterative design
- Expanded to include
  1. People's tasks and goals drive development
  2. People's behaviour and context is studied
  3. People's characteristics are designed for
  4. Stakeholders are consulted throughout
  5. Design decisions are taken in this context
  - Specific usability and user experience goals
  - Iteration through all interaction design activities

Four Basic Activities of ID

1. Discovering requirements
2. Designing alternatives
3. Prototyping alternative designs
4. Evaluating product and its user experience throughout

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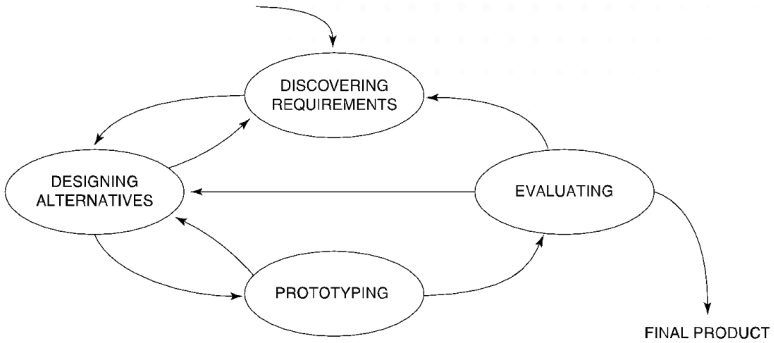
A Simple ID Lifecycle Model

Exemplifies a people-centered design approach

Focus:
Interaction design & usability



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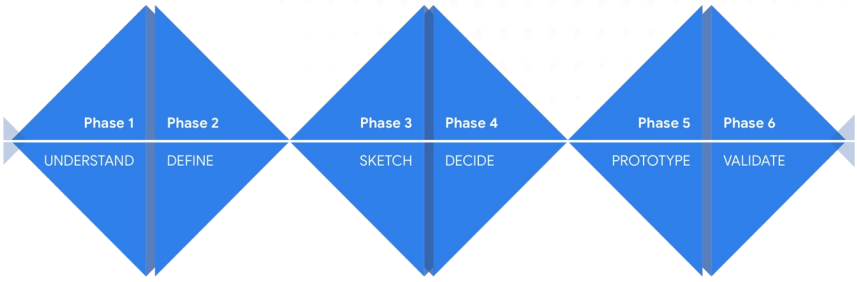
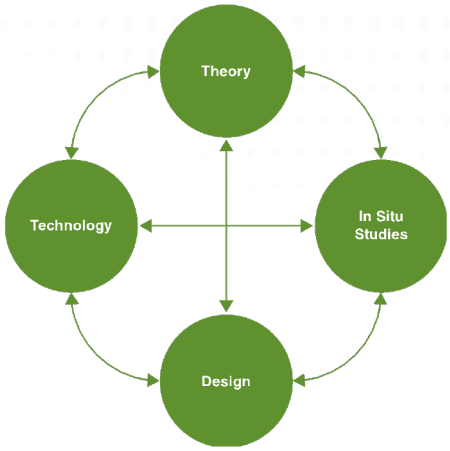
Another Lifecycle Model: Design Sprints



Source: designsprintkit.withgoogle.com/methodology/overview

13

Another Lifecycle Model: Research in the Wild



A framework for research in the wild studies

Source: Rogers and Marshall, 2017, p6. (used courtesy of Morgan and Claypool)

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Comparison of HCI Lifecycle Models

Aspect	Simple ID Lifecycle	Double Diamond	Design Sprint	Research in the Wild
Main Goal	Usable interactive systems	Find problem & solution	Rapid prototype & test	Real-world understanding
Origin	HCI	Design Council (UK)	Google Ventures	HCI Research
Structure	Req → Design → Proto → Eval	Discover → Define → Develop → Deliver	Understand → Sketch → Decide → Proto → Test	Design → Deploy → Observe → Analyze
Duration	Flexible	Flexible	5 days	Long-term
Iteration	Yes	Yes	After sprint	Continuous
Focus	Usability	Creativity & framing	Fast validation	Real context & behavior
Environment	Controlled/lab	Flexible	Test sessions	Real world
Users	Participants	Stakeholders	5-7 users	Real users
Data	Usability metrics	Insights	Feedback	Contextual data
Output	Improved interface	Defined direction	Tested prototype	Research insights
Strength	Structured, usability	Broad thinking	Fast & collaborative	High realism
Limitation	Slow	Abstract	Shallow depth	Hard to control
Best For	HCI systems	Complex design	Early testing	Real-world use

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SOME PRACTICAL ISSUES

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Some Practical Issues

- How to find out what people need
- How to decide what to design
- How to generate alternative designs
- How to choose among alternatives
- How to integrate interaction design activities with other lifecycle models

How to find out what people need

- Users rarely know what is possible
- Instead:
  - Explore the problem space
  - Investigate potential users and their activities
  - Try out ideas with potential users
- Focus on peoples' goals, usability, and user experience goals, rather than expect stakeholders to articulate requirements

How to decide what to design

- Explore
  - What is the current user experience?
  - Why is a change needed?
  - How will this change improve the situation?

- Articulating the problem space
  - Team effort
  - Explore different perspectives
  - Avoid incorrect assumptions and unsupported claims

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How to generate alternative designs

- Humans tend to stick with something that works
- Considering alternatives helps identify better designs
- Where do alternative designs come from?
  - 'Flair and creativity': research and synthesis
  - Cross-fertilisation of ideas from different perspectives
  - Users can generate different designs
  - Prompts to provoke thinking, e.g. SCAMPER
  - Seek inspiration: other products and domain
  - Different perspectives
- Balancing constraints and trade-offs

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How to choose among alternatives

- Interaction design focuses on externally-visible and measurable behavior
- Technical feasibility
- Evaluation with users or peers
  - Prototypes not static documentation because behavior is key
- A/B Testing
  - Online method to inform choice between alternatives
  - Nontrivial to set appropriate metrics and choose user group sets
- Quality thresholds
  - Different stakeholder groups have different quality thresholds
  - Usability and user experience goals lead to relevant criteria

How to integrate ID activities within other models

- Integrating interaction design activities in lifecycle models from other disciplines requires careful planning
- Software development lifecycle models are prominent
- Integrating with agile software development:
  - It incorporates tight iterations
  - It champions early and regular feedback
  - It handles emergent requirements
  - It aims to strike a balance between flexibility and structure

AGILEUX

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Technical Debt in UX

- Making compromises that are expedient in short term, but cause problems in long term
- Refactoring to correct pragmatic trade-offs
- UX debt arises from trade-offs
  - A history of neglecting UX
  - Large portfolio of products developed independently

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Agile Development

- Short (one to three week) timeboxes of iterative development (sprint, iteration, cycle)
- Early and repeated customer/user feedback
- Re-prioritization of work based on customer or user feedback so that emergent requirements can be handled
- Many approaches, for example, eXtreme Programming (XP), Scrum, Kanban, and DSDM

AgileUX

- Integrates techniques from interaction design and agile methods
- AgileUX requires balancing research and reflection for good UX with rapid iterations incorporating user feedback
- In Agile iterations, requirements are elaborated and re-prioritised, rather than specified up-front

AgileUX (cont.)

- All techniques in UX are still relevant, but when to use and how much needs careful planning
- Focus on product, not design, as deliverable
- Cross-functional teams
- Three practical areas: user research, aligning work practices, and documentation

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User Research

- Seeks to characterise users, tasks, and context through data gathering and analysis
- Detailed user research cannot be fitted within a limited timebox
  - Some user research can be performed in iteration 0 (zero), before implementation starts
  - Ongoing program of user research over a longer period of time

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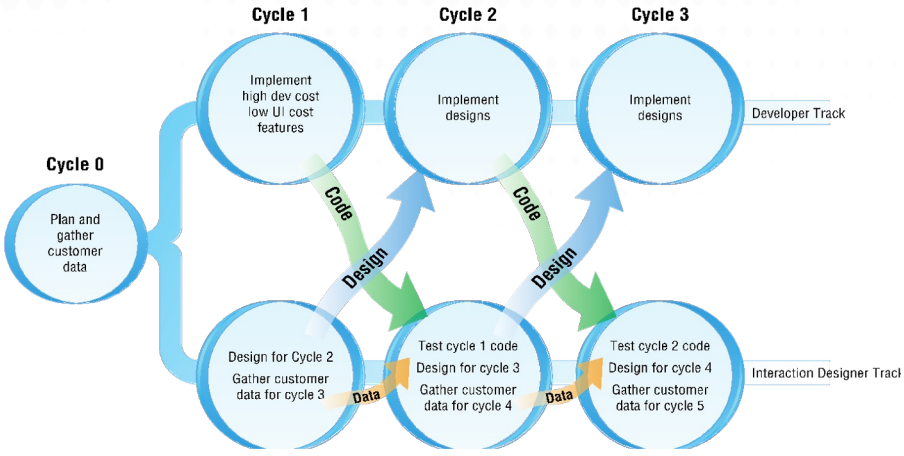
Aligning Work Practices

- Designing a complete product upfront causes problems because of re-prioritization
- Some upfront work is needed (technical and UX)
- Use a dual tracks approach:
  - Create product vision before development starts
  - Do design work one iteration ahead of development
  - Some teams work two iterations ahead

Dual Tracks Approach to AgileUX



Source: Sy, D. (2007) Adapting usability investigations for development, Journal of Usability Studies 2(3), May, 112–130.

Aligning Work Practices (cont.)

- Advantages of dual tracks approach:
  - No design time wasted on features not implemented
  - Usability testing and contextual inquiry could be done on the same customer visit, saving time
  - Timely feedback on the designs was received from developers and customers
- Agile flexibility supports schedule changes if a problem is found
- Parallel tracks is commonly used

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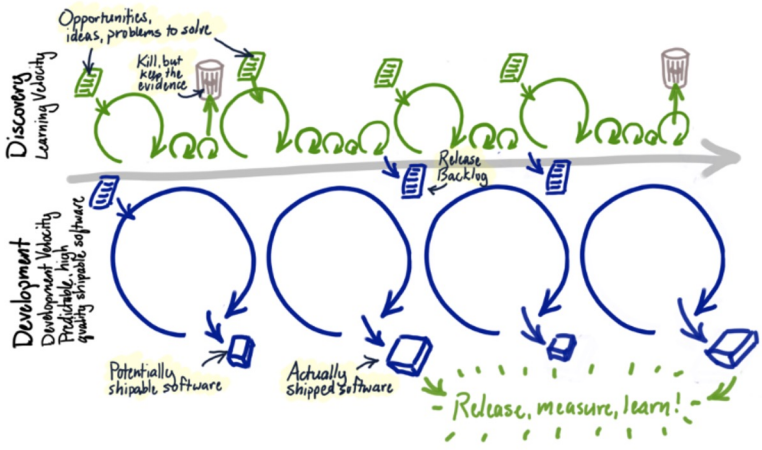
Dual Tracks: Discovery & Development



Source: www.jpattonassociates.com/dual-track-development

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Documentation

- Common communication approach for UX designers
- Agile encourages minimal documentation so more time can be spent on design and discussion
- Only use documentation where needed. Ask:
  - Can the amount of time spent on documentation be reduced?
  - Who uses documentation?
  - What is the minimum needed by readers?
  - How efficient is the sign-off process?
  - Is there duplication anywhere?
  - How polished does documentation need to be?

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DESIGN PATTERNS

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Design Patterns

- Capture design experience:
  - A solution to a problem in a context
  - Can be instantiated in many ways: generative
- Patterns may be individual, in languages, in catalogues, galleries, or libraries
- Patterns often are associated with software components, for example, GitHub or platform websites

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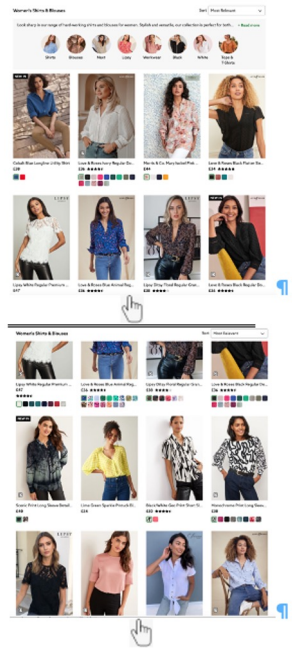
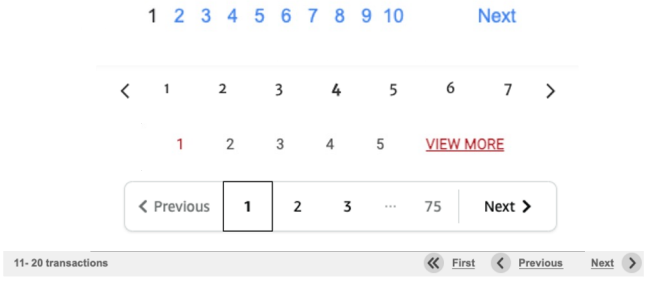
Design Patterns



Continuous scrolling

Pagination



40

Bad Design Patterns

- Capture design experience — but that doesn't necessarily mean good design:
  - Anti-patterns: Don't do it this way! (classing e.g.: "click here")
  - Dark patterns: deliberate tricks

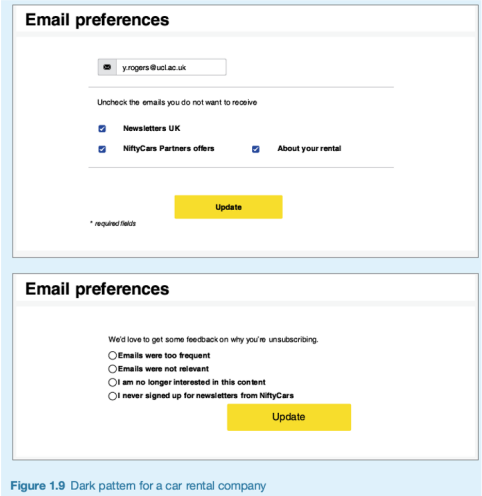


Figure 1.9 Dark pattern for a car rental company

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Design Language and Systems

- Structured collection of patterns and components
- May also include brand or accessibility guidelines
- Reduce effort, support learning between designers and increase collaboration
- Larger chunks of design can be reused
- User flows, e.g. overflow.io or uxarchive.com)

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OPEN SOURCE RESOURCES & TOOLS FOR ID

....

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Open Source Resources

- Components, frameworks, systems available free of charge
- Community-driven
- Available for interaction design:
  - Design pattern libraries
  - Bootstrap framework

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Tools for Interaction Design

- Tools support all aspects of the design process:
  - Creativity, sketching, simulation, brainstorming, library search, mind mapping, and video capture
- Tools integrate together to speed up prototyping
- Tooling landscape changes all the time
- Interactive wireframes or mockups can be produced using, for example:
  - Sketch©
  - Balsamiq©
  - Axure©
  - Figma©
- Higher-fidelity prototypes can be produced by linking interactive wireframes to design pattern library with software components

Summary (1)

- Four basic activities in interaction design process
  - Discovering requirements
  - Designing alternatives
  - Prototyping
  - Evaluating
- People-centered design rests on three principles
  - Early focus on users and tasks
  - Empirical measurement
  - Iterative design
- Understand problem space before starting design
- Involve stakeholder groups throughout development
- Generate alternative designs and evaluate them

Summary (2)

- AgileUX refers to approaches that integrate UX design and agile development
  - Needs a balance between research and reflection, and rapid iterations
  - Requirements are repeatedly re-prioritized, which seeks to avoid wasted effort
  - UX design activities need careful planning: when, how much, and how to take forward
- Design patterns and implementation libraries support good user experience design (look out for anti- and dark patterns)
- Open source resources, for example, on GitHub, make development of standard applications easier and quicker
- A range of digital tools to support interaction design in practice is available

Topic 2: User Behaviour (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

COGNITIVE ASPECT

1

01 WHAT IS COGNITION?

02 COGNITIVE PROCESSES & DESIGN IMPLICATIONS

Attention, Perception, Memory

03 COGNITIVE FRAMEWORKS

2

WHAT IS COGNITION

3

What is Cognition?

- Thinking, remembering, learning, daydreaming, decision-making, seeing, reading, talking, writing…

- Ways of classifying cognition at a higher level:
  - Experiential vs. reflective cognition (Norman, 1993)
  - Fast vs slow thinking (Kahneman, 2011)

4

Which involves fast vs slow thinking?

- 2 + 2 =
- 21 × 29 =
- What colour eyes do you have?
- How many colours are there in the rainbow?
- How many months in the year have 31 days?
- What is the name of the first school you attended?

5

How can understanding cognition help?

- Provides knowledge about what people can and cannot be expected to do
- Identifies and explains the nature and causes of problems that people encounter when using technology
- Provides theories, modeling tools, guidance, and methods that can lead to the design of better interactive products

6

COGNITIVE PROCESSES & DESIGN IMPLICATIONS

7

Cognitive Processes

- **Attention**
- **Perception**
- **Memory**
- Learning
- Reading, speaking and listening
- Problem-solving, planning, reasoning and decision-making

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Cognitive Process: Attention

- Selecting things on which to concentrate at a point in time from the mass of stimuli around us
- Allows us to focus on information that is relevant to what we are doing
- Involves audio and/or visual senses
  - Focused and divided attention
- Enables us to be selective in terms of the mass of competing stimuli, but limits our ability to keep track of all events
- Design recommendation
  - Information at the interface should be structured to capture peoples' attention, for example, use perceptual boundaries (windows), colour, reverse video, sound, and flashing lights

Activity

Find the price for a double room at the Quality Inn in Pennsylvania

Pennsylvania
Bedford Motel/Hotel: Crinaline Courts
(814) 623-9511 S: $118 D: $120
Bedford Motel/Hotel: Holiday Inn
(814) 623-9006 S: $129 D: $136
Bedford Motel/Hotel: Midway
(814) 623-8107 S: $121 D: $126
Bedford Motel/Hotel: Penn Manor
(814) 623-8177 S: $119 D: $125
Bedford Motel/Hotel: Quality Inn
(814) 623-5189 S: $123 D: $128
Bedford Motel/Hotel: Terrace
(814) 623-5111 S: $122 D: $124
Bradley Motel/Hotel: De Soto
(814) 362-3567 S: $120 D: $124
Bradley Motel/Hotel: Holiday House
(814) 362-4511 S: $122 D: $125
Bradley Motel/Hotel: Holiday Inn
(814) 362-4501 S: $132 D: $140
Breezewood Motel/Hotel: Best Western Plaza
(814) 735-4352 S: $120 D: $127
Breezewood Motel/Hotel: Motel 70
(814) 735-4385 S: $116 D: $118

Activity

Find the price of a double room at the Holiday Inn in Columbia

South Carolina

City	Motel/Hotel	Area code	Phone	Rates Single	Rates Double
Charleston	Best Western	803	747-0961	$126	$130
Charleston	Days Inn	803	881-1000	$118	$124
Charleston	Holiday Inn N	803	744-1621	$136	$146
Charleston	Holiday Inn SW	803	556-7100	$133	$147
Charleston	Howard Johnsons	803	524-4148	$131	$136
Charleston	Ramada Inn	803	774-8281	$133	$140
Charleston	Sheraton Inn	803	744-2401	$134	$142
Columbia	Best Western	803	796-9400	$129	$134
Columbia	Carolina Inn	803	799-8200	$142	$148
Columbia	Days Inn	803	736-0000	$123	$127
Columbia	Holiday Inn NW	803	794-9440	$132	$139
Columbia	Howard Johnsons	803	772-7200	$125	$127
Columbia	Quality Inn	803	772-0270	$134	$141
Columbia	Ramada Inn	803	796-2700	$136	$144
Columbia	Vagabond Inn	803	796-6240	$127	$130

Findings based on Activity

- Tullis (1987) found that the two screens produced quite different results
  - 1st screen: Took an average of 5.5 seconds to search
  - 2nd screen: Took 3.2 seconds to search
- Why, since both displays have the same density of information (31percent)?
- Spacing
  - In the 1st screen, the information is bunched up together, making it hard to search
  - In the 2nd screen, the characters are grouped into vertical categories of information making it easier

Multitasking and Attention

- Is it possible to perform multiple tasks without one or more of them being detrimentally affected?
- Multitasking can cause people to lose their train of thought, make errors, and need to start over
- Ophir et al. (2009) compared heavy vs light multitaskers
  - Heavy multitaskers were more prone to being distracted than those who infrequently multitask
  - Heavy multitaskers are easily distracted and find it difficult to filter irrelevant information

Multitasking at Work

- It is increasingly common for workers to multitask
  - For example, hospital workers have to attend to multiple screens in an operating room that provide new kinds of real-time information
  - This requires clinician's constant attention to check if any data is unusual or anomalous
  - Need to develop new attention and scanning strategies

Apps to Help People Refocus

- Apps have been designed to help people get back on track or avoid being distracted
  - many are designed to block or limit the distracting sources, such as notifications, newsfeeds and social media

- An example is FocusMe that claims to ""wall off online temptation"
  - help people improve their willpower so they can develop better digital habits

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Is it OK to use phone when driving?

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No!

- Driving is very demanding and drivers are prone to being distracted and causing accidents
  - drivers' reaction times are longer to external events when talking on the phone in a car (Caird et al., 2018)
- Drivers often try to imagine what the other person's face is like - the person to whom they are speaking
  - Doing so competes with the processing resources needed to enable them to notice and react to what is in front of them
  - when using their phones they rely more on their expectations about what is likely to happen next
- But some drivers persist despite the dangers!

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Are hands-free phones safer to use when driving?

- No, as same type of cognitive processing is happening when talking
- The same thing happens when talking with front seat passenger
  - But both can stop in mid-sentence if a hazard is spotted allowing the driver to switch immediately to the road
  - So, it's less dangerous talking to a front seat passenger than a remote person
  - A remote person on the end of a phone is not privy to what the driver is seeing and will carry on the conversation when there is a hazard
  - This makes it difficult for the driver to switch all their attention to the road

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Design Implications for Attention

- Context: Make information salient when it needs to be attended to at a given stage of a task
- Use techniques to achieve this:
  - For example, colour, ordering, spacing, underlining, sequencing, and animation
- Avoid cluttering visual interfaces with too much information
- Consider designing different ways to support effective switching and returning to an interface

Cognitive Process: Perception

- How information is acquired from the world and transformed into experiences
- Obvious implication is to design representations that are readily perceivable, for instance:
- Text should be legible
- Icons should be easy to distinguish and read

Activity

Is colour contrast good?

Find Italian

Black Hills Forest	Peters Landing	Jefferson Farms	Devlin Hall
Cheyenne River	Public Health	Psychophysics	Positions
Social Science	San Bernardino	Political Science	Hubard Hall
South San Jose	Moreno Valley	Game Schedule	Fernadino Beach
Badlands Park	Altamonte Springs	South Addision	Council Bluffs
Juvenile Justice	Peach Tree City	Cherry Hills Village	Classical Lit
Results and Stats	Highland Park	Creative Writing	Sociology
Thousand Oaks	Manchesney Park	Lake Havasu City	Greek
Promotions	Vallecito Mts.	Engineering Bldg	Wallace Hall
North Palermo	Rock Falls	Sports Studies	Concert Tickets
Credit Union	Freeport	Lakewood Village	Public Radio FM
Wilner Hall	Slaughter Beach	Rock Island	Children's Museum
Performing Arts	Rocky Mountains	Deerfield Beach	Writing Center
Italian	Latin	Arlington Hill	Theater Auditions
Coaches	Pleasant Hills	Preview Game	Delaware City
McKees Rocks	Observatory	Richland Hills	Scholarships
Glenwood Springs	Public Affairs	Experts Guide	Hendricksville
Urban Affairs	Heskett Center	Neff Hall	Knights Landing
McLeansboro	Brunswick	Grand Wash Cliffs	Modern Literature
Experimental Links	East Millinocket	Indian Well Valley	Studio Arts
Graduation	Women's Studies	Online Courses	Hughes Complex
Emory Lindquist	Vacant	Lindquist Hall	Cumberland Flats
Clinton Hall	News Theatre	Fisk Hall	Central Village
San Luis Obispo	Candlewood Isle	Los Padres Forest	Hoffman Estates

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Activity

Are borders and white space better?

Find French

Webmaster	Curriculum	Student Life	Dance
Russian	Emergency (EMS)	Accountancy	Gerontology
Athletics	Statistics	McKnight Center	Marketing
Go Shockers	Award Documents	Council of Women	College Bylaws
Degree Options	Language Center	Commute	Why Wichita?
Newsletter	Future Shockers	Small Business	Tickets
Geology	Intercollegiate	Thinker & Movers	Career Services
Manufacturing	Bowling	Alumni	Doers & Shockers
Management	Wichita Gateway	Foundations	Core Values
UCATS	Transfer Day	Corbin Center	Grace Wilkie Hall
Alumni News	Job Openings	Jardine Hall	Strategic Plan
Saso	Live Radio	Hugo Wall School	Medical Tech
Educational Map	Beta Alpha Psi	Staff	Softball, Men's
Physical Plant	Liberal Arts	Aerospace	McKinley Hall
Graphic Design	Counseling	Choral Dept.	Email
Non Credit Class	Biological Science	Alberg Hall	Dental Hygiene
Media Relations	Duerksen Fine Art	French	Tenure
Advertising	EMT Program	Spanish	Personnel Policies
English	Religion	Parents	Instrumental
Graduate Complex	Art Composition	Wrestling	Nursing
Music Education	Physics	Philosophy	Opera
Advising Center	Entrepreneurship	Wichita Lyceum	Sports History
Medical School	Koch Arena	Fairmount Center	Athletic Dept.
Levitt Arena	Roster	Women's Museum	Health Plan

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Findings based on Activity

- Weller (2004) found people took less time to locate items for information that was grouped
  - Using a border (2nd screen) compared with using color contrast (1st screen)
- Some argue that too much white space on web pages is detrimental to search process
  - Makes it hard to find information

- Do you agree?

Activity

Which is the easiest to read and why?

What is the time?

What is the time?

What is the time?

What is the time?

What is the time?

Design Implications for Perception

- Icons should enable users to distinguish their meaning readily
- Bordering and spacing are effective visual ways of grouping information
- Sounds should be audible and distinguishable
- Research proper color contrast techniques when designing an interface:
  - Yellow on black or blue is fine
  - Yellow on green or white is a no-no
- Haptic feedback should be used judiciously

Cognitive Process: Memory

- Involves recalling various kinds of knowledge that allow people to act appropriately
  - For example, recognising someone's face or remembering someone's name
- First encode and then retrieve knowledge
- We don't remember everything-it involves filtering and processing what is attended to
- Context is important as to how we remember (that is, where, when, how, and so on)
- We recognise things much better than being able to recall things
  - We remember less about objects that we have photographed than when we observe them with the naked eye (Henkel, 2014)

Processing in Memory

- Encoding is first stage of memory
  - Determines which information is attended to in the environment and how it is interpreted
- The more attention paid to something…
- The more it is processed in terms of thinking about it and comparing it with other knowledge…
- The more likely it is to be remembered
  - For example, when learning about HCI, it is much better to reflect upon it, carry out exercises, have discussions with others about it, and write notes than just passively read a book, listen to a lecture or watch a video about it

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What is the difference between working & long-term memory?

- Working memory refers to our ability to recall a small amount of information from a recent time period
  - e.g. what someone said last during a conversation.
- Long-term memory is the capacity to recall memories from a longer time ago
  - e.g a tune someone heard from two decades ago

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Context is Important

- Context affects the extent to which information can be subsequently retrieved
- Sometimes it can be difficult for people to recall information that was encoded in a different context:
  - "You are on a train and someone comes up to you and says hello. You don't recognise him for a few moments, but then realise it is one of your neighbours. You are only used to seeing your neighbour in the hallway of your apartment building, and seeing him out of context makes him difficult to recognise initially"

Activity

- Try to remember the dates of your grandparents' birthday
- Try to remember the cover of the last two books you read
- Which was easiest? Why?

- People are very good at remembering visual cues about things
  - e.g. the colour of items, the location of objects
- They find it more difficult to learn and remember arbitrary material, e.g. birthdays and phone numbers

Recognition vs. Recall

- Command-based interfaces require people to recall from memory a name from a possible set of 100s of names
- Graphical interfaces provide visually-based options (menus, icons) that people need only browse through until they recognize one
- Web browsers provide tabs and history lists of visited URLs that support recognition memory

The problem with the classic 7+or-2 memory phenomenon

- George Miller's (1956) theory of how much information people can remember
- People's immediate memory capacity is very limited to 7+or-2
- Has been applied in interaction design when considering how many options to display
- But is it a good use of a theory in HCI?
- Is it helpful?

When creating an interface, should the designer…

- Present only 7 options on a menu
- Display only 7 icons on a tool bar
- Have no more than 7 bullets in a list
- Place only 7 items on a pull down menu
- Place only 7 tabs on the top of a website page?
- Not necessarily…

The reason is…

- People can scan lists of bullets, tabs, and menu items for the one they want
- They don't have to recall them from memory, having only briefly heard or seen them
- So you can have more than nine at the interface
  - For instance, history lists of websites visited
- Sometimes a small number of items is good
  - For example, smart watch displays
- Depends on task and available screen estate

Saving and Searching Files

- The number of documents written, images created, music files recorded, videoclips downloaded, emails with attachments saved, etc. is huge
  - Where and how to save them all?
  - Then remembering what they were called and where to find them again?
- Naming most common means of encoding them
- But can be difficult to remember, especially when you have 10,000s
- How might such a process be facilitated taking into account people's memory abilities?

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Personal Information Management

- Bergman and Whittaker (2016) suggest helping people manage their "digital stuff" better
  - by deciding what personal information to keep
  - how to organise that information when storing it
  - which strategies to use to retrieve it later
- Provide richer metadata tools, such as time stamping, categorizing, tagging, and attribution (for example colour, text, icon, sound, or image)
- However, trying to remember which metadata was created some time back may also prove to be difficult!
- What strategy do you find works for you best?

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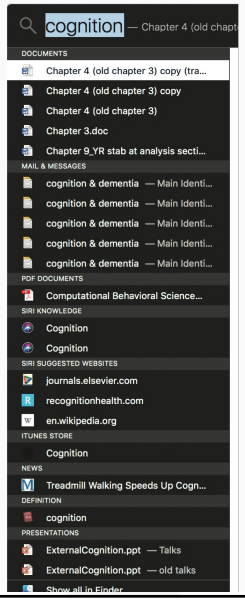
Apple's Spotlight Search Tool



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Memory Load

- Online/mobile and phone banking now require users to provide multiple pieces of information to access their account
  - For instance, ZIP code, birthplace, a memorable date, first school attended
  - Known as multifactor authentication (MFA)
- Why?
  - Increased security concerns
- Password managers, such as LastPass, have been developed that require only one master password
  - reduces stress and memory load on users
- Passwords could become extinct with the widespread use of biometrics and computer vision algorithms

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Digital Forgetting

- When might you wish to forget something that is online?
  - When you break up with a partner
  - Emotionally painful to be reminded of them through shared photos, social media, and so on.

- Sas and Whittaker (2013) suggest ways of harvesting and deleting digital content
  - For example, making photos of ex into an abstract collage
  - Helps with closure

Memory Aids

- SenseCam, developed by Microsoft Research Labs (now Autographer)
  - A wearable device that intermittently takes photos without any user intervention while worn
  - Digital images taken are stored and revisited using special software
  - Has been found to improve people's memory, especially those suffering from dementia

- Other aids include RemArc, which triggers long-term memory using old BBC materials

SenseCam

Figure 3.5 The SenseCam device and a digital image taken with it
Source: ©Microsoft Research Cambridge.

Design Implications for Memory

- Reduce cognitive load by avoiding long and complicated procedures for carrying out tasks
- Design interfaces that promote recognition rather than recall
- Provide users with various ways of labelling digital information to help them easily identify it again
  - For example, folders, categories, colour, flagging, and time stamping

COGNITIVE FRAMEWORKS

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Cognitive Frameworks

- These are used to explain and predict user behaviour at the interface
  - Based on theories of behaviour
  - Focus is on mental processes that take place
  - Also use of artifacts and representations
- Most well known are:
  - Mental models
  - Gulfs of execution and evaluation
  - Distributed cognition
  - External and embodied cognition

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Mental Models

- People develop an understanding of a system through learning about and using it
- This knowledge is sometimes described as a mental model:
  - How to use the system (what to do next)
  - What to do with unfamiliar systems or unexpected situations (how the system works)
- People make inferences using mental models of how to carry out tasks

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More on Mental Models

- Craik (1943) described mental models as:
  - Internal constructions of some aspect of the external world enabling predictions to be made
- Involves unconscious and conscious processes
  - Imagery and analogies are activated
- Deep versus shallow models
  - For example, how to drive a car and how it works

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Everyday Reasoning and Mental Models

- 1️⃣ You arrive home on a hot afternoon, and your house feels like an oven. Do you set the air conditioner to the lowest temperature (e.g., 16°C) thinking it will cool the room faster, or do you set it to a comfortable temperature (e.g., 24°C)?

- 2️⃣ You want to iron your clothes quickly before going out. The iron's instructions suggest setting it to medium heat for your fabric type, but you turn it to the highest heat instead, thinking it will remove wrinkles faster.

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Reasoning

- Many people when asked (a) choose the first option
- Why?
  - They think it will cool the room up quicker
  - General valve theory, where 'more is more' principle is generalised to different settings (for instance, gas pedal, gas cooker, tap, radio volume)
  - But it is a wrong mental model for thermostats based on on-off switch model
- Many people when asked (b) choose the first option
  - Iron work on the same principle as thermostats
- Most of us have erroneous mental models (Kempton, 1996)

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Erroneous Mental Models

- Lots of people hit the button for elevators and pedestrian crossings at least twice
  - Why? Think it will make the lights change faster or ensure that the elevator arrives!

- What kinds of mental models do users have for understanding how interactive devices work?
  - Poor, often incomplete, easily confusable, based on inappropriate analogies and superstition (Norman, 1983)

How can UX be designed to help people build better mental models?

- Clear and easy to use instructions
- Appropriate tutorials and contextual sensitive guidance
- Provide online videos and chatbot windows when needing help
- Transparency: to make interfaces intuitive to use
- Affordances of what actions an interface allows
  - For example, swiping, clicking, or selecting

Limitations

- Based on modelling mental activities that happen exclusively inside the head
- Do not adequately account for how people interact with computers and other devices in real world
- Nowadays the model is rarely used as other models have superseded it
- Instead, a move towards understanding cognitive activities in the context in which they occur and analysing how we interact with technologies in the wild

Design Implication

- Provide external representations at the interface that can reduce memory load and facilitate computational offloading
  - For example, information visualizations have been designed to allow people to make sense and rapid decisions about masses of data

Summary

- Cognition involves many processes including attention, memory, perception, and learning
- The way an interface is designed can greatly affect how well people can perceive, attend, learn, and remember how to do their tasks
- Theoretical frameworks, such as mental models, provide ways of understanding how and why people interact with products
- This can lead to thinking about how to design better products

Topic 2: User Behaviour (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

EMOTIONAL INTERACTION

1

01 EMOTIONS AND BEHAVIOUR

02 EXPRESSIVE INTERFACES

03 AFFECTIVE COMPUTING & EMOTIONAL AI

04 PERSUASIVE TECHNOLOGY & BEHAVIOURAL CHANGE

2

EMOTIONS & BEHAVIOUR

3

Emotions and Behaviour

- HCI has traditionally been about designing efficient and effective systems
- Now more about how to design interactive systems that make people respond in certain ways
  - For example, to be happy, to be trusting, to learn, or to be motivated
- *Emotional interaction* is concerned with how we feel and react when interacting with technologies
- *Affective computing* is improving with better recognition software and machine learning algorithms

4

Emotional Interaction

- What makes us happy, sad, annoyed, anxious, frustrated, motivated, delirious, and so on
  - Translating this into different aspects of the user experience
- Why people become emotionally attached to certain products (for instance, virtual pets)
- Can social robots help reduce loneliness and improve well-being?
- How to change human behaviour through the use of emotive feedback

5

Activity

- Try to remember the emotions you went through when buying a big-ticket item online (for example, a refrigerator, a vacation, a computer)

- How many different emotions did you go through?

6

Why has this simple way of obtaining visitor feedback been so effective?

8

8

Pulling at the heart strings with an emotive message

9

Pulling at the heart strings with an emotive message

10

Is it possible to design an interface to match or change how we are feeling?

- Should an interface be designed to improve how we feel?
  - If so, how?
- Our moods and feelings are continuously changing
  - How does the interface keep track and know when to do something?
- What moods match which kinds of interfaces?
- How would you design an interface for when someone is happy, angry, sad, bored, or focused?

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How do emotions affect behaviour and vice versa?

- Does being angry make you concentrate better or more distracted?
- When you are happy do you take more risks such as spend more money or buy more?
- Baumeister et al (2007) argue the relationship is more complex than a single cause-and-effect model

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Automatic (affect) vs. Conscious Emotions

- Emotions can be short-lived (for instance, a fit of anger) or complex and long-lasting (for example, jealousy)
- Emotions have been categorised as automatic or conscious:
  - Automatic ones are rapid and dissipate quickly
  - Conscious ones develop slowly and take a long time to go (for instance, reflection)

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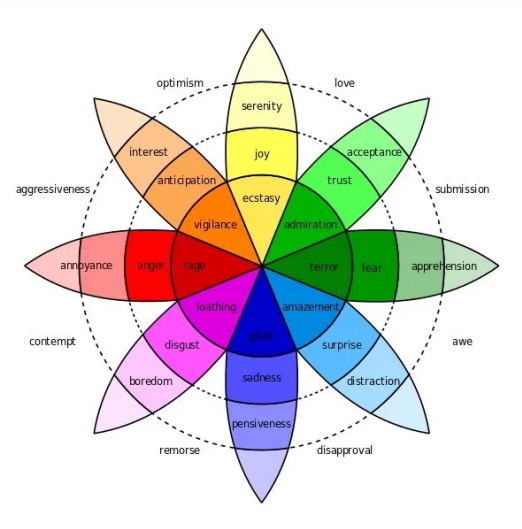
Designing with the 3-levels in mind

- Visceral design refers to making products look, feel, and sound good
- Behavioural design is about use, and it equates with traditional values of usability
- Reflective design is about considering the meaning and personal value of a product

Analysing a Swatch watch design using the model

- Cultural images and graphical elements designed at the reflective level
- Affordances of use at the behavioral level
- Brilliant colours and wild design attract user's attention at the visceral level

Plutchik's (1980) Wheel of Emotions



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Plutchik's Wheel

- Categorises human emotions into 7 well known emotions
  - Anger, Disgust, Fear, Sadness, Anticipation, Joy and Surprise and Trust (not usually considered an emotion)
- Emotional labels are added to these
  - optimism, love, submission, awe, disproval, remorse, contempt, aggressiveness
- Colours used in the wheel reflect the intensity of an emotion
  - the darker the shade, the more intense the emotion is
- Can be used as a 'colour palette' akin to a UX mood board
  - elicit different kinds and levels of emotional response
- It does not provide prescriptive advice on how to instruct design for a selection of emotions

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EXPRESSIVE INTERFACES

18

Expressive Interfaces

- Provide reassuring feedback that can be both informative and fun
- Can also be intrusive, however, causing people to become annoyed and even angry
- Colour, icons, sounds, graphical elements, and animations are used to make the 'look and feel' of an interface appealing
  - Conveys an emotional state
- In turn, this can affect the usability of an interface
  - People are prepared to put up with certain aspects of an interface (for instance, slow download rate) if the end result is appealing and aesthetic

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The Appearance of an Interface

(a) Emotional icons were used in the 1980s to indicate rebooting or crashed computer

- Smiling apple face

(b) Nowadays, computers use more impersonal but aesthetically-pleasing icons to indicate that the user needs to wait

- Beachball

(a)

(b)

The Design of Thermostats

(a) The Nest thermostat has a minimalist and aesthetically-pleasing design

- Round face and simple dial
- Large font and numbers

(b) It is very different from earlier thermostat designs

- Utilitarian and dull

(a)

(b)

Design of Website to be Aesthetics

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AFFECTIVE COMPUTING & EMOTIONAL AI

....

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Affective Computing & Emotional AI

- Affective computing is concerned with how to use computers to recognise and express emotions as humans do (Picard, 1998)
- It involves designing ways for people to communicate their emotional state
- It uses sensing technologies to measure GSR, facial expressions, gestures, and body movement
- Explores how affect influences personal health
- *Emotional AI* aims to automate the measurement of feelings and behavior using AI to infer them from facial expressions and voice
- The goal is to predict user's emotions and aspects of their behavior
  - For example, what is someone most likely to buy online when feeling sad, bored, or happy

Techniques Used

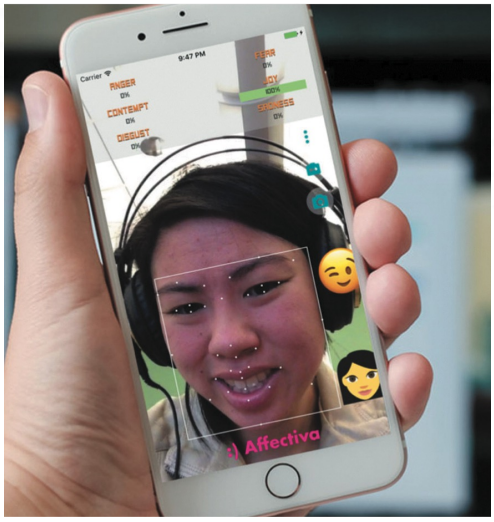
- Cameras for measuring facial expressions
- Biosensors placed on fingers or palms to measure GSR
- Affective expression in speech (for example, intonation, pitch, and loudness)
- Body movement and gestures using accelerometers and motion capture systems

Classification of Emotions

- Six core expressions typically measured:
  - Sadness, disgust, fear, anger, contempt, and joy

- Type of facial expression chosen by AI through detecting presence of absence of:
  - Smiling, eye widening, brow raising, brow furrowing, raising a cheek, mouth opening, upper-lip raising, and wrinkling of the nose

Facial Coding using Affdex software



How is this Emotional Data used?

- If user screws up their face when an ad pops up
  - ➡ Feel disgust
- If user starts smiling
  - ➡ They are feeling happy
- Website can adapt its ad, movie storyline, or content to match user's emotional state
- If system used in a car, it might detect an angry driver and suggest they take a deep breath
- Eye-tracking, finger pulse, speech, and words/phrases also analyzed when tweeting or posting to Facebook

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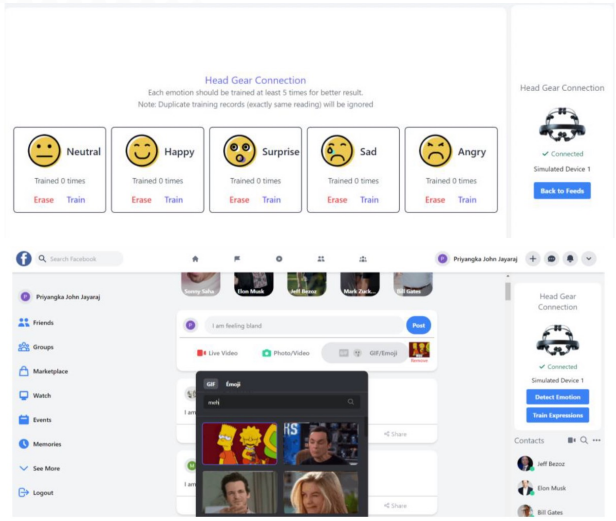
Detecting & Expressing Emotions



Publication:

Priyangka John Jayaraj and Masitah Ghazali. User's Perspectives on Expressing Emotions on Social Media Platforms with Mobile Brain-Computer Interface. In Fusion 2021 Proceedings of 3rd National Symposium on Human-Computer Interaction, 19th August 2021, pp. 8-17. eISBN: 978-967-18511-1-1

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Indirect Emotion Detection

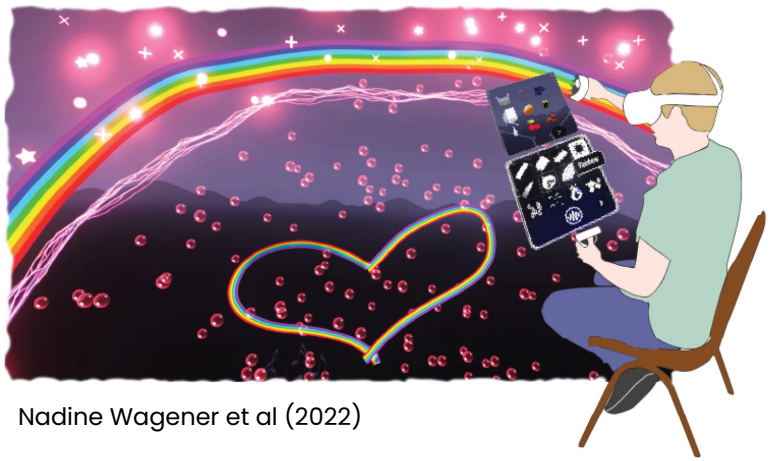
- Also used to infer or predict someone's behaviour
  - e.g, determining a person's suitability for a job or how they will vote in an election

- Do you think it is ethical that technology can read your emotions from your facial expressions or from your tweets?

Detecting and Reflecting on Moods

- How can technology be designed to help people understand more about their moods and mood swings?
- We can be in a good mood one day and then a bad mood the next day. How does this happen and why?
- Not possible to detect different moods in the way tech is used for emotion detection
  - often not expressed through obvious physiological responses
- Mobile mood tracker apps (e.g., Moodnotes, Daylio)
  - intended to help people keep track of their moods
  - to reflect more on why they might be feeling gloomy or cheerful
  - understanding their moods is assumed to help improve well-being
- Virtual reality is also used to enable people to explore their moods (e.g. Mood Worlds)

Mood Worlds: a VR exploratory tool



Nadine Wagener et al (2022)

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PERSUASIVE TECHNOLOGIES & BEHAVIOURAL CHANGE

....

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Persuasive Technologies & Behavioural Change

- Interactive computing systems designed to change people's attitudes and behaviors (Fogg, 2003)
- A diversity of techniques now used to change what they do or think
  - Pop-up ads, warning messages, reminders, prompts, personalized messages, recommendations, or Amazon 1-click
  - Commonly referred to as *nudging*

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Nintendo's Pocket Pikachu

- Developed to change bad habits and improve well being
  - Designed to motivate children to be more physically active on a regular basis
  - Owner of the digital pet that 'lives' in the device is required to walk, run, or jump
  - If owner does not exercise, the virtual pet becomes angry and refuses to play anymore

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How Effective?

- Can interactive technologies that monitor, nag, or behave like a human keep them interested in looking after it and in doing so make themselves more fit?
- How does looking after a virtual pet change a child's behavior?
  - Emotional attachment
  - Happy Pokemon makes them feel good
  - Sulking Pokemon makes them feel bad

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Dilemma: Should voice assistants teach kids good manners?

- Many children talk to Alexa as if she was their friend
- They also learn that it is not necessary to say **please** and **thank you** to her when asking questions
- Is this lack of using etiquette a problem?
- Would it transfer over to real life situations?
  - For example, demanding "Auntie, get me my drink."
- Should parents or voice assistants teach their kids good manners
- How much parental control should voice assistants be given?
  - Would children find it weird or creepy that their Alexa (who is their friend) nags them to clean their teeth?
- Recent research has shown that children know to treat humans differently compared with the way they talk to a voice assistant (Alexis Hiniker et al., 2021)

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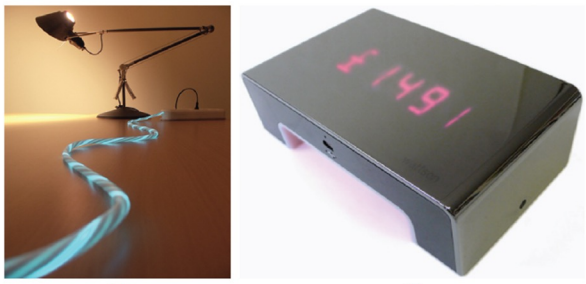
Tracking Devices

- Mobile apps designed to help people monitor and change their behaviour (for instance, fitness, sleeping, or weight)
- Can compare with online leader boards and charts to show how they have done in relation to their peers and friends
- Also apps that encourage reflection, which in turn increase well-being and happiness

Sustainable HCI

- Focus on designing tech interventions to help people reduce their energy consumption
- An effective technique is to provide homeowners with feedback on their consumption
- Simple infographics and emoticons are often most powerful:
  - Encourage people to reflect and see what they can change to reduce their energy consumption
- Peer pressure and social norms are also powerful methods

Energy Reduction



(a) (b)

Figure 5.12 (a) The Power Aware Cord consists of an electrical power strip in which the cord is designed to visualize the energy rather than hiding it. Increase and decrease in use is conveyed through showing glowing pulses, flow, and intensity of light. (b) The Waatson (now a commercial product available in many countries) measures in watts or cost how much electricity someone is using in their home at any moment. This is conveyed in LEDs on the top side. On the underside are colored lights: when they glow blue it shows you are using less than normal; when it changes to purple it indicates that your usage is average: and when it is red it indicates you are using more than normal

Source: (a) Photo taken from the Interactive Institute's research program "Static!" and reproduced with permission. (b) Reproduced with permission from DIY Kyoto Ltd. www.diykyoto.com.

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Energy Reduction via Peripheral Display

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The Tidy Street Project

- Large-scale visualization of the street's electricity usage
  - Stenciled display on the road surface using chalk
  - Provided real-time feedback that all could see change each day
  - Reduced electricity consumption by 15 percent

(Bird and Rogers, 2010)

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Phishing Scams

- Web used to deceive people into parting with personal details
  - For example, PayPal, eBay, and "you won the lottery" emails
- Allows Internet fraudsters to access their bank accounts and draw money from them
- Many vulnerable people fall for it
- The art of deception is centuries old but internet allows ever more ingenious ways to trick people

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Summary

- Emotional aspects of interaction design are concerned with how to facilitate certain states (for example, pleasure) or avoid reactions (for instance, frustration)
- Well-designed interfaces can elicit good feelings in people
- Aesthetically-pleasing interfaces can be a pleasure to use
- Expressive interfaces can provide reassuring feedback to users
- Badly designed interfaces make people frustrated, annoyed, or angry
- Emotional AI and affective computing use AI and sensor technology for detecting people's emotions by analyzing their facial expressions and conversations
- Emotional technologies can be designed to persuade people to change their behaviors

Topic 3: Conceptualising Interaction (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 CONCEPTUAL MODELS

02 INTERFACE METAPHORS

03 INTERACTION TYPES

2

CONCEPTUAL MODEL

3

Conceptual Model

- A conceptual model is:

  "...a high-level description of how a system is organised and operates" (Johnson and Henderson, 2002, p26)

- It enables:

  "...designers to straighten out their thinking before they start laying out their widgets" (Johnson and Henderson, 2002, p28)

- Provides a working strategy and framework of general concepts and their interrelations

4

Components

- Metaphors and analogies
  - Understand what a product is for and how to use it for an activity
- Concepts that people are exposed to through the product
  - Task–Domain objects, their attributes, and operations (for example, saving, revisiting, organising)
- Relationship and mappings between these concepts
  - E.g. whether one object contains one another, and, whether one can revisit through looking at a list of visited sites

5

5

...what concepts are needed to understand how to interact with this interface..?

Mappings?
Insert screwdriver
Turning screwdriver

6

6

First Steps in Formulating a Conceptual Model

- What will people be doing when carrying out their tasks?
- How will the proposed model support these?
- What kind of interface metaphor, if any, will be appropriate?
- What kinds of interaction modes and styles to use?
- Keep in mind when making design decisions how the user will understand the underlying conceptual model

Conceptual Models

- Many kinds and ways of classifying them

- The best conceptual models are often those that appear:
  - Obvious and simple
  - The operations they support are intuitive to use

INTERFACE METAPHORS

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Interface Metaphors

- Interface designed to be similar to a physical entity but also has own properties
  - e.g, desktop metaphor, web portals, search engine
- Can be based on activity, object, or a combination of both
- Exploit user's familiar knowledge, helping them to understand 'the unfamiliar'
- Conjures up the essence of the unfamiliar activity, enabling users to leverage this to understand more aspects of the unfamiliar functionality

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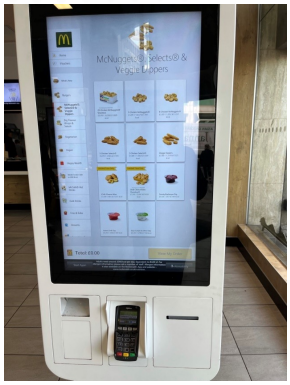
Examples of Interface Metaphors

- Conceptualising what users are doing
  - e.g, surfing the Web
- A conceptual model instantiated at the interface
  - e.g., the desktop metaphor
- Visualising an operation
  - e.g., an icon of a shopping cart into which the user places items

The Card Metaphor

- The card is a very popular metaphor. Why?
  - It has familiar form factor
  - It can easily be flicked through, sorted, and themed
  - Cards have strong associations, providing an intuitive way of organising content that is "card sized"
  - Structures content into meaningful chunks



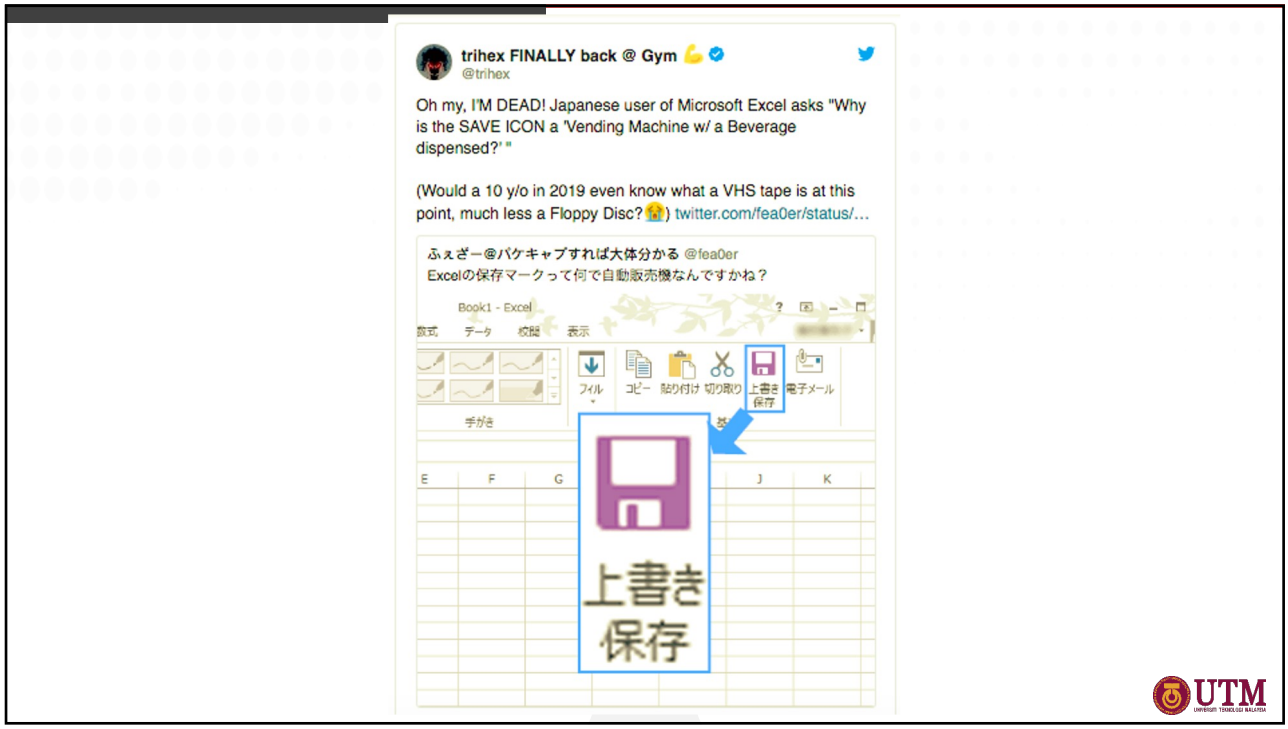
A fast food display

Benefits of Interface Metaphors

- Makes learning new systems easier
- Familiar artefact or action helps people understand the underlying conceptual model
- Can be innovative and enable apps to be made accessible to a greater diversity of people

Problems with Interface Metaphors

- Break conventional and cultural rules
  - e.g, recycle bin placed on desktop
- Can constrain designers in the way that they conceptualize a problem space
- Can conflict with design principles
- Designers can inadvertently use bad existing designs and transfer the bad parts over
- Limits designers' imagination in coming up with new conceptual models
- Often don't scale very well, especially when trying to describe how complex systems work (Cooper, 2020)


trihex FINALLY back @ Gym
@trihex
Oh my, I'M DEAD! Japanese user of Microsoft Excel asks "Why is the SAVE ICON a 'Vending Machine w/ a Beverage dispensed?' "
(Would a 10 y/o in 2019 even know what a VHS tape is at this point, much less a Floppy Disc?) twitter.com/fea0er/status/...
ふぇざー@パケキャプすれば大体分かる @fea0er
Excelの保存マークって何で自動販売機なんですかね？
Book1 - Excel
上書き保存

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INTERACTION
TYPES
17

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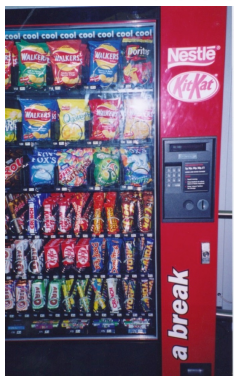
Interaction Types

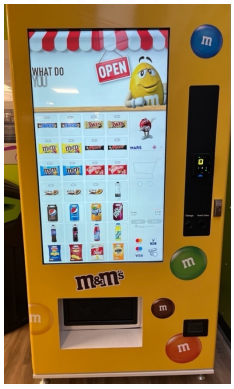
1. Instructing
   - Issuing commands and selecting options
2. Conversing
   - Interacting with a system as if having a conversation
3. Manipulating
   - Interacting with objects in a virtual or physical space by manipulating them
4. Exploring
   - Moving through a virtual environment or a physical space
5. Responding
   - The system initiates the interaction and the person chooses whether to respond

1. Instructing

- Where someone instruct a system and tells it what to do
  - e.g.Tell the time, print a file, or save a file
- Very common conceptual model underlying a diversity of devices and systems
  - e.g. Word processors, VCRs, vending machines
- The main benefit is that instructing supports quick and efficient interaction
  - Good for repetitive kinds of actions performed on multiple objects

Which is easiest and why?



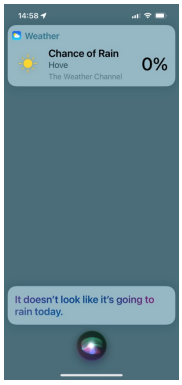


2. Conversing

- Underlying model of having a conversation with another human
- Ranges from simple voice recognition menu-driven systems to more complex 'natural language' dialogs
- Examples include timetables, search engines, advice-giving systems, help systems, virtual agents, chatbots, toys, and pet robots designed to converse with people

Pros and Cons of Conversational Model

- Allows people to interact with a system in a way that is familiar to them
  - can make them feel comfortable, at ease, and less scared
- Misunderstandings can arise when the system does not know how to parse what someone says
  - e.g, voice assistants can misunderstand what children say



Siri's response when asked if a person needs an umbrella today

Pros and Cons of Conversational Model

**"If you'd like to press 1, press 3.
If you'd like to press 3, press 8.
If you'd like to press 8, press 5..."**

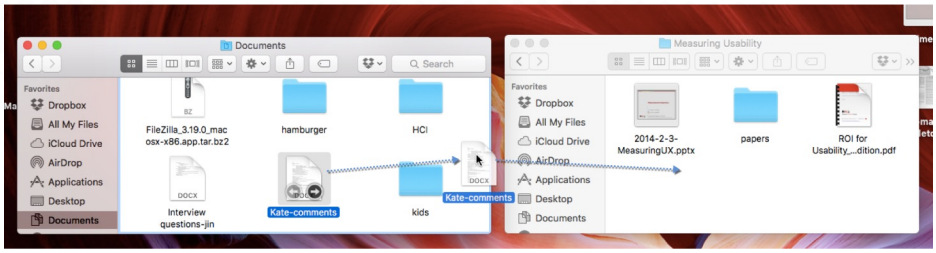
3. Manipulating

- Involves dragging, selecting, opening, closing and zooming actions on virtual objects
- Exploit's someone's knowledge of how they move and manipulate in the physical world
- Can involve actions using physical controllers or air gestures to control the movements of an on-screen avatar
- Tagged physical objects (for instance, balls) that are manipulated in a physical world result in physical/digital events (such as animation)

Direct Manipulation

- Ben Shneiderman (1983) coined the term

- Three core properties:
  - Continuous representation of objects and actions of interest
  - Physical actions and button pressing instead of issuing commands with complex syntax
  - Rapid reversible actions with immediate feedback on object of interest



Moving a file on MacOS using direct manipulation involves dragging that file from the source folder and moving it into the destination folder.

https://www.nngroup.com/articles/direct-manipulation/

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Benefits of Direct Manipulation

- Novices can learn the basic functionality quickly
- Experienced users can carry out a wide range of tasks efficiently and rapidly
- Intermittent users can retain operational concepts over time
- Error messages rarely needed
- Users can immediately see if their actions are furthering their goals, and if not, do something else
- Users experience less anxiety
- Users gain confidence and mastery and feel in control

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Disadvantages

- Some people take the metaphor of direct manipulation too literally
- Not all tasks can be described by objects, and not all actions can be done directly
- Some tasks are better achieved through delegating, for example, spell checking
- Can become screen space 'gobblers'
- Moving a cursor using a mouse or touchpad can be slower than pressing function keys to do the same actions

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4. Exploring

- Involves moving through virtual or physical environments
  - People can explore aspects of a 3D environment
  - Physical environments can also be embedded with sensors that when detect the presence of someone will trigger digital or physical events to happen
- Many examples of virtual environments, including cities, parks, buildings, rooms, and datasets
  - enable people to fly over them and zoom in and out of different parts

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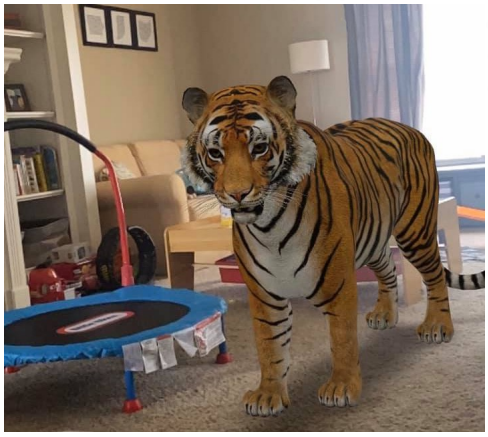
Seeing things larger than life and in situ in AR



A virtual lion appearing in someone's living room created with Google 3D object

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Exploring Data in VR

Someone using Immersive data analytics allowing them to explore Tableau data in 3D

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5. Responding

- This type suggested by Christopher Lueg et al. (2018)
- System takes the initiative to alert user to something that it "thinks" is of interest
- System does this by:
  - detecting the location and-or presence of someone in a vicinity and notifying them on their phone or smartwatch
  - what it has learned from their repeated behaviors
- Examples:
  - alerts the user of a nearby coffee bar where some friends are meeting
  - A wearer's fitness tracker notifies them of a milestone reached
- Automatic system response without any requests made by the user

Potentials Cons of System-initiated Notifications

- Can get tiresome or frustrating if too many notifications or the system gets it wrong

- What does it do when it gets something wrong?
  - Does it apologise?
  - Does it allow the user to correct the advice or information?

Choosing an Interaction Type

- Direct manipulation is good for 'doing' types of tasks, for example, designing, drawing, flying, driving, or sizing windows
- Issuing instructions is good for repetitive tasks, for example, spell-checking and file management
- Having a conversation is good for finding out information, requesting music/TV or learning
- Hybrid conceptual models are good for supporting multiple ways of carrying out the same actions

Difference between Interaction Types and Interface Styles

- Interaction type:
  - A description of what a person is doing when interacting with a system, for example, instructing, talking, browsing, or responding
- Interface style:
  - The kind of interface used to support the interaction, for instance, command, menu-based, gesture, or voice

Many Kinds of Interface Styles Available (more next topic)

- Command
- Speech
- Data-entry
- Form fill-in
- Query
- Graphical
- Web
- Pen
- Augmented reality
- Gesture

Which Interaction Type to Choose?

- Need to determine requirements and user needs
- Take budget and other constraints into account
- Also will depend on suitability of technology for activity being supported

Summary

- A conceptual model is a high-level description of a product in terms of:
  - What users can do with it and the concepts they need to understand how to interact with it
- Interaction types provide a way of thinking about how to support user's activities

Topic 3: Trends in Interfaces (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 INTERFACE TYPES

02 NATURAL USER INTERFACES AND BEYOND

03 WHICH INTERFACE?

2

INTERFACE TYPES

3

Overview

22 interface types covered

1. Command
2. Graphical
3. Multimedia
4. Virtual reality
5. Web
6. Mobile
7. Appliance
8. Voice
9. Pen
10. Touch
11. Touchless
12. Haptic
13. Multimodal
14. Shareable
15. Tangible
16. Augmented reality
17. Wearables
18. Robots and drones
19. Brain-computer
20. Smart
21. Shape-changing
22. Holographic

- Provide an overview of the diversity of interfaces
- Highlight the main design and research considerations for each of the interfaces

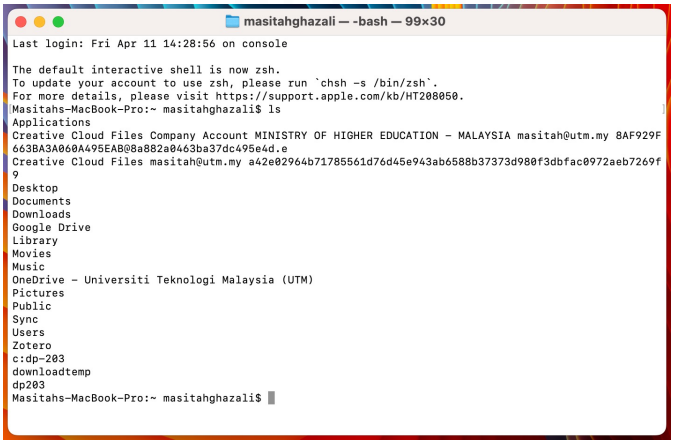
4

1. Command Line Interfaces

- Commands such as abbreviations (for instance, **ls**) typed in at the prompt to which the system responds (for example, by listing current files)
- Some are hard wired at keyboard, while others can be assigned to keys
- Efficient, precise, and fast
- Large overhead to learning set of commands

5

Command ls – lists what is on the laptop drive



6

Research and Design Considerations

- Form, name types and structure are key research questions
- Consistency is most important design principle
  - For example, always use first letter of command
- Command interfaces popular for web scripting and shared document editing (e.g. Overleaf)

2. Graphical User Interfaces (GUIs)

- Xerox Star first WIMP gave rise to GUIs
- Windows
  - Sections of the screen that can be scrolled, stretched, overlapped, opened, closed, and moved around the screen using the mouse
- Icons
  - Pictograms that represent applications, objects, commands, and tools that were opened when clicked on
- Menus
  - Lists of options that can be scrolled through and selected
- Pointing device
  - A mouse controlling the cursor as a point of entry to the windows, menus, and icons on the screen

Example of First-Generation GUI

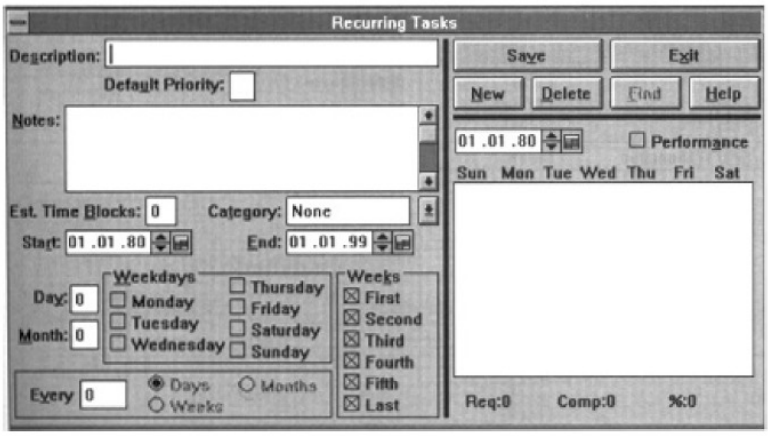


Figure 6.2 The boxy look of the first generation of GUIs. The window presents several check boxes, notes boxes, and options as square buttons

Source: Mullet, Kevin; Sano, Darrell, *Designing Visual Interfaces: Communication Oriented Techniques*, 1st, © 1995. Reproduced by permission of Pearson Education, Inc., Upper Saddle River, New Jersey.

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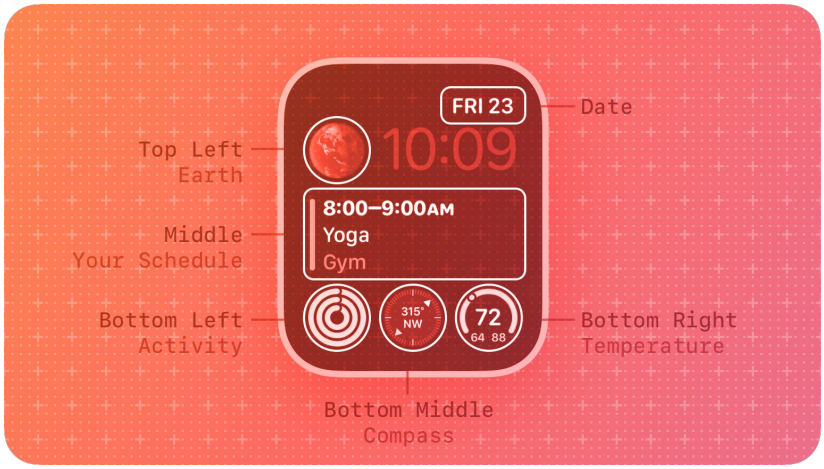
Smartwatch Complication Display



An Apple watchOS complication display that include the features "circulars" (three kinds shown in the bottom line) and "inlines" (shown in the middle and the upper right-hand corner of the display)

10

10

Window Design

- Windows were invented to overcome the physical constraints of a computer display
- They enable more information to be viewed and tasks to be performed
- Scroll bars within windows enable more information to be viewed
- Multiple windows can make it difficult to find desired one
- Listing, tabbing, and thumbnails are techniques that can help

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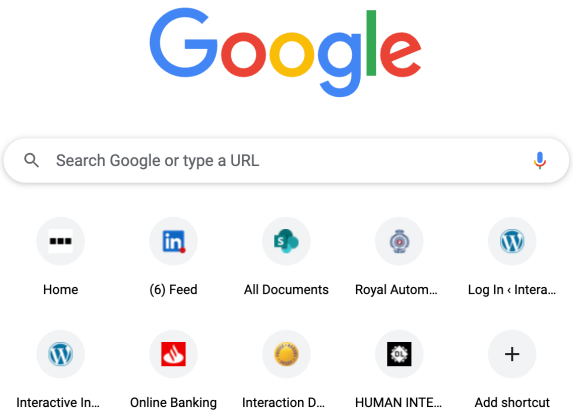
Window Design: favicons of top sites visited below the Google search bar



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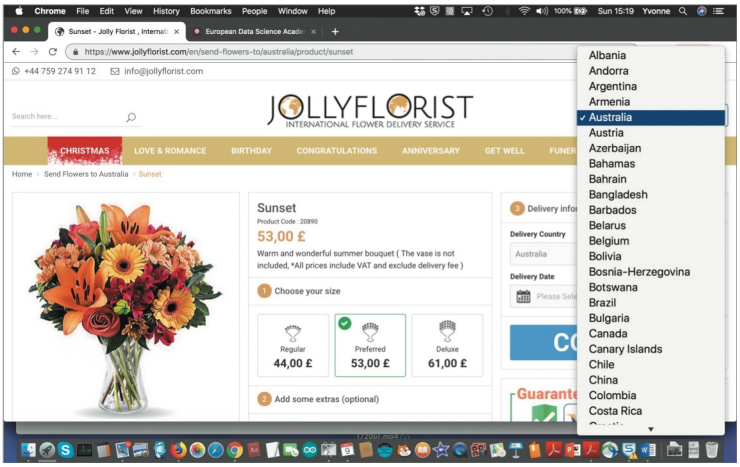
Selecting a country from a scrolling window



13

13

Is this method better? If so, why?

All Countries

A B C D E F G H I J K L M N O P Q R S T U V W XYZ

Algeria	Angola	Antigua
Argentina	Armenia	Aruba
Australia	Austria	Azerbaijan
Azores		

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14

Menu Styles

- **Flat list:** Good for showing large number of options at the same time when display is small
- **Drop down:** Shows more options on same screen (for example, cascading)
- **Pop-up:** When pressed, command key for relevant options
- **Contextual:** Provides access to often-used commands associated with a particular item
- **Collapsible:** Toggles between + and - icons on a header to expand or contract its contents
- **Mega:** All options shown using 2D drop-down layout

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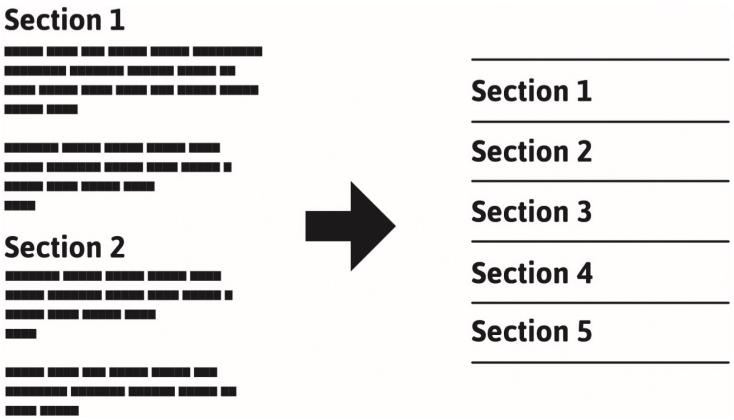
Template for a Collapsible Menu



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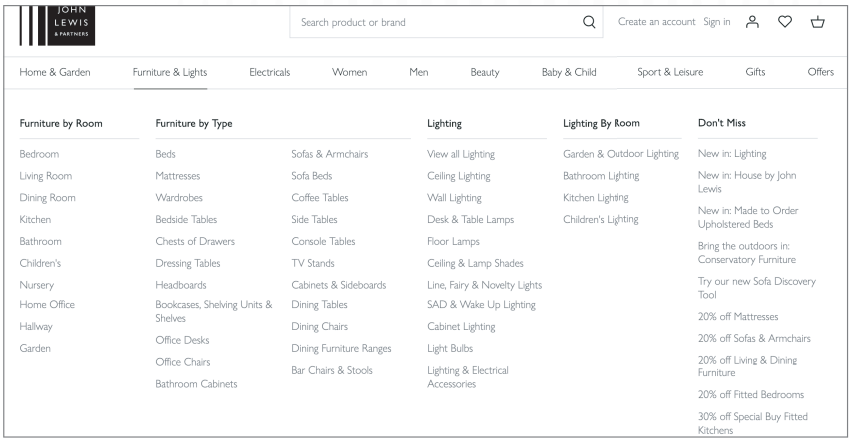
A Mega Menu



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Expanding Menus

- Enables more options to be shown on a single screen than is possible with a single flat menu
- More flexible navigation, allowing for selection of options to be done in the same window
- Most popular are cascading ones
  - primary, secondary and even tertiary menus
  - downside is that they require precise mouse control
  - can result in overshooting or selecting wrong options

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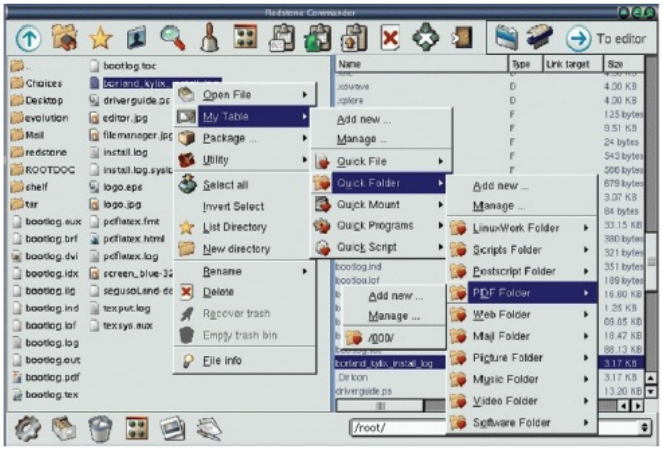
A Cascading Menu



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Contextual Menus

- Provide access to often-used commands that make sense in the context of a current task
- Appear when the user presses the Control key while clicking on an interface element
  - e.g., clicking on a photo in a website together with holding down the Control key results in options 'open it in a new window,' 'save it,' or 'copy it'
- Helps overcome some of the navigation problems associated with cascading menus

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Windows Jump List Menu

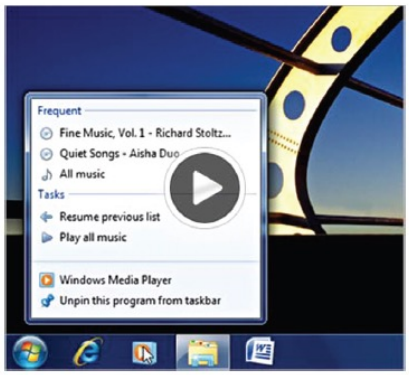

Figure 6.8 Windows jump list
Source: http://windows.microsoft.com/en-US/windows7/products/features/jump-lists.

Research and Design Considerations

- Window management
  - Enables users to move fluidly between different windows (and monitors)
- How to switch attention between windows without getting distracted
- Design principles of spacing, grouping, and simplicity should be used
- Which terms to use for menu options (for example, "front" versus "bring to front"
- Mega menus easier to navigate than drop-down ones

Icon Design

- Icons are assumed to be easier to learn and remember than commands
- Icons can be designed to be compact and variably positioned on a screen
- Now pervasive in every interface
  - For example, they represent desktop objects, tools (for example, a paintbrush), applications (for instance, a web browser), and operations (such as cut, paste, next, accept, and change)

Icons

- Since the Xerox Star days, icons have changed in their look and feel:
  - black and white → Color, shadowing, photorealistic images, 3D rendering, animation, flat
- Many designed to be very detailed and animated making them both visually attractive and informative
- Can be highly inviting, emotionally appealing, and feel alive

Icon Forms

- The mapping between the representation and underlying referent can be:
  - Similar (for example, a picture of a file to represent the object file)
  - Analogical (for instance, a picture of a pair of scissors to represent 'cut')
  - Arbitrary (such as the use of an X to represent 'delete')

- The most effective icons are similar ones
- Many operations are actions making it more difficult to represent them
  - Use a combination of objects and symbols that capture the salient part of an action

2 types of Apple Aqua icon styles

Flat 2D icons for a smartphone and a smartwatch

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Research and Design Considerations

- There is a wealth of resources for creating icons
  - Guidelines, style guides, icon builders, libraries, online tutorials
- Text labels can be used alongside icons to help identification for small icon sets
- For large icon sets (for instance, photo editing or word processing) can use the hover function

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3. Multimedia

- Combines different media within a single interface with various forms of interactivity
  - Graphics, text, video, sound, and animation
- Users click on links in an image or text
  - Another part of the program
  - An animation or a video clip is played
  - Users can return to where they were or move on to another place
- Can provide better ways of presenting information than a single media can

Pros and Cons

- Facilitates rapid access to multiple representations of information
- Can provide better ways of presenting information than can any media alone
- Can enable easier learning, better understanding, more engagement, and more pleasure
- Can encourage users to explore different parts of a game or story
- Tendency to play video clips and animations while skimming through accompanying text or diagrams

Multimedia learning app designed for tablet

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Research and Design Considerations

- How to design multimedia to help users explore, keep track of, and integrate the multiple representations
  - Provide hands-on interactivities and simulations that the user has to complete to solve a task
  - Provide quizzes, electronic notebooks, and games

- Multimedia good for supporting certain activities, such as browsing, but less optimal for reading at length

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4. Virtual Reality

- Computer-generated graphical simulations providing:
  - "the illusion of participation in a synthetic environment rather than external observation of such an environment" (Gigante, 1993)
- Provide new kinds of experience, enabling users to interact with objects and navigate in 3D space
- Create highly-engaging user experiences

Pros and Cons

- Can have a higher level of fidelity with objects that they represent compared to multimedia
- Induces a sense of presence where someone is totally engrossed by the experience
  - "a state of consciousness, the (psychological) sense of being in the virtual environment" (Slater and Wilbur, 1999)
- Provides different viewpoints: first and third person
- Early head-mounted displays were uncomfortable to wear and could cause motion sickness and disorientation
- Lighter VR headsets are now available (for example, HTC Vive) with more accurate head tracking

Examples

- Video games
- Arcade games for social groups
- Well being and therapy for fears
- Experience how others feel emotions
  - For example, empathy and compassion
- Enrich user's planning experience for travel destinations
- Architecture, design, and education

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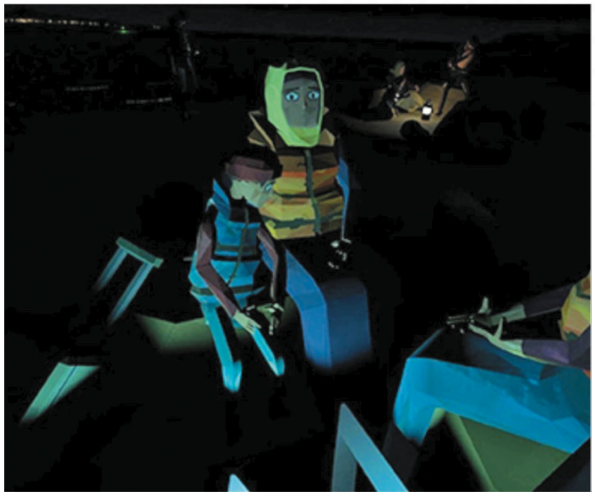
Polygon graphics used to represent avatars for the We Wait VR experience



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UTM Classroom of the Future

Offshore experience
Source: https://magicx.my/classroom-of-the-future/

Research and Design Considerations

- Much research on how to design safe and realistic VRs to facilitate training
  - For example, flying simulators
  - Help people overcome phobias (for example, spiders or talking in public)
- Design issues
  - How best to navigate through them (for instance, first versus third person)
  - How to control interactions and movements (for example, by using head and body movements)
  - How best to interact with information (for instance by using keypads, controllers, pointing, and joystick buttons)
  - Level of realism to aim for to engender a sense of presence

5. Web

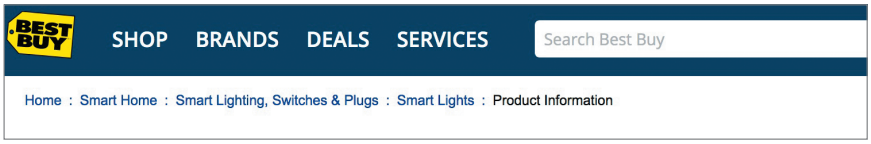
- Early websites were largely text-based, providing hyperlinks
- Focus was on how best to structure information to enable users to navigate and access them easily and quickly
- Nowadays, more emphasis is on making pages distinctive, striking, and aesthetically pleasing
- Need to think of how to design information for different platforms—keyboard or touch?
  - For example, smartphones, tablets, and PCs

Usability versus Aesthetics

- Vanilla or multi-flavour design?
  - Ease of finding something versus aesthetic and enjoyable experience
- Web designers are:
  - "thinking great literature"
- Users read the web like a:
  - "billboard going by at 60 miles an hour" (Krug, 2014)
- Need to determine how to brand a web page to catch and keep 'eyeballs'

Breadcrumbs for Navigation

- Breadcrumbs are category labels:
  - Enable users to look at other pages without losing track of where they have come from
  - Very usable
  - Enable one-click access to higher site levels
  - Attract first time visitors to continue to browse a website having viewed the landing page



Web Design Styles

- Responsive website design
  - the browser automatically resizes the layout, and changes the graphic design, fonts, and appearance depending on the screen size (smartphone, tablet, or PC) on which it is being displayed.
  - downsizing the content in this way makes it more time-consuming as more pages need to be loaded.
  - can also make it more fiddly navigating multiple pages and menus
- Infinite scrolling
  - websites are designed to enable browsing content on one long page
  - avoids a visitor needing to wait for pages to load when clicking on them
  - navigation is largely done by swiping across or down the page until the end is reached
  - a side effect is the tendency to glance while scrolling without focusing on individual items

In-your-face Web Ads

- Web advertising is often intrusive and pervasive
- Flashing, aggressive, persistent, and annoying
- Often requires action to get rid of
- What is the alternative?
  - Use of ad blockers

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Research and Design Considerations

- Many books and guidelines on website design
- Veen's (2001) three core questions to consider when designing any website:
  - Where am I?
  - Where can I go?
  - What's here?

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6. Mobile

- Handheld devices intended to be used while on the move
- Have become pervasive, increasingly used in all aspects of everyday and working life
  - For example, phones, fitness trackers, and smartwatches
- Larger-sized tablets used in mobile settings
  - Including those used by flight attendants, marketing professionals, and at car rental returns

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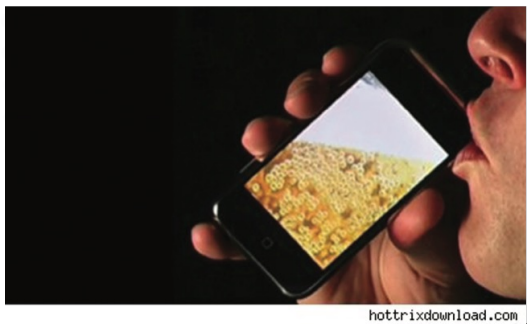
The Advent of the iPhone App

- A whole new user experience that was designed primarily for people to enjoy
  - many apps not designed for any need, want or use but purely for idle moments to have some fun
  - e.g. iBeer developed by magician Steve Sheraton
  - ingenious use of the accelerometer that is inside the phone



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QR Codes and Smartphones

Mobile Challenges

- Smaller screens, small number of physical keys and restricted number of controls
- Innovative physical designs including:
  - roller wheels, rocker dials, up/down 'lips' on the face of phones, 2-way and 4-way directional keypads, softkeys, silk-screened buttons
- Usability and preference varies
  - depends on the dexterity and commitment of the user
- Smartphones overcome mobile physical constraints through using multi-touch displays

Research and Design Considerations

- Mobile interfaces can be cumbersome to use for those with poor manual dexterity or 'fat' fingers
- Key concern is hit area:
  - Area on the phone display that the user touches to make something happen, such as a key, an icon, a button, or an app
  - Space needs to be big enough for all fingers to press accurately
  - If too small, the user may accidentally press the wrong key
  - Fitts' law can be used to help design right spacing
    - Minimum tappable areas should be 44 points x 44 points for all controls

7. Appliances

- Everyday devices in home, public places, or car
  - For example, washing machines, remotes, toasters, printers, and navigation systems)
- And personal devices
  - For instance, digital clock and digital camera
- Used for short periods
  - For example, starting the washing machine, watching a program, buying a ticket, changing the time, or taking a snapshot
- Need to be usable with minimal, if any, learning

Simple Toaster Control

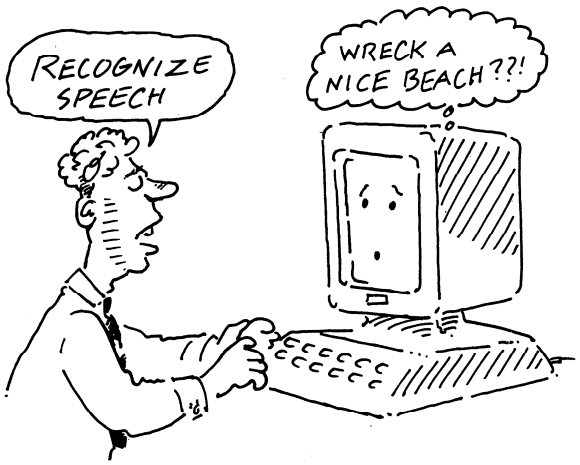
Research and Design Considerations

- Need to design as transient interfaces with short interactions
- Simple interfaces
- Consider trade-off between soft and hard controls
  - e.g. buttons or keys, dials or scrolling
- Understand the innate human ability to manipulate physicality (Ghazali and Dix)
  - The relationship between the physical controllers with the digital underlying functionalities

8. Voice

- Involves a person talking with a spoken language app, for example, timetable, travel planner, or phone service or chatbots
- Used mainly for inquiring about specific information, for example, flight times or to perform a transaction, such as buying a ticket
- Mostly a voice system is reactive responding to a person's queries
- Also used by people with visual impairments
  - For example, speech recognition word processors, page scanners, web readers, and home control systems

Have speech interfaces come of age?



Modelling Human Conversations

- People often interrupt each other in a conversation
  - Especially when ordering in a restaurant, rather than let the waiter go through all of the options
- Speech technology has a similar feature called 'barge-in'
  - Users can choose an option before the system has finished listing all of the options available

Structuring Voice Dialogs

- Directed dialogs are where the system is in control of the conversation
  - Where it asks specific questions and requires specific responses
- More flexible systems allow the user to take the initiative:
  - For example, "I'd like to go to Paris next Monday for two weeks."
- Guided prompts can help callers back on track
  - For example, "Sorry I did not get all that. Did you say you wanted to fly next Monday?"

Voice Assistants (for e.g. Alexa)

- Allow all to use rather than being single use
- Support families playing games, interactive storytelling, jokes, and so forth
- Can encourage social and emotional bonding
- Young children (under 4), however, find it difficult to be understood by the voice assistants
  - Frustrating for them

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Research and Design Considerations

- How to design systems that can keep conversation on track
  - Help people navigate efficiently through a menu system
  - Enable them to recover easily from errors
  - Guide those who are vague or ambiguous in their requests for information or services
- Type of voice actor (for example, male, female, neutral, or dialect)
  - Do people prefer to listen to and are more patient with a female or male voice, a northern or southern accent?

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9. Pen-based Devices

- Enable people to write, draw, select, and move objects at an interface using light pens or styluses
  - Capitalise on the well-honed drawing skills developed from childhood
- Smartpens use a combination of ordinary ink pen with digital camera that digitally records everything written with the pen on special paper

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The LiveScribe Echo 2 Smartpen

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Pros and Cons

- Allows people to annotate existing documents quickly and easily
- Can be used to fill in paper-based forms that can readily be converted to a digital record using standard typeface
- Can be used by remote teams to communicate and work on the same documents

- Can be difficult to see options on the screen because a user's hand can occlude part of it when writing
- Can have lag and feel clunky

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10. Touchscreens

- Single touchscreens are used in walk-up kiosks (such as ticket machines and ATMs) to detect the presence and location of a person's touch on the display
- Multi-touch surfaces support a range of more dynamic finger tip actions, for example, swiping, flicking, pinching, pushing, and tapping
- They do so by registering touches at multiple locations using a grid
- Now used for many kinds of displays, such as smartphones, iPods, tablets, and tabletops
  - Supports one and two hand gestures, including tapping, zooming, stretching, flicking, dwelling, and dragging

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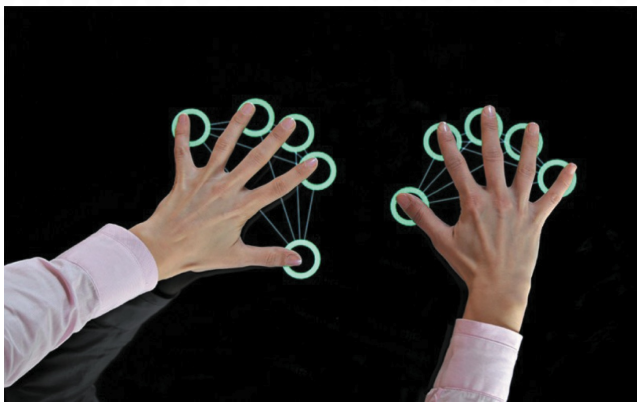
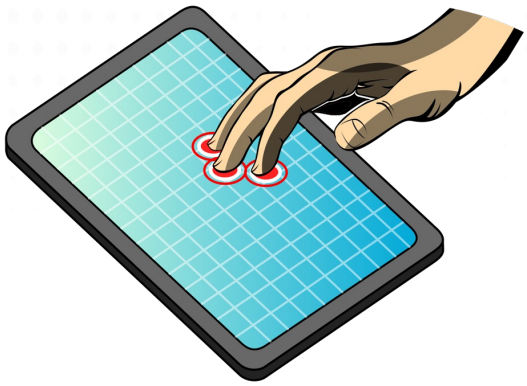
A Multi-touch Surface





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Research and Design Considerations

- Provides fluid and direct styles of interaction involving freehand and pen-based gestures for certain tasks
- Core design concerns include whether size, orientation, and shape of touch displays effect collaboration
- Much faster to scroll through wheels, carousels, and bars of thumbnail images or lists of options by finger flicking
- Gestures need to be learned for multi-touch, so a small set of gestures for common commands is preferable
- More cumbersome, error-prone, and slower to type using a virtual keyboard on a touch display than using a physical keyboard

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11. Touchless

- Gestures involve moving arms and hands to communicate
- Uses camera recognition, sensor, and computer vision techniques
  - Recognise people's arm and hand gestures in a room
  - Gestures need to be presented sequentially to be understood (compare with the way sentences are constructed)

Touchless interface in the operating theater

Recognises core gestures for manipulating MRI or CT images using Microsoft Kinect

Entertainment (public space)

MaGICX Johor Port 2015
Source: https://magicx.my/

Car Dashboards using Gestures

- BMW's iDrive enables drivers to control certain functions with the use of hand gestures captured by a 3D camera
  - e.g. turning up or down the audio volume, For audio control these comprise of rotating the index finger clockwise (up) or anti-clockwise (down)
- Gesture control sensors are also positioned in the roof lining of the car
- To ensure they are safe, the gestures were developed and tested in the simulator

Research and Design Considerations

- How does computer recognize and delineate user's gestures?
  - Start and end points?
  - Difference between deictic and hand waving
- How realistic must the mirrored graphical representation of the user be in order for them to be believable?

12. Haptic

- Provide tactile feedback
  - By applying vibration and forces to a person's body, using actuators that are embedded in their clothing or a device they are carrying, such as a smartphone
- Vibrotactile feedback can be used to simulate the sense of touch between remote people who want to communicate
- Ultrahaptics creates the illusion of touch in midair using ultrasound to make the illusion of 3D shapes

Realtime Vibrotactile Feedback

- Provides nudges when playing violin incorrectly
- Uses motion capture to sense arm movements that deviate from model
- Nudges are short vibrations on arms and hands

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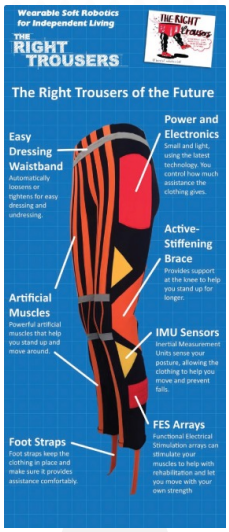
Exoskeleton with artificial muscles that uses bubble haptic feedback



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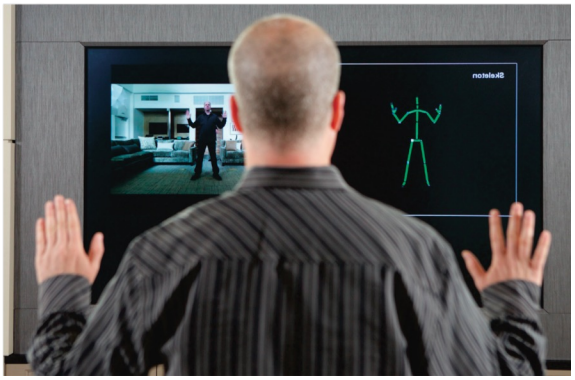
Research and Design Considerations

- Where best to place actuators on body
- Whether to use single or sequence of 'touches'
- When to buzz and how intense
- How does the wearer feel it in different contexts?
- What kind of new smartphone/smartwatch apps can use vibrotactile creatively?
  - For example, slow tapping to feel like water drops meant to indicate that it is about to rain, and heavy tapping to indicate a thunderstorm is looming

13. Multimodal Interfaces

- Provide enriched user experiences
  - By multiplying how information is experienced and detected using different modalities, such as touch, sight, sound, and speech
  - Support more flexible, efficient, and expressive means of human-computer interaction
  - Most common combination is speech and vision
- Can be combined with multi-sensor input to enable other aspects of the human body to be tracked
  - For example, eye gaze, facial expression, and lip movements
  - Provides input for customising user interfaces

Tracking a person's movement



- Kinect camera can detect multimodal input in real time using RGA camera for facial recognition and gestures, depth camera for movement tracking, and microphones for voice recognition
- Used to build model of person and represented as avatar on display programmed to move just like them

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Sending modalities via smartphone

Send taste, smell, touch via smartphone:
https://learningenglish.voanews.com/a/smartphones-may-be-able-to-send-tastes-smeel-touches/2634824.html

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Research and Design Considerations

- Need to recognise and analyse user behavior, for example, speech, gesture, handwriting, or eye gaze
- Much harder to calibrate these than single modality systems
- What is gained from combining different input and outputs
- Is talking and gesturing, as humans do with other humans, a natural way of interacting with a computer?

14. Shareable Interfaces

- Designed for more than one person to use:
  - Provide multiple inputs and sometimes allow simultaneous input by co-located groups
  - Large wall displays where people use their own pens or gestures
  - Interactive tabletops where small groups interact with information using their fingertips
    - For example, DiamondTouch, Smart Table, and Surface

A smartboard and an interactive tabletop interface

(a)

(b)

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Benefits

- Provide a large interactional space that can support flexible group working
- Can be used by multiple users
  - Can point to and touch information being displayed
  - Simultaneously view the interactions and have the same shared point of reference as others
- Can support more equitable participation compared with groups using single PC

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Research and Design Considerations

- Core design concerns include whether size, orientation, and shape of the display have an effect on collaboration
- Horizontal surfaces compared with vertical ones support more turn-taking and collaborative working in co-located groups
- Providing larger-sized tabletops does not improve group working but encourages more division of labor
- Having both personal and shared spaces enables groups to work on their own and in a group
- A design challenge is how to bridge across devices, applications and time such that the peoples' interactions are seamless

15. Tangible Interfaces

- Type of sensor-based interaction, where physical objects, for example, bricks, are coupled with digital representations
- When a person manipulates the physical object/s, it causes a digital effect to occur, for example, an animation
- Digital effects can take place in a number of media and places, or they can be embedded in the physical object

Examples

- Flow Blocks
  - Depict changing numbers and lights embedded in the blocks
  - Vary depending on how they are connected together
- Urp
  - Physical models of buildings moved around on tabletop
  - Used in combination with tokens for wind and shadows → Digital shadows surrounding them to change over time
- MagicCubes
  - Connect physical electronic components and sensors to make digital events occur (for example, change color depending on how much shaken)

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Learning to code and create with the tangible MagicCubes

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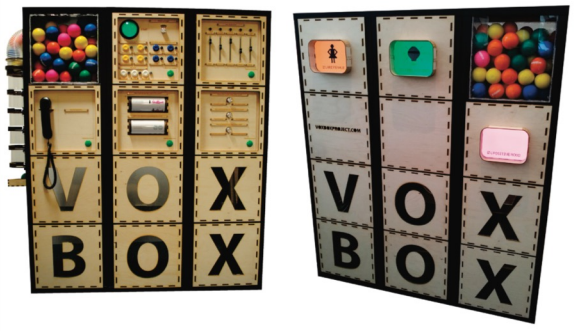
Benefits

- Can be held in one or both hands and combined and manipulated in ways not possible using other interfaces
  - Allows for more than one person to explore the interface together
  - Objects can be placed on top of each other, beside each other, and inside each other
  - Encourages different ways of representing and exploring a problem space
- People are able to see and understand situations differently
  - Can lead to greater insight, learning, and problem-solving than with other kinds of interfaces
  - Can facilitate creativity and reflection

VoxBox

A tangible system that gathers opinions at events through playful and engaging interaction (Goldsteijn et al., 2015)



Research and Design Considerations

- What kinds of conceptual frameworks to use to help identify novel and specific features
- What kind of coupling to use between the physical action and digital effect
  - If it is to support learning, then an explicit mapping between action and effect is critical
  - If it is for entertainment, then it can be better to design it to be more implicit and unexpected
- What kind of physical artifact to use
  - Bricks, cubes, and other component sets are most commonly used because of flexibility and simplicity
  - Stickies and cardboard tokens can also be used for placing material onto a surface
- With what kinds of digital outputs should tangible interfaces be combined?

16. Augmented Reality

- Digital representations are superimposed on physical objects or the environment
- Pokémon Go made it a household game
  - Used smartphone camera and GPS to place virtual characters onto objects in the environment as if they really are there
- Many other applications including medicine, navigation, air traffic control, games, and everyday exploring

Other Examples

- In medicine
  - Virtual objects, for example, x-rays and scans, are overlaid on part of a patient's body
  - Aid the physician's understanding of what is being examined or operated

- In air traffic control
  - Dynamic information about aircraft overlaid on a video screen showing the real planes, and so on landing, taking off, and taxiing
  - Helps identify planes difficult to make out

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Augmented reality overlay on a car windshield

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Devices for Viewing AR

- AR can be viewed through headsets, smartphones and glasses (e.g., SnapChat's AR spectacles).
- Headsets can provide a more immersive experience, but the downside is they are cumbersome to wear and fiddly to calibrate
- Convenient and easy-to-wear AR glasses are beginning to appear

Mobile AR BoboiBoy

AR that uses forward facing camera

- Enables virtual try-ons (for example, Snapchat filters, Zoom filters)
- AT mirrors set up in retail stores for trying on make-up, sunglasses, jewelry
  - Convenient, engaging, and easy to compare more choices
  - But cannot feel the weight, texture, or smell of what is being tried on
- Can be used to enable users to step into a character (for example, David Bowie, Queen Victoria)

Singers trying on the virtual look of two characters from the opera Akhnaten

(a) (b)

Research and Design Considerations

- What kind of digital augmentation?
  - When and where in physical environment?
  - Needs to stand out but not distract from ongoing task
  - Needs to be able to align with real world objects
  - What happens if the AR is slightly off?
- What kind of device?
  - Smartphone, tablet, head up display or other?

17. Wearables

- First developments were head- and eyewear-mounted cameras that enabled user to record what was seen and to access digital information
- Since then, jewelry, head-mounted caps, smart fabrics, glasses, shoes, and jackets have all been used
  - Provides the user with a means of interacting with digital information while on the move
- Applications include automatic diaries, tour guides, cycle indicators, and fashion clothing

Google Glass (2014)

Why was there so much excitement and concern about people filming what they could see right in front of them?

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Other Examples

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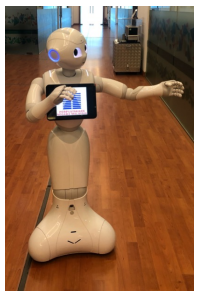
Research and Design Considerations

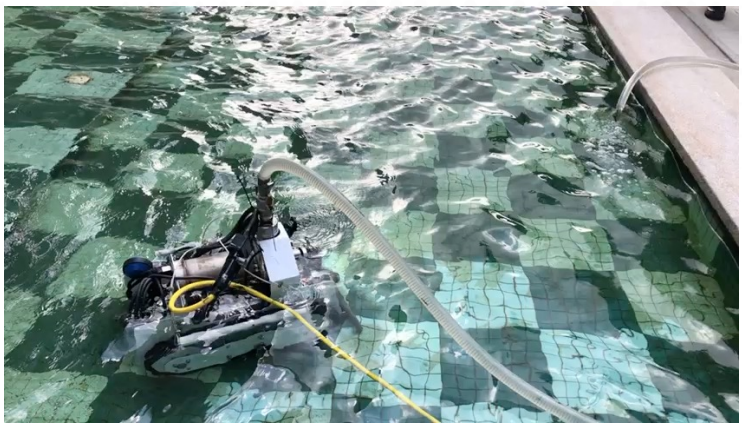
- Comfort
  - Needs to be light, small, not get in the way, fashionable, and preferably hidden in the clothing
- Hygiene
  - Is it possible to wash or clean the clothing once worn?
- Ease of wear
  - How easy is it to remove the electronic gadgetry and replace it?
- Usability
  - How does the user control the devices that are embedded in the clothing?

18. Robots …

- Main types
  - Remote robots used in hazardous settings
    - Can be controlled to investigate bombs and other dangerous materials
  - Domestic robots helping around the house
    - Can pick up objects and do daily chores like vacuuming
  - Pet robots as human companions
    - Have therapeutic qualities, helping to reduce stress and loneliness
  - Sociable robots that work collaboratively with humans
    - Encourage social behaviors



Cleaning Robot



UTM CAIRO: ROV Pleco to clean tank, pool

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Social Robots: Mel and Paro

- Cute and cuddly
- Can open and close eyes and make sounds and movements

Source: Images courtesy of Mitsubishi Electric Research Labs.

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18. … and Drones

- Unmanned aircraft that are controlled remotely and used in a number of contexts
  - Medical supplies, groceries and carrying drinks and food to people at festivals and parties
  - Construction and agricultural applications, such as flying them over vineyards and fields to collect data about crops
  - Firework displays
  - Helping to track poachers in wildlife parks in Africa
- Can fly low and and stream photos to a ground station where images can be stitched together into maps
- Can be used to determine the health of a crop, or when it is the best time to harvest the crop

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Drone being used to survey the state of a vineyard

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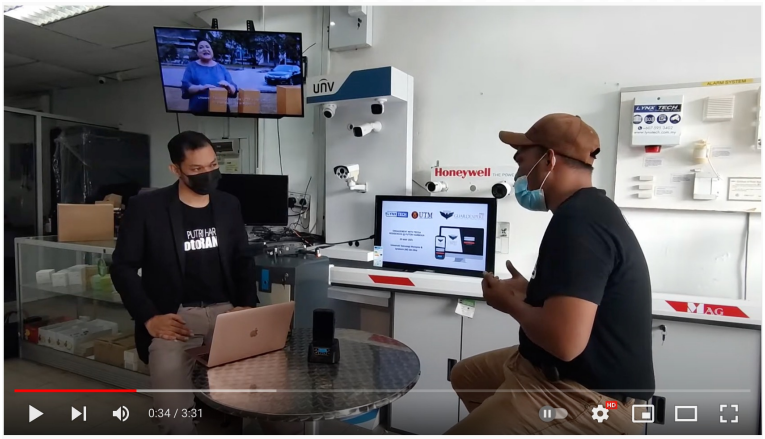
Drone to patrol surrounding



GuardExpert Pro, an Internet-Of-Thing (IoT) based security guard patrolling system. GUARD EXPERT PRO

Research and Design Considerations

- How do humans react to physical robots designed to exhibit behaviors (for example, making facial expressions) compared with virtual ones?
- Should robots be designed to be human-like or look like and behave like robots that serve a clearly-defined purpose?
- Should the interaction be designed to enable people to interact with the robot as if it was another human being or more human-computer-like (for example, pressing buttons to issue commands)?
- Is it acceptable to use unmanned drones to take a series of images or videos of fields, towns, and private property without permission or people knowing what is happening?

19. Brain-Computer Interfaces

- Brain-computer interfaces (BCI) provide a communication pathway between a person's brain waves and an external device, such as a cursor on a screen
- Person is trained to concentrate on the task, for example, moving the cursor
- BCIs work through detecting changes in the neural functioning in the brain
- BCIs apps:
  - Games (for example, Brain Ball)
  - Enable people who are paralysed to control robots

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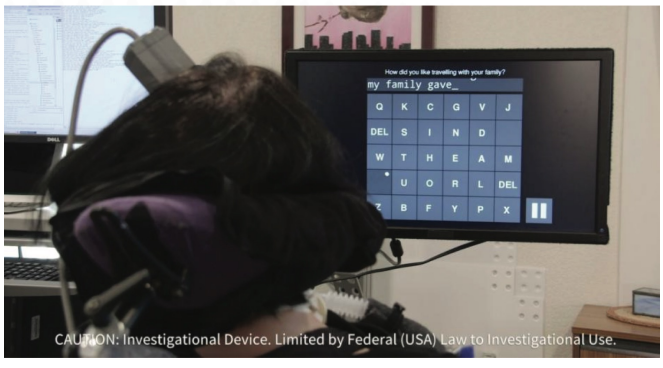
Examples

Brainball



A brain-computer interface being used by a woman who is paralyzed to select letters on the screen

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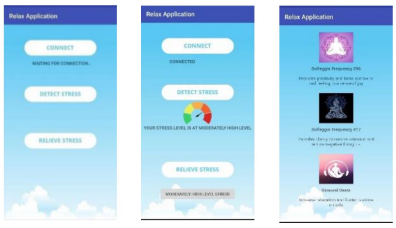
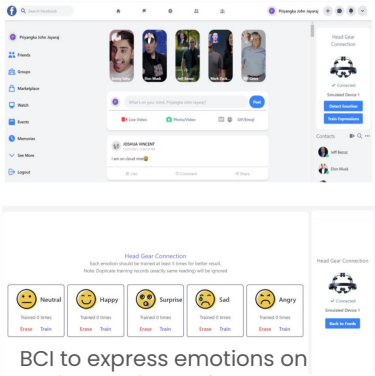
Other Examples

Using BCI to control and manipulate different types of mobile applications
Source: N.A.Iahad, A.B.G. Ahmed-Gaber, M. Ghazali. 2018. A Conceptual Model for Mobile Interaction using Brain Computer Interface. International Journal of Future Generation Communication & Networking (IJFGCN)



Mobile BCI to reduce stress among students
Source: P.J. Jayaraj, M. Ghazali and A.Gaber. Relax: Mobile Brain-Computer Interface App to Reduce Stress among Students. In Special Proceedings of 5th Asian CHI Symposium 2021.



BCI to express emotions on social media platform
Source: P. J. Jayaraj and M. Ghazali. User's Perspectives on Expressing Emotions on Social Media Platform with mobile Brain-Computer Interface. In Proceedings of 3rd Fusion Symposium 2021

Research and Design Considerations

- What is the best way to lay out letters on a digital screen so that the target space for each letter is sufficiently wide enough to allow for it to be selected using BCI?
- Ethical concerns are raised by brain-computer interfaces that are being developed to work out what someone is thinking
  - Our thoughts have always been private and making them interpretable by machines, which in turn could be accessed and read by other people, raises privacy concerns
- How accurate is BCI in what it infers to be someone's thoughts?

20. Smart Interfaces

- Smart: phones, speakers, watches, cars, buildings, cites
- Smart refers to having some intelligence and connected to the internet and other devices
- Context-aware
  - Understand what is happening around them and execute appropriate actions, for example, a Nest thermostat
- Human-building interaction
  - Buildings are designed to sense and act on behalf of the inhabitants but also allow them to have some control and interaction with the automated systems

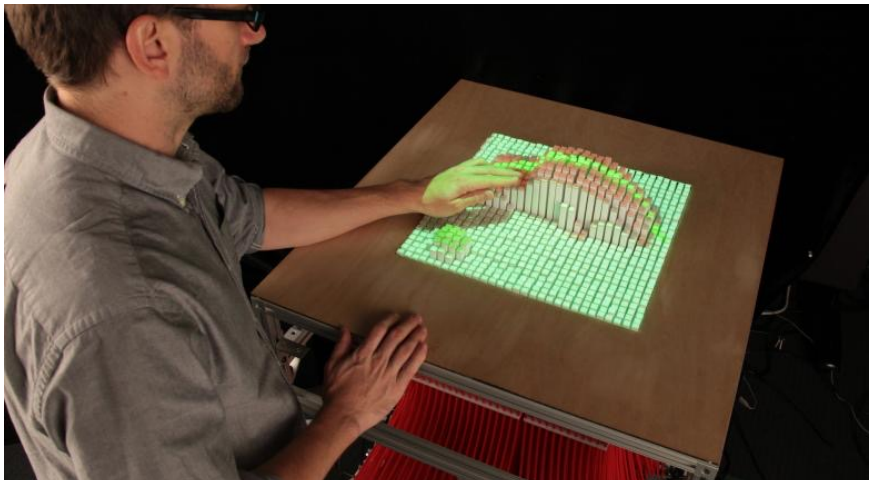
Research and Design Considerations

- Much current research is concerned with human values, needs, and priorities when addressing people's interactions with smart environments
- Do smart interfaces amplify our actions and remain attentive to our goals or are they becoming more autonomous?
- Do we need new metaphors to make understanding and using smart interfaces easier?
  - What might these be?
- Another approach is to imbue smart technologies, like speakers and robots, with personalities
  - Can promote user trust and acceptance

21. Shape Changing Interfaces

- Use physical shape change as input and output to systems
  - e.g. physical 3D bar chart that is positioned in a grid where a matrix of 3D rods move up and down to convey changes in a digital dataset
- These kinds of dynamic physical representations have been found to help people with visualisation tasks
- Shape-changing interfaces have also been developed as a form of dynamic material to explore novel interaction possibilities.
- Provide a different way of interacting with content compared with reading and touching digital displays

InFORM: a shape changing interface



Physicalisations

- Physical artefacts that are designed to encode data in specific materials
- An example is Physikit (Houben et al., 2016) which is a physical-digital system comprising a set of physical cubes that convey digital properties and are programmed to visualize real-time environmental data in the home
- A main benefit is to make data more accessible and enable people to more readily connect with the context in which the data is being collected or is changing over time (Sauvé et al, 2022)

Research and Design Considerations

- Provide new opportunities for experiencing and exploring data using the sense of touch as well as sight
- Does this facilitate enhanced understanding and engagement with a dataset?
- Design considerations include what is the optimal size for a grid of physical rods and how many physicalisations to design in a set that people can learn and remember what they mean

22. Holographic Interfaces

- Create the illusion of a 3D person being present through using various forms of trickery
- Advances in projection and display technology have enabled these kinds of digital representations to appear quite convincing.
  - e.g. the Proto system lets people beam themselves to a remote location and interact with the people there
- The show Abba Voyage, first launched in 2022 feature avatars (dubbed 'ABBAtars'), depicting the group as they appeared in the late 70s
- The technology used includes a 65million pixel screen and in motion capture technology

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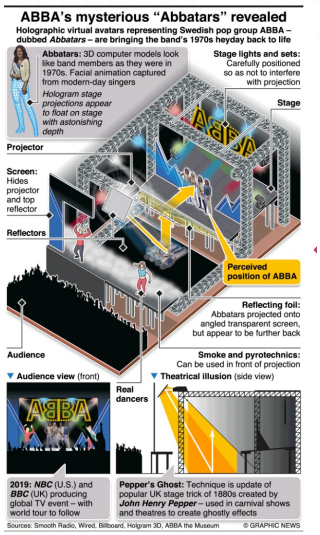
Examples



ABBA's holographic projections

UTM's Holo Professor

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Research and Design Considerations

- What is the best way to represent people in virtual spaces so that they feel comfortable, are engaging to interact with, feel natural and do not appear creepy?
- Design considerations include what size the holographs should be and how other's viewing them can interact and communicate with those being projected into their space

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NUI AND BEYOND

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From GUIs to Natural Interfaces

- Interfaces have shifted from screen-based to more embodied, intuitive forms. NUIs aim to reduce the cognitive load by leveraging skills we already use in the real world.
- Traditional interfaces from GUI → mobile → touch → wearables
- Emerging trends include holographic and smart interfaces
- Future direction requires more **natural** interactions
- Natural User Interfaces (NUIs):
  - Interact using voice, hands, gestures, facial expressions
  - Aim to replicate real-world human interaction

The Challenges with Naturalness

- Even 'natural' gestures can feel unnatural depending on context. Simplicity works best. More complex tasks may need a hybrid of natural and traditional inputs.
- NUIs should be intuitive—but context matters:
  - Is it easier to say 'open' or flick a switch?
  - Raising arms to change a TV channel vs pressing a button
- Don Norman's naturalness depends on:
  - Required learning
  - Task complexity
  - Need for speed & precision
- Example: sensor-based faucets work well with limited functions
  - But complex gestures for flow/temp? Less intuitive

Beyond Natural: Smart, Inclusive, Adaptive

- Interfaces of the future won't just be natural—they'll be smart, inclusive, and adaptive.
  - Gestural/voice interfaces enhance accessibility
    - e.g., iPhone VoiceOver for visually impaired
  - Brain-Computer Interfaces (BCIs): powerful but unnatural
    - Require significant training
  - Combining sensors (brain, body, environment) enables:
    - Real-time emotional/cognitive adaptation
    - Continuous & discrete inputs
- Merging sensor tech with AI opens up entirely new possibilities for human-computer collaboration.
    - AI empowers systems to collaborate with users

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WHICH INTERFACE?

....

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Which Interface?

- Which interface to use will depend on task, users, context, cost, robustness, and so on
- Is multimedia better than tangible interfaces for learning?
- Is speech as effective as a command-based interface?
- Is a multimodal interface more effective than a mono-modal interface?
- Will wearable interfaces be better than mobile interfaces for helping people to find information in foreign cities?
- Are virtual environments the ultimate interface for playing games?
- Are shareable interfaces better at supporting communication and collaboration compared with using networked desktop PCs?

Summary

- Many innovative interfaces have emerged in the last 30 years, including speech, wearable, mobile, brain, and tangible
- This raises many design and research questions as to decide which to use
  - For example, how best to represent information to the user so that they can carry out ongoing activity or task
- New smart interfaces that are context-aware and monitor people
- Raising new ethical issues concerned with what data is being collected and what it is used for

Topic 3: Social Mechanism (part 3)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 BEING SOCIAL

02 FACE-TO-FACE CONVERSATIONS

03 REMOTE COLLABORATION AND COMMUNICATION

04 CO-PRESENCE

05 SOCIAL GAMES

2

BEING SOCIAL

3

Social Interaction

- We live together, work together, play together, talk to each other, and socialise
- Social technologies developed to enable us to persist in being social when apart
  - They differ in how they support us
  - Some encourage social interactions (for example, family games with Alexa)
  - Others have a negative impact on everyday conversations (Turkle, 2015)

4

Are we spending too much time in our own digital bubbles?

5

5

Questions raised by social tech

- Are in person conversations being superseded by social media interactions?
- How many friends do you have on Facebook, LinkedIn, WhatsApp, and so on versus real life?
- How much do they overlap?
- How are the ways that we live and interact with one another changing?
- Are the established rules and etiquette still applicable to online and offline?

6

6

FACE-TO-FACE CONVERSATIONS

7

Conversational Mechanisms

- Various mechanisms and 'rules' are followed when holding a conversation face-to-face such as mutual greetings

A: Hi there

B: Hi!

C: Hi

A: All right?

C: Good, how's it going?

A: Fine, how are you?

C: OK

B: So-so. How's life treating you?

8

Conversational Rules

- Sacks et al. (1978) conversation analysis of conversations propose three basic rules:

  Rule 1: The current speaker chooses the next speaker by asking an opinion, question, or request

  Rule 2: Another person decides to start speaking

  Rule 3: The current speaker continues talking

More Conversational Rules

- Turn-taking used to coordinate conversation

  A: Shall we meet at 8:00?
  B: Um, can we meet a bit later?

  A: Shall we meet at 8:00?
  B: Wow, look at him?

  A: Yes what a funny hairdo!
  B: Um, can we meet a bit later?

  Gets embedded

- Back channeling to signal to continue and following

  Uh-uh, umm, ahh

Further Conversational Rules

- Farewell rituals
  - Bye then, see you, yeah bye, see you later….
- Implicit and explicit cues
  - For instance, looking at watch or fidgeting with coat and bags
  - Explicitly saying, "Oh dear, look at the time, I must go, I'm running late…"

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Breakdowns in Conversation

- When someone says something that is misunderstood:
  - Speaker will repeat with emphasis:

    A: "This one?"

    B: "No, I meant that one!"
  - Also use tokens:

    Eh? Quoi? Huh? What?

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What happens in online conversations?

- Do the same conversational rules apply?
- Are there different kinds of breakdowns?
- How do people repair them for:
  - Email?
  - Instant messaging?
  - Texting?
  - Skype or other videoconferencing software?

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New Social Rules during COVID-19

- For how to behave and interact with others when communicating via video conferencing
- Muting yourself when not talking
- Raising a yellow hand when wanting to speak
- Other emoji reactions became commonly used to signify various forms of praise, emotions and non-verbal feedback, e.g.
  - clicking on the red heart and party hat icons that momentarily appear in someone's window and then disappear after 10 seconds

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Online Collaboration and Communication

- Much research on how to support conversations when people are remote
- Many applications have been developed
  - email, videoconferencing, instant messaging, chatrooms
- To what extent do they mimic or move beyond existing ways of conversing?

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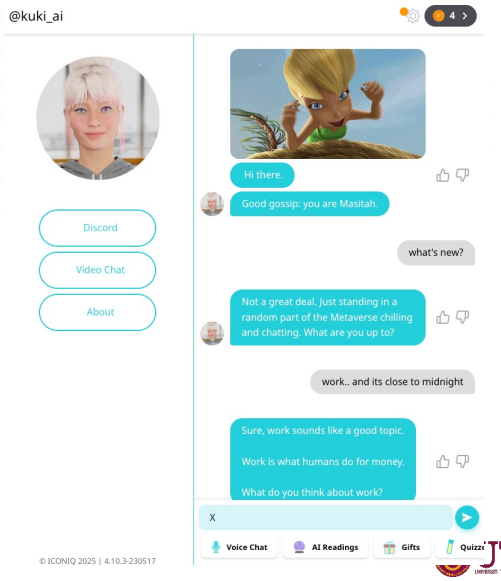
Talking with Chatbots

- Conversational User Interfaces, such as chatbots, are more sophisticated in how they emulate turn-taking in conversations
  - e.g. Kuki.ai
- Recent developments in AI are changing how we converse and interact with AI
  - e.g, ChatGTP



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Dilemma: Is it OK to talk with a dead person using a chatbot?

- Eugenia Kuyda lost a close friend in a car accident who was only in his 20s
- She took all his texts sent over the course of his life and made a chatbot using them
- Chatbot responds to text messages so that Eugenia can talk to her friend as if he was alive
- Is this a creepy or comforting way to deal with grief?
  - Is it respectful of the dead person?

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REMOTE COLLABORATION & COMMUNICATION

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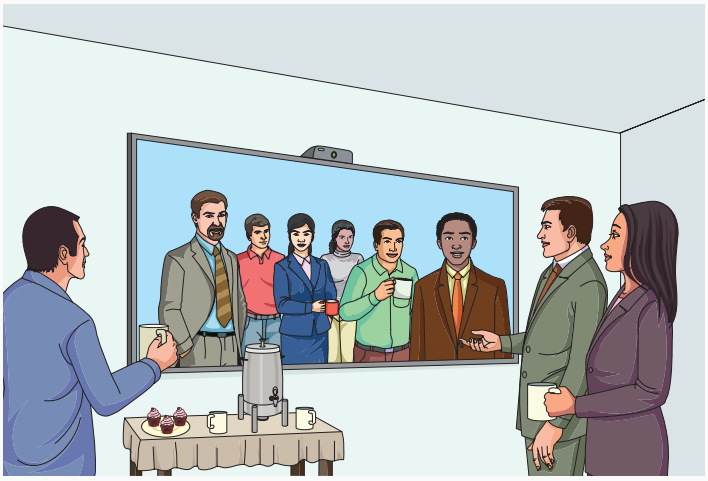
Early Videoconferencing: VideoWindow (Bellcore, 1989)

- Shared space that allowed people 50 miles apart to carry on a conversation as if in same room drinking coffee together
- 3 x 8 foot 'picture-window' between two sites with video and audio
- People did interact via the window, but strange things happened (Kraut, 1990)

Diagram of VideoWindow in Use



Findings of how VideoWindow System was used

- Talked constantly about the system
- Spoke more to other people in the same room rather than in other room
- When trying to get closer to someone in the other place it had opposite effect—participants went out of range of the camera and microphone
- No way of monitoring this

Videoconferencing and Telepresence Rooms

- Many to choose from to connect multiple people (e.g. Zoom, Teams, Google Meet, Webex)
- Customised telepresence rooms for groups

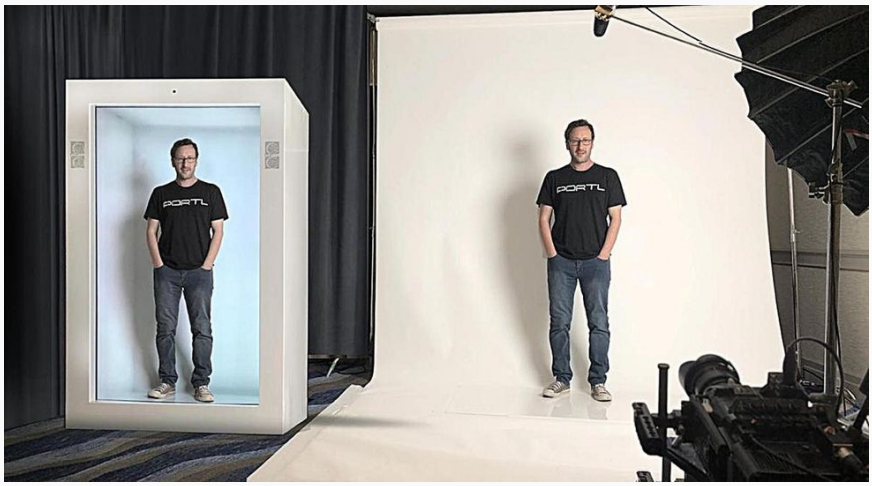
Current Videoconferencing

- Webex, Zoom etc., have greatly extended how we communicate while providing tools to make is easier to switch between talking and working together

- Zoom fatigue came into being (Bailenson, 2021)
  - excessive amounts of close-up eye gaze
  - intense cognitive load
  - increased self-evaluation from staring at a video of oneself
  - physically being in the same place for hours on end

Microsoft Prototype of a Technology-enhanced Hybrid Meeting

The future: 3D person in a box?



David Nussbaum demonstrating how they capture and present the Proto person in a box

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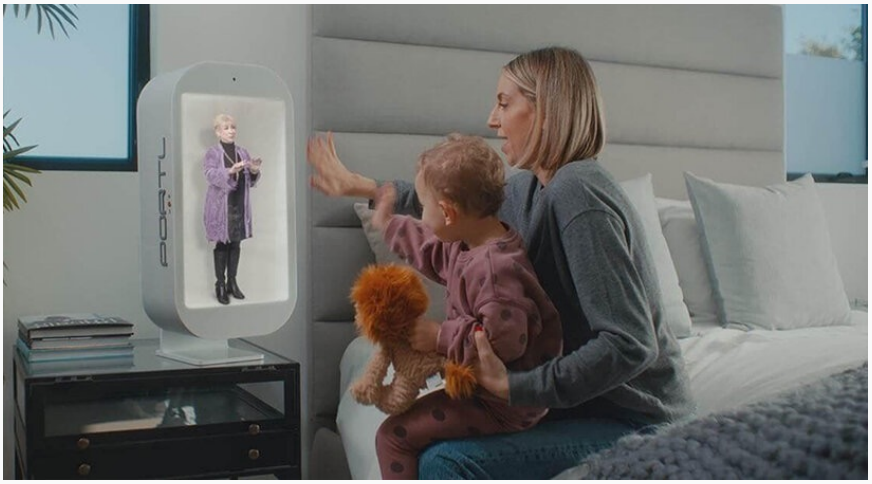
Proto M



Talking with a 3D video of granny in a box. The embedded camera at the top of the box faces the mother and child so that Granny can see and hear them in real time.

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Telepresence and Social Presence

- *Telepresence* refers to one party being present with another party, who is present in a physical space, such as a meeting room
- *Social presence* refers to the feeling of being there with a real person when in virtual reality (VR)

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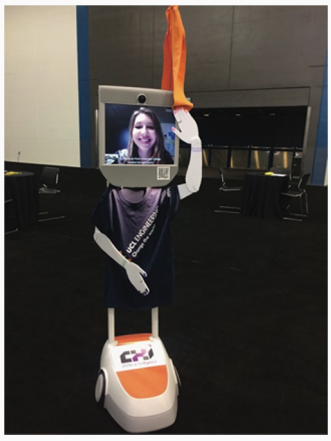
Telepresence Robots

Enable people to attend events who could not do so, such as by controlling their robot remotely

- In places such as schools, conferences, and museums
- Early example: Beam+
- Often dressed up to appear like the person to others at the event
- Positive experience of being there



Susan Lechelt at ACM CHI

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The Metaverse

- Meta's vision of three friends socializing in a 3D world represented as torso avatars
- Users experience each other through donning VR headsets

How much realism and immersion are necessary…?

- …in telepresence to make it compelling?
- *Telepresence rooms* try to make remote people appear to be life-like
  - Use multiple high-definition cameras with eye-tracking features and directional microphones
- Does FaceTime have as much presence as more high-definition settings?

Online Collaborative Tools

- Now more commonplace in our everyday and working lives
  - e.g. the sharing of calendars, word processing, design and project management tools like Slack
- Places for sharing knowledge
- Tools like Miro, Figjam and Google Docs enable online collaborative creation and editing of reports, designs, etc.,

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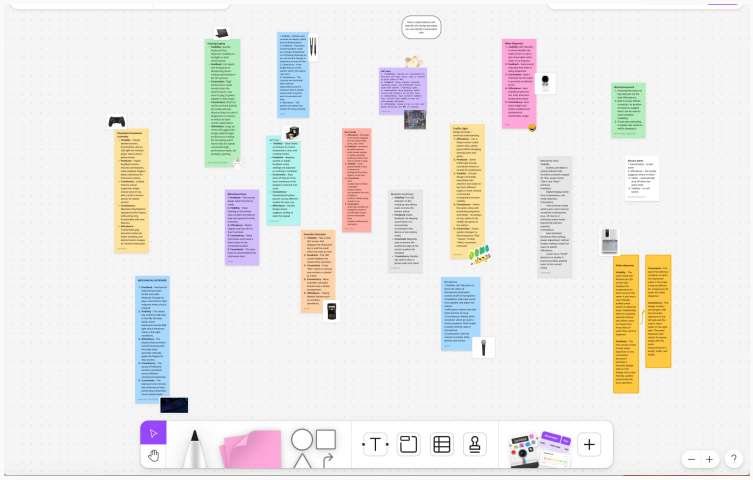
Collaborative Online Tools



Screen shot of a Figjam board used in an online class on Human-Computer Interaction where students upload their examples

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CO-PRESENCE

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What is Co-presence?

- Co-located groups who want to collaborate
- Many technologies have been designed to:
  - Enable groups to work, learn and socialise more effectively together
  - For example, tabletops, whiteboards, and public displays

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Coordination Mechanisms

- When a group of people act or interact together, they need to coordinate themselves
  - For example, when playing football or navigating a ship
- To do so, they use:
  - Verbal and non-verbal communication
  - Schedules, rules, and conventions
  - Shared external representations

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In Person Coordinating Mechanisms

- Talk is central
- Non-verbal also used to emphasize and as a substitute
  - e.g., nods, shakes, winks, glances, gestures, and hand-raising
- Formal meetings
  - Explicit structures such as agendas, memos, and minutes are employed to coordinate the activity

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Awareness Mechanisms

- Involves knowing who is around, what is happening, and who is talking with whom (Dourish and Bly, 1992)
- Peripheral awareness
  - Keeping an eye on things happening in the periphery of vision
  - Overhearing and overseeing—allows tracking of what others are doing without explicit cues
- Situational awareness
  - Being aware of what is happening around you in order to understand how information and your actions will affect ongoing and future events
    - For example, air traffic control or an operating theatre

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Shareable Interfaces

- Designed to capitalize on existing forms of coordination and awareness mechanisms

- Several studies investigating whether they help people to work together better, have found:
  - More equitable participation
  - More natural to work around
  - More comfortable sitting around a table than standing in front of a display

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The Reflect Table

- LEDs lit up to reflect how much each member of the group spoke
- Used microphones in front of each individual to do this
- Study showed those who spoke the most changed their behaviour the most
- Those who spoke the least did not change their behaviour
- Why do you think this is?

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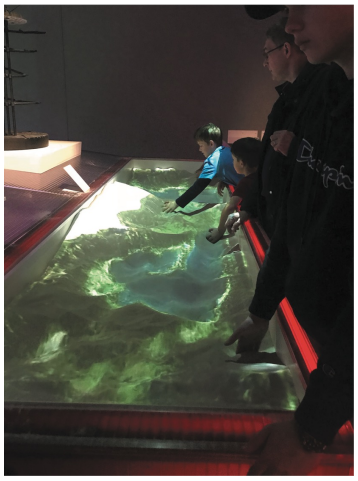
Playing Together in Same Space

- Visitors using an AR sandbox at the V&A museum in London
- Visitors sculpt landscapes out of sand
- System reacts with changing superimposed digital colored landscape
- Enables creative forms of collaboration



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PeopleLens: A head mounted device that enhances a blind child's spatial awareness of those around them

Video See PeopleLens in action: https://youtu.be/astmNfJHT4A.

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SOCIAL GAMES

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Social Games

- Designed to facilitate social interaction
- Can be played indoors or outdoors, with and without technology.
  - examples include board games, tabletop games, and videogames (such as Minecraft)
- Each player is aware of other players' presence, their actions and how well they are playing

Social Online Games

- Can involve creating a community, where competition, collaboration, peer pressure, rebellion, jealousy, and so on are all played out in their various forms
- Matt Richetti (2022) has proposed three heuristics for evaluating them:
  - Does the social game involve synchronous or asynchronous player interaction, where players either chat with each other or they take turns.
  - Is the social interaction symmetrical or asymmetrical, in the sense that does forming a relationship require input from both parties or can they be formed unilaterally by a single party?
  - Does the social relationship involve strong or weak ties in the sense of whether the relationships between players become deep and long lasting or are they transitory?
- By asking these questions, game designers can think about the kinds of social interactions they want to support

Summary

- Social interaction is central to our everyday lives
- Social mechanisms, like turn-taking, enable us to collaborate and coordinate our activities
- Keeping aware of what others are doing and letting others know what you are doing are important aspects of collaborative working and socialising
- Many technologies have been built to support telepresence, social presence, and co-presence
- Social online games media has brought about significant changes in how online games can support social interaction

Topic 5: Data Gathering (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

Aims

- Discuss how to plan and run successful data gathering sessions.
- Enable you to plan and run an interview.
- Enable you to design a simple questionnaire.
- Enable you to plan and carry out an observation.

2

01 ISSUES AND CAPTURING DATA CHALLENGES

02 INTERVIEWS

03 QUESTIONNAIRES

04 OBSERVATION

05 PUTTING THE TECHNIQUES TO WORK

3

3

ISSUES AND CAPTURING DATA CHALLENGES

....

4

4

Six Key Issues

1. Setting goals
   - Decide how to analyze data once collected
2. Identifying participants
   - Decide from whom to gather data
   - Determine how many participants are needed
3. Relationship with participants
   - Clear and professional
   - Informed consent when appropriate
4. Ethical considerations of collection and storage
   - Data collection is easy with today's lightweight devices
   - Personal data is protected by regulations, consent is needed
   - Storage of data must be secure
5. Triangulation
   - Investigate phenomena from more than one perspective
   - Collect data from different sources, with different investigators, using different theoretical frameworks and different techniques
6. Pilot studies
   - Small trial of main study

5

Capturing Data

- Notes, audio, video, and photographs can be used individually or in combination:
  - Notes plus photographs
  - Audio plus photographs
  - Video
- Different challenges and advantages with each type of data recording

6

INTERVIEWS

7

Interviews

- **Unstructured:** Not directed by a script. Rich but not replicable.
- **Structured:** Tightly scripted, often like a questionnaire. Replicable but may lack richness.
- **Semi-structured:** Guided by a script, but interesting issues can be explored in more depth. Can provide a good balance between richness and replicability.
- **Focus groups:** A group interview

8

Interviews Questions

- Two types:
  - 'Closed questions' have a predetermined answer format, for example, 'yes' or 'no'
  - 'Open questions' do not have a predetermined format
- Closed questions are easier to analyse
- Avoid:
  - Long questions
  - Compound sentences — split them into two
  - Jargon and language that the interviewee may not understand
  - Leading questions that make assumptions, for example, why do you like …?
  - Unconscious biases, for instance, gender stereotypes

9

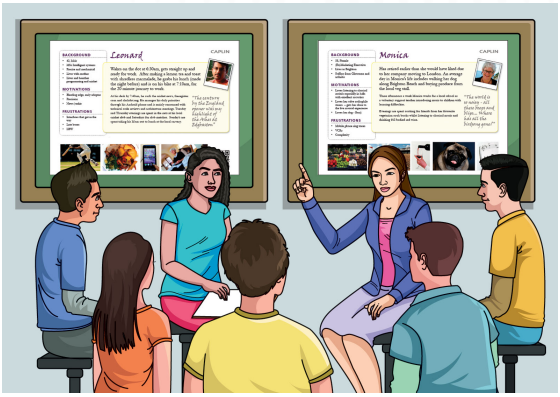
Running the Interviews

- **Introduction:** Introduce yourself, explain the goals of the interview, reassure about the ethical issues, ask to record, and present the informed consent form.
- **Warm-up:** Make first questions easy and non-threatening.
- **Main body:** Present questions in a logical order
- **A cooling-off period:** Include a few easy questions to defuse tension at the end
- **Closure:** Thank interviewee, signal the end, for example, switch recorder off.

10

Enriching the Interview Process

- **Props:** Devices for prompting interviewee, for example, use personas, prototypes or scenarios



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QUESTIONNAIRES

12

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Questionnaires

- Questions can be closed or open
- Closed questions are easier to analyze, and may be distributed and analyzed by computer
- Disseminated online so can be administered to large populations
- Sampling can be a problem when the size of a population is unknown as is common in online evaluations

Questionnaire Design

- The impact of a question can be influenced by question order.
- Different versions of the questionnaire may be needed for different populations.
- Provide clear instructions on how to complete the questionnaire.
- Consider whether the questionnaire is too long
- If the questionnaire is long consider allowing participants to opt out at certain stages.
- Think about layout and pacing.

Question and Response Format

- Closed-ended responses with predefined list:
  - radio buttons (single response)
  - check boxes (multiple responses)
- Rating scales
  - Likert scales
  - Semantic differential scales
  - 3, 5, 7 or more points
- Open-ended responses

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Encouraging a Good Response

- Make the purpose of study clear
- Promise anonymity
- Design the questionnaire carefully and run a pilot study
- Offer a short version for those who do not have time to complete a long questionnaire
- Follow-up with prompting messages
- Provide an incentive, e.g. voucher
- 40% response rate is generally acceptable but much lower rates are common

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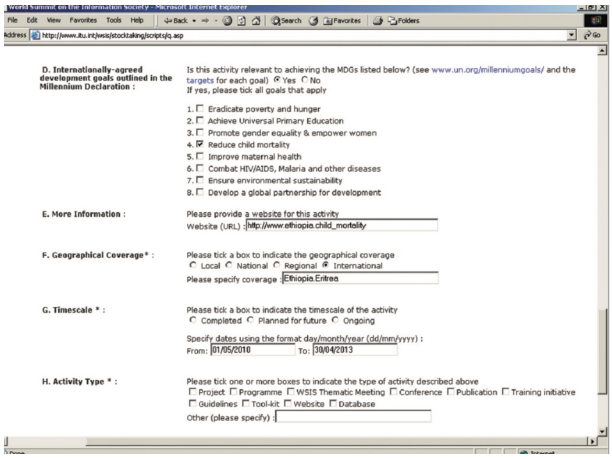
Example Web-based Questionnaire



Questionnaire showing check boxes, radio boxes, and pull-down menus

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Administering Questionnaires

- Plan the timeline
- Design offline
- Program/complete online template
- Test the survey to make sure that it behaves as you would expect
- Test it with a group that will not be part of the survey to check that the questions are clear
- Recruit participants

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OBSERVATION

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Observation

- Direct observation in the wild
  - Structuring frameworks
  - Degree of participation (passive or participant)
  - Ethnography
- Direct observation in controlled environments
  - The Think-aloud technique
- Indirect observation: tracking users' activities
  - Diaries
  - Interaction logging, web analytics and data scraping
  - Video and photographs collected remotely, e.g. by drones
  - Wearable sensors and social media

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Structuring Frameworks to Guide Observation

- Three easy-to-remember parts:
  - The person: Who?
  - The place: Where?
  - The thing: What?
- A more detailed framework (Robson & McCarten, 2016):
  - Space: What is the physical space like and how is it laid out?
  - Actors: What are the names and relevant details of the people involved?
  - Activities: What are the actors doing and why?
  - Objects: What physical objects are present, such as furniture
  - Acts: What are specific individual actions?
  - Events: Is what you observe part of a special event?
  - Time: What is the sequence of events?
  - Goals: What are the actors trying to accomplish?
  - Feelings: What is the mood of the group and of individuals?

Planning and Conducting Observation in the Wild

- Decide on how involved you will be: from passive observer to active participant
- How to gain acceptance
- How to handle sensitive topics, for example, culture, private spaces, and so on
- How to collect the data:
  - What data to collect
  - What equipment to use
  - When to stop observing

Ethnography

- Ethnography is a philosophy with a set of techniques that include participant observation and interviews
- Debate about differences between participant observation and ethnography
- Ethnographers immerse themselves in the culture that they study
- A researcher's degree of participation can vary
- Analysing video and data logs can be time-consuming
- Comments, incidents, and artifacts are collected

Ethnography (cont.)

- Co-operation of people being observed is required
- Data analysis is continuous
- Interpretivist technique
- Questions are refined as understanding grows
- Reports usually contain examples

Observations and Materials that might be Collected (Crabtree, 2003)

- Activity or job descriptions
- Rules and procedures that govern particular activities
- Descriptions of activities observed
- Recordings of the talk taking place between parties
- Informal interviews with participants explaining the detail of observed activities
- Diagrams of the physical layout
- Photographs, videos and descriptions of artifacts
- Workflow diagrams showing the sequential order of tasks
- Process maps showing connections between activities

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Direct Observation in a Controlled Environment

- The Think-aloud Technique

*"I'm typing in **www.lycos.com**, as you told me." <types>*
*"Now I am typing **child's ebike** and then clicking the search button**.** <pause and silence>*
"It's taking a few seconds to respond."
"Oh! Now I have a choice of other websites to go to. Hmm, I wonder which one I should select. Well, I need some help in deciding, so perhaps I should start by looking at a review of ebikes for kids. This one has an up-to-date list of current ebikes. <He clicks on The 13 BEST Electric Bikes for Kids [2022]>
"Gosh, there's a lot of models to select from, and I need to know what size wheels would suit a 10-year-old. I guess that depends on how long their legs are! Hmm, maybe some of them recommend ages in their specifications."
<pauses and looks at the screen>
I guess I should scroll through them and identify those that might be appropriate."
<silence . . . >

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Indirect Observation: Tracking User's Activities

- Diaries
- Interaction logs
- Web analytics
- Data scraping

- Video, audio, photos, and notes are used to capture data in both direct and indirect observation

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PUTTING THE TECHNIQUES TO WORK

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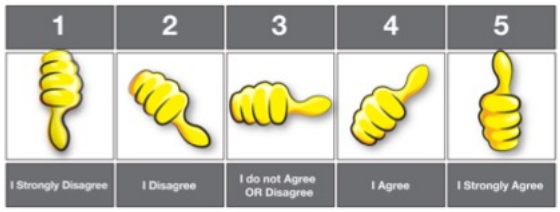
Choosing Techniques

- Focus of the study
- Participants involved
- Nature of the technique(s)
- Resources available

* Adapting techniques to suit different participants

Adapting the Techniques for Different Participants



Visual representation of a Likert scale for children (Putnam et al, 2020)

GPS tracker on a cat (Paci et al, 2020)

Gathering Data Remotely

Rapid Transitioning to Remote UCD Activities
1. Establish remote access for as many systems as possible.
2. Include remote access in IRB protocols.
3. Run pilot tests before conducting sessions with participants.
4. Have backup plan(s) in case of technological issues.
Interacting with Participants
5. Inform participants ahead of time about any technical requirements.
6. Use technologies that will be familiar and common to participants.
7. Use retrospective questioning if facing issues with the think-aloud method.
8. Gather information about the field site before running sessions.
Interacting with Other Researchers
9. Define the roles for each research member before the session.
10. Introduce the research team members and their various roles at the beginning of the session.

Best practices for remote data gathering activities (Mastrianni et al, 2021)

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Key Points

- Data gathering sessions should have clear goals
- An informed consent and other permissions may be needed
- Six key issues of data gathering are: goals, identifying participants, participant relationship, ethics of collection and storage, triangulation and pilot studies
- Data may be captured as notes, audio and/or video recording, photographs, or any combination of these
- Interviews may be structured, semi-structured, or unstructured
- Questionnaires may be on online, sent by email or paper-based
- Observation may be direct or indirect; in the wild or in controlled settings
- Techniques are commonly combined for any one study
- Techniques may need to be adapted for participants and their context

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Topic 5: Discovering Requirements (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 WHAT, HOW AND WHY?

02 WHAT ARE REQUIREMENTS?

03 DATA GATHERING FOR REQUIREMENTS

04 BRINGING REQUIREMENTS TO LIFE: PERSONAS AND SCENARIOS

04 CAPTURING INTERACTION WITH USE CASES

2

WHAT, HOW, & WHY?

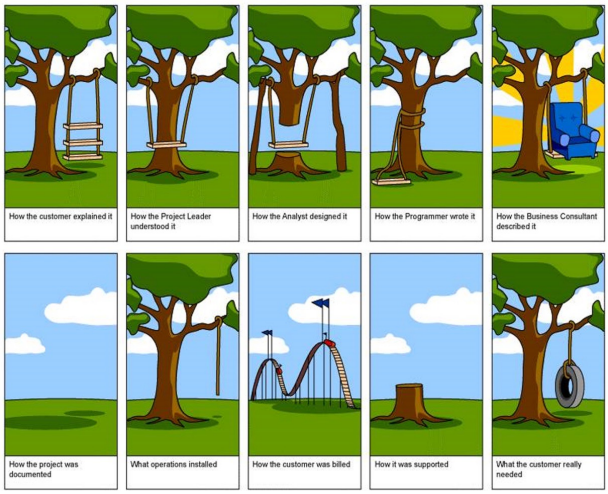
3

What, How?

- What is the purpose of the requirements activity?
  - Explore the problem space
  - Establish the design challenge to be addressed
- How can requirements be captured once discovered
  - In prototypes or operational product
  - Through structured or rigorous notations
  - Criteria to show when requirements are fulfilled
  - Different physical and digital representations emphasise and de-emphasise different aspects

4

Why are requirements important?



Requirements activity is the stage where miscommunication occurs most commonly

5

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WHAT ARE REQUIREMENTS?

6

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What are Requirements?

- A statement about an intended product that specifies what it is expected to do or how it will perform
- Different forms and different levels of abstraction
- User stories (prevalent in agile development contexts)
- Format:

  As a <role>, I want <behavior> so that <benefit>
- Example user stories for a travel organiser might be:

  As a <traveler>, I want <to save my favorite airline for all my flights> so that <I will be able to collect air miles>

  As a <travel agent>, I want <my special discount rates to be displayed to me> so that <I can offer my clients competitive rates>

7

7

Volere Shell

Requirement #: **75** Requirement Type: **9** Event/use case #: **6**

Description: **The product shall issue an alert if a weather station fails to transmit readings.**

Rationale: **Failure to transmit readings might indicate that the weather station is faulty and needs maintenance, and that the data used to predict freezing roads may be incomplete.**

Source: **Road Engineers**
Fit Criterion: **For each weather station the product shall communicate to the user when the recorded number of each type of reading per hour is not within the manufacturer's specified range of the expected number of readings per hour.**

Customer Satisfaction: **3** Customer Dissatisfaction: **5**
Dependencies: **None** Conflicts: **None**
Supporting Materials: **Specification of Rosa Weather Station**
History: **Raised by GBS, 28 July**

Volere
Copyright © Atlantic Systems Guild

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Different kinds of Requirements

- Functional:
  - What the system should do
- Data:
  - What kinds of data need to be stored?
  - What are the characteristics of the data?

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Different kinds of Requirements (cont.)

- Environment or context of use:
  - **Physical**: dusty? noisy? vibration? light? heat? humidity? …. (for example, in a hospital)
  - **Social:** collaboration and co-ordination, data sharing, distributed, synchronous or asynchronous, privacy
  - **Support:** user support, communications structure and infrastructure, availability of training
  - **Technical:** on which technologies will it run or with which it needs to be compatible?

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Different kinds of Requirements (cont.)

- User characteristics — Who are they?
  - **Characteristics:** educational background, personal circumstances, abilities or skills
  - **System use**: novice, expert, casual, frequent
    - **Novice**: prompted, constrained, clear
    - **Expert**: flexibility, access/power
    - **Frequent:** shortcuts
    - **Casual/infrequent**: clear menu paths
  - User profile

Different kinds of Requirements (cont.)

- Usability goals
- User experience goals
- Different products have different requirements and may be implemented in different ways, for example, security

Usable Security

- How to make security robust without detracting from user experience
  - If the usability of security is ignored, then security mechanisms will be circumvented
  - What might affect usable security
    - Making security mechanisms visible or not
    - Sonification of password strength
    - Good design may have undesirable effect on security

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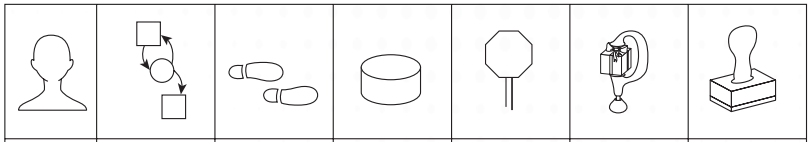
The Seven Product Dimensions



User	Interface	Action	Data	Control	Environment	Quality Attribute
Users interact with the product	The product connects to users, systems, and devices	The product provides capabilities for users	The product includes a repository of data and useful information	The product enforces constraints	The product conforms to physical properties and technology platforms	The product has certain properties that qualify its operation and development

Source: Gottesdiener and Gorman (2012), p.58. Used courtesy of Ellen Gottesdiener

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DATA GATHERING FOR REQUIREMENTS

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Data Gathering for Requirements

- Interviews, observation, and questionnaires
- Studying documentation:
  - Procedures and rules are often written down in manuals
  - Good source of data about the steps involved in an activity and any regulations governing a task
  - Not to be used in isolation
  - Good for understanding legislation and getting background information
  - No stakeholder time, which is a limiting factor for other techniques
- Researching similar products:
  - Good for prompting requirements

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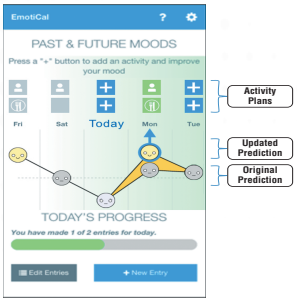
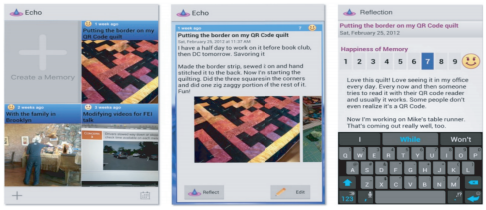
Combining Data Gathering

Direct observation, indirect observation, interviews, diaries, and surveys



(a)



(b)

Source: Hollis et al (2017), Figure 1.
Used courtesy of Taylor and Francis

Combining Data Gathering

- *Diaries and interviews*: multiple information devices
- *Interviews, think aloud evaluation, questionnaire, evaluation of working prototype*: memory aid for traumatic brain injury
- *Studying documentation, evaluating other systems, user observation, and focus groups*: ship's maneuvering system
- *Questionnaire, focus group, design probe, and user study*: smart meters for power consumption and privacy

Using Probes with Stakeholders

- Many types of probe:
  - Designed to prompt stakeholders into action
  - For researchers to learn about participants
  - All rely on data logging: automatic or manual
- Design probe:
  - Form relates specifically to particular question and context
- Cultural probe:
  - Participants asked to answer questions using postcards, maps, camera, photo album, and diary
- Technology probe:
  - A working prototype placed in an in-the-wild context
- Provocative probe:
  - Designed to challenge existing norms and attitudes

Contextual Inquiry

- Part of Contextual Design, and used on its own to discover requirements
- One-on-one field interviews (contextual interviews)
  - 1.5 to 2 hours long
  - Focus on daily life at home or work relevant to the project
  - Uses a model of master (participant) and apprentice (researcher)
- Four main principles
  - **Context:** Going to the participant, wherever they are, and seeing what they do as they do it
  - **Partnership:** Participant & interviewer explore participant's life together
  - **Interpretation:** Observations interpreted by participant & interviewer together
  - **Focus:** Project focus guides what should be paid attention to

Contextual Inquiry (cont.)

- Interview guided by "cool concepts" divided into two groups
- Joy of life concepts:
  - How products make our lives richer and more fulfilling
  - Accomplish, connection, identity, and sensation
- Joy of use concepts:
  - Describe impact of using the product
  - Direct in action, the hassle factor, and the learning delta
- Interview in four parts
  - Overview, transition, main interview, and wrap-up
- Following interview, interpretation session
  - Contextual design models are created or consolidated
  - Most relevant models are chosen by team, out of 10 suggested

Brainstorming for Innovation

- Alex Osborn's (1930s) four rules:
  - Quantity over quality
  - Criticisms should be withheld
  - Encourage out-of-the-box thinking
  - Combine, refine and improve ideas
- Brainstorming requires careful facilitation

BRINGING REQUIREMENTS TO LIFE

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Bringing Requirements to Life

- Augmenting the basic requirements expressed as stories, in Volere template, or in other form
- Personas
  - Characterises someone who might use the product, not specific people
- Scenarios
  - An informal narrative story, simple, 'natural', personal, and not generalisable

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Scenarios and Personas

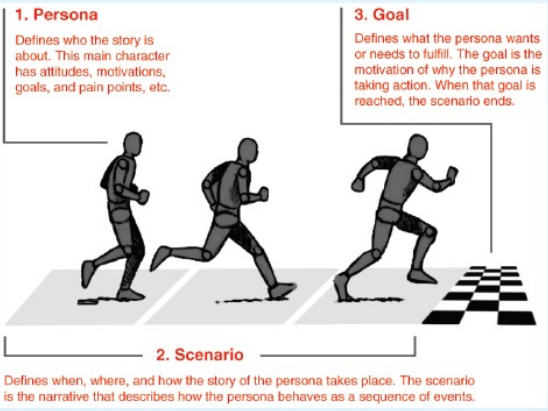

Figure 10.10 The relationship between a scenario and its associated persona
Source: http://www.smashingmagazine.com/2014/08/06/a-closer-look-at-personas-part-1/

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Personas

- Capture a set of user characteristics (user profile)
- Synthesised from real people based on research
- Typical, not idealised
- Bring to life with name, characteristics, goals, and personal background
  - Relevant to product under development
- Good persona helps designer with design decisions and reminds team about who will use the product
- Develop a small set of personas with one primary

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Example Persona #1

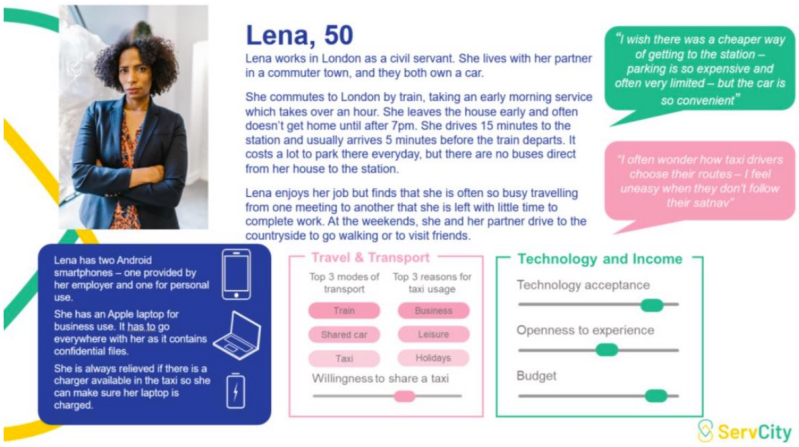

Figure 5. Example persona "Lena." Photo by RODNAE Productions from Pexels.

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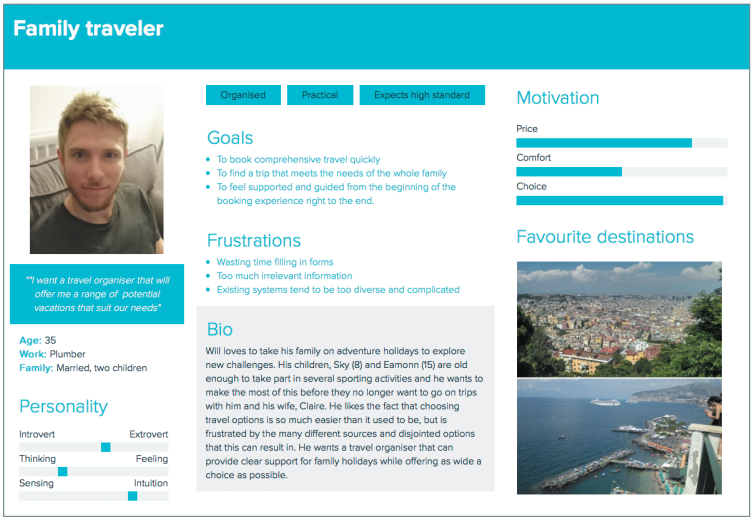
Example Persona #2



Developed using Xtensio Templates

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Scenario for Group Travel Organiser

"The Thomson family enjoy outdoor activities and want to try their hand at sailing this year. There are four family members: Sky (8 years old), Eamonn (12 years old), Claire (32), and Will (35). One evening after dinner they decide to start exploring the possibilities. They want to discuss the options together but Claire has to visit her elderly mother so will be joining the conversation from her mother's house down the road. As a starting point, Will enters an idea they had been discussing over dinner – a sailing trip for four novices in the Mediterranean. The system supports users to log on from different locations and use different devices so that all members of the family can interact easily and comfortably with it wherever they are. The system's initial suggestion is a flotilla, where several crews (with various levels of experience) sail together on separate boats. Sky and Eamonn aren't very happy at the idea of going on vacation with a group of other people, even though the Thomson's would have their own boat. The travel organizer shows them descriptions of flotillas from other children their ages and they are all very positive, so eventually, everyone agrees to explore flotilla opportunities. Will confirms this recommendation and asks for detailed options. As it's getting late, he asks for the details to be saved so everyone can consider them tomorrow. The travel organizer messages them a summary of the different options available."

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Scenarios

- May be textual descriptions, animations, audio or video
- Example animation scenarios

Source: Tommy Nilsson, et al (2020) Visions, Values, and Videos: Revisiting Envisionings in Service of UbiComp Design for the Home, DIS '20: Pages 827–839
ACM Publications.

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Design Fiction

- Communicate a vision with future technologies
- Fictional world in which ethics, emotions, and context can be explored without concrete constraints
- Examples:
  - Privacy and surveillance
  - Exploring ethics
- Scenarios are about "overcoming the monster," while design fiction is about "quest"

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CAPTURING INTERACTION USING USE CASES

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Use Cases

- Focus on functional requirements and capture interaction
- Can be used in design or to capture requirements
- Use cases are step-by-step descriptions of interactions
- Two styles:
  - Essential use cases: division of tasks, no implementation detail
  - Use case with normal and alternative courses: more detail

Example Essential Use Case for Travel Organiser

retrieveVisa

USER INTENTION	SYSTEM RESPONSIBILITY
Find visa requirements	
	Request destination and nationality
Supply required information	
	Obtain appropriate visa info
Obtain copy of visa info	
	Offer info in different formats
Choose suitable format	
	Provide info in chosen format

Note: The user intention and system responsibility are offset vertically, showing a sequence of interactions

Use Case for Travel Organiser

1. The product asks for the name of the destination country
2. The user provides the country's name
3. The product checks that the country is valid
4. The product asks the user for their nationality
5. The user provides their nationality
6. The product checks the visa requirements of that country for a passport holder of the user's nationality
7. The product provides the visa requirements
8. The product asks whether the user wants to share the visa requirements on social media
9. The user provides appropriate social media information

Alternative Courses for Travel Organiser

Some alternative courses:

4. If the country name is invalid:
    - 4.1: The product provides an error message
    - 4.2: The product returns to step 1
6. If the nationality is invalid:
    - 6.1: The product provides an error message
    - 6.2: The product returns to step 4
7. If no information about visa requirements is found:
    - 7.1: The product provides a suitable message
    - 7.2: The product returns to step 1

Summary

- A requirement is a statement about an intended product that specifies what it is expected to do or how it will perform.
- Articulating requirements avoids miscommunication, supports technical developers and allows stakeholders to contribute.
- Different kinds of requirements: functional, data, environmental (context of use), user characteristics, usability goals, and user experience goals.
- Scenarios are a story-based narrative to explore existing behavior, potential of new products, and futuristic visions of use.
- Personas capture realistic profiles of people who might use the product and are synthesized from data gathering.
- Scenarios and personas can be used together throughout the product lifecycle.
- Use cases capture details about an existing or imagined interaction between users and the product.

Topic 6: More on Persona, Scenario & HTA

SCSM2113 Human-Computer Interaction Fundamentals

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1

01 CREATING A PERSONA

02 TASKS ANALYSIS

03 WRITING A SCENARIO

2

CREATING A PERSONA

3

Recap: What is a Persona?

- A persona is a fictional character representing a major user group for a system being designed.
- Captures the primary needs and expectations of key user segments.
- Derived from user background, tasks, behaviors, attitudes, and motivations.
- Informed by user interviews.
- Documented in a concise, CV-style format.
- Persona development is a relatively fast process.

4

Why are Personas Needed?

- Traditionally user-centred design involved researching the needs of as many users as possible and collecting all of their requirements. This resulted in a long list of needs with no sense of priority. This lack of direction typically translated into designs that tried to serve all users but ended up serving no user particularly well.

- Creating Personas allow you to identify discrete sets of users and create typical users to represent each group.

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Benefits

- Simplifies design decision-making.
- Clarifies persona behaviours and goals.
- Prevents assumptions about users.
- Ensures diverse user needs are represented.
- Avoids designing for wants instead of actual needs.
- Enables ongoing design evaluation against personas.
- Resolves design disagreements by referencing personas.

6

6

Limitations

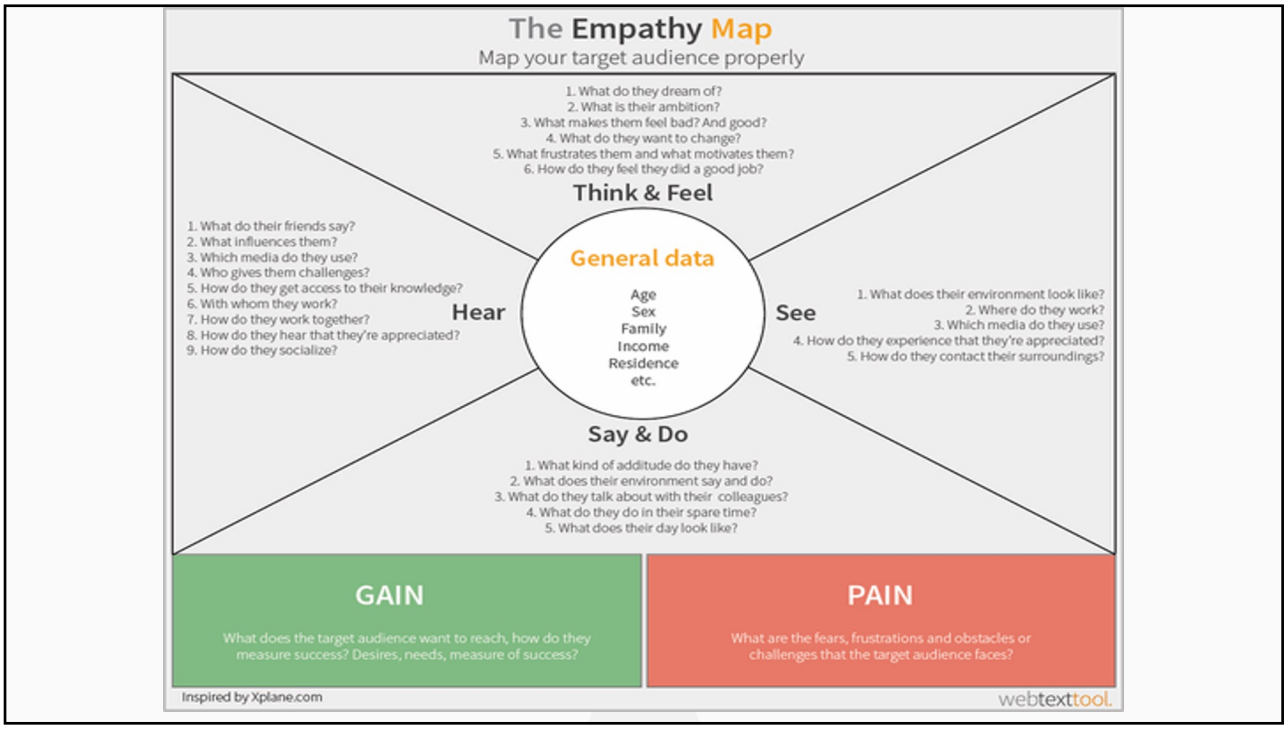
- Personas must address a specific design problem.
- They are not reusable across different projects.
- Context-specificity is critical.
- Behaviours and goals must align with the domain.
- Unique personas are required for each product.

7

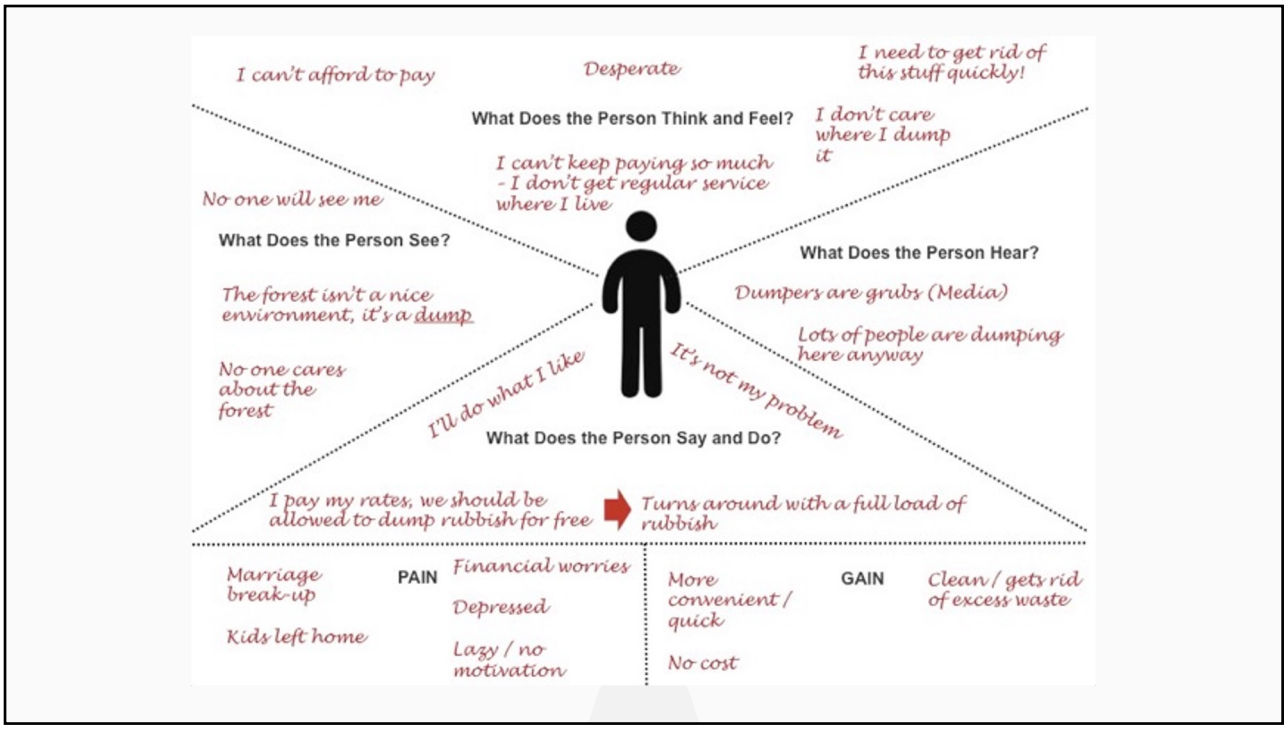
How to Create a Persona?

- Collect background data from your target users
- Take primarily tasks, behaviour, goals, attitudes and environment into account
  - Use **empathy map**
- Add personal and motivational details
- Make the personas come to life by including real photos
- Check back with different stakeholders and people who have regular customer contact
- Check them in context with market segments

8


The Empathy Map
Map your target audience properly
1. What do they dream of?
2. What is their ambition?
3. What makes them feel bad? And good?
4. What do they want to change?
5. What frustrates them and what motivates them?
6. How do they feel they did a good job?
Think & Feel
1. What do their friends say?
2. What influences them?
3. Which media do they use?
4. Who gives them challenges?
5. How do they get access to their knowledge?
6. With whom they work?
7. How do they work together?
8. How do they hear that they're appreciated?
9. How do they socialize?
Hear
General data
Age
Sex
Family
Income
Residence
etc.
See
1. What does their environment look like?
2. Where do they work?
3. Which media do they use?
4. How do they experience that they're appreciated?
5. How do they contact their surroundings?
Say & Do
1. What kind of additude do they have?
2. What does their environment say and do?
3. What do they talk about with their colleagues?
4. What do they do in their spare time?
5. What does their day look like?
GAIN
What does the target audience want to reach, how do they measure success? Desires, needs, measure of success?
PAIN
What are the fears, frustrations and obstacles or challenges that the target audience faces?
Inspired by Xplane.com
webtexttool.

9


I can't afford to pay
Desperate
I need to get rid of this stuff quickly!
What Does the Person Think and Feel?
I don't care where I dump it
I can't keep paying so much - I don't get regular service where I live
No one will see me
What Does the Person See?
The forest isn't a nice environment, it's a dump
No one cares about the forest
What Does the Person Hear?
Dumpers are grubs (Media)
Lots of people are dumping here anyway
I'll do what I like
It's not my problem
What Does the Person Say and Do?
I pay my rates, we should be allowed to dump rubbish for free
Turns around with a full load of rubbish
PAIN
Marriage break-up
Kids left home
Financial worries
Depressed
Lazy / no motivation
GAIN
More convenient / quick
No cost
Clean / gets rid of excess waste

10

https://conceptboard.com/blog/create-a-customer-empathy-map-in-6-easy-steps/

11

Example of a Person (1)

Nikki is 50 and works as a Doctor in the Royal Melbourne Hospital. She is a specialist in cardiology and has over 18 years experience in that field of research. She often uses a computer tomography or ultrasound machine. Nikki makes judgments rationally. She wants the software that comes with the medical devices to give her results fast and in graphical format with the opportunity to zoom things to the max. Nikki has a positive attitude in terms of technology as she has experienced it as a great support. She is confident using a computer even though he's not an expert.

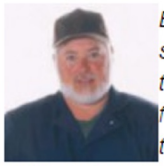
12

..from Nikki's Persona Profile

- Nikki's goals when using the software, she
  - wants the software that comes with the medical devices to give her results fast and in graphical format with the opportunity to zoom things to the max.

13

Example of a Persona (2)



Bob is 52 years old and works as a mechanic with an organisation offering road service to customers when their car breaks down. He has worked in the job for the past 12 years and knows it well. Many of the younger mechanics ask Bob for advice when they meet up in the depot as he always knows the answer to tricky mechanical problems. Bob likes sharing his knowledge with the younger guys, as it makes him feel a valued part of the team.

Bob has seen many changes over the years with the company and has tried his best to move with the times. However he found it a bit daunting when a new computer was installed in his van several years ago, and now he has heard rumours that the computer is going to be upgraded to one with a bigger screen that's meant to be faster and better.

Bob's been told that he will be able to access the intranet on the new computer. He has heard about the intranet and saw once in an early version on his manager's computer. He wonders if he will be able to find out want's going on in the company more easily, especially as customers' seem to know more about the latest company news than he does when he turns up at a job. This can be embarrassing and has been a source of frustration for Bob throughout his time with the company.

14

..from Bob's Persona Profile

- Bob's goals when using the internet
  - wants to avoid feeling stupid
  - would like to retain status as mentor
  - make him more informed when interacting with customers

15

TASKS ANALYSIS

16

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What is a Task?

- A task is a set of user actions to achieve a specific user goal, supported by the system.
- The scope of a task depends on the user's objective and the design context.
- Tasks vary in size and complexity based on the goal and system requirements.

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Task Analysis

- Task descriptions: often used to envision new systems or devices
- Task analysis: used mainly to investigate an existing situation
- Means of analysing and describing the jobs people do
  - Method and notation
- A method/set of methods for understanding the tasks users carry out with a product/system
- To analyse the underlying rationale and purpose of what people are doing; what are they trying to achieve, why are they trying to achieve it, and how are they going about it
- Focus on
  - User Analysis – involves learning about the user
  - peoples' goals and the actions they carry out
  - things people know
  - the things they act on
- Can be used for many different purposes within design and evaluation activities
- Many techniques, the most popular is **Hierarchical Task Analysis (HTA)**

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Benefits of Task Analysis

- Determine:
  - The tasks the system has to support
  - The scope of the system
- Develop a
  - Streamlined interaction method.
  - Logical structured user interface
- Create a system that reflects
  - Users goals
  - Tasks
  - Workflow

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What can we learn from task analysis?

- User's goals
- How the user achieves their goals
- Personal, social, and cultural characteristics that influence the user in doing their tasks
- How the physical environment affects them in completing their tasks
- How previous experience knowledge influence their thinking
- How they approach work and the workflow they use to complete tasks

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Data Collection: Where do tasks come from?

- How can we find out about what people do this?
  - Watch them
  - Talk to them
  - Read documentation, training material, etc.
  - Do the task ourselves
  - Become familiar with existing systems

Visiting Real Users

- Important to analyse real tasks
  - Not imaginary ones
  - Not the task of the designer
- Contextual inquiries
  - Genuine studies of what real users do in the workplace
- Task analysis is about **gathering and analysing data**

Question and Things to Consider

- Questions to ask the user:
  - What are the overall tasks you are trying to achieve?
  - What process do you follow to complete the task?
  - What likes and dislikes you have in the way you complete tasks?
- Things to consider:
  - Does the user efficiently complete the task?
  - What other ways are their to complete the task?
  - Is there a more efficient way?
- Who are the users?
  - Diversity
  - Question people who *don't* use the system
- What do they do?
  - What do they *really* do?
  - Don't give false goals
  - Observe goals as well as tasks

Structured Task Analysis Process

- Identify the task to be analysed.
- Break this down into between 4 and 8 subtasks.
- Draw the subtasks as a layered diagram
- Decide upon the level of detail into which to decompose.
- Validate the analysis with team members

Task Decomposition

- state of the system that a human wants to accomplish.
- activities required, used, or deemed necessary to achieve a goal.
- steps required to complete the task.

Hierarchical Task Analysis (HTA)

- Involves breaking a task down into subtasks, then sub-sub-tasks and so on. These are grouped as plans which specify how the tasks might be performed in practice
- HTA focuses on physical and observable actions, and includes looking at actions not related to software or an interaction device
- Start with a user goal which is examined and the main tasks for achieving it are identified
- Tasks are sub-divided into sub-tasks
- Represent the goals, sub goals, operations and plans using either:
  - graphical views (boxes and arrows)
  - non-graphical methods (e.g. tabulation, outlines, textual)

Example of HTA (Graphical)

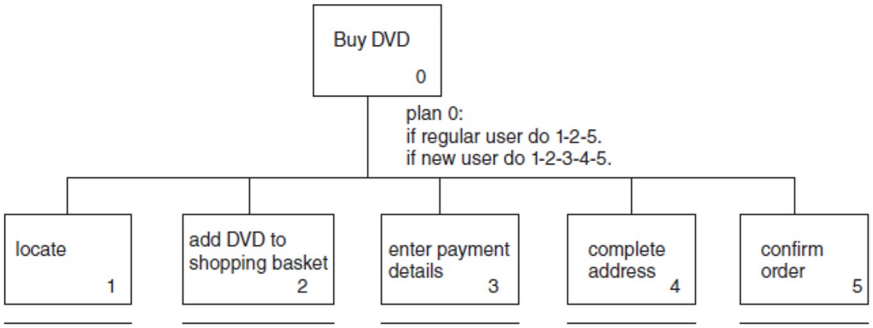

Figure 10.15 A graphical representation of the task analysis for buying a DVD

Example of Textual HTA

0. In order to buy a DVD
1. locate DVD
2. add DVD to shopping basket
3. enter payment details
4. complete address
5. confirm order

plan 0: If regular user do 1-2-5.
If new user do 1-2-3-4-5.

Adding Plans

- Hierarchical diagram / text specifies what subtasks are part of a task
- Does not specify how the subtasks are carried out
- Plans are used to describe
  - order of subtasks
  - conditional or optional subtasks
  - repetition
  - etc.
- A plan needed for each decomposed task

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HTA Example 1: withdrawing cash from an ATM

- Withdrawing cash from an ATM requires tasks from the user…
- What are those tasks?

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HTA Example 1 Textual Presentation

0 Withdraw cash
- 1 Check machine will work
  - 1.1 Look at status indicator
  - 1.2 Look for card logo
- 2 Insert card
- 3 Enter PIN number
- 4 Initiate withdrawal transaction
  - 4.1 Select withdraw cash
  - 4.2 Enter amount
- 5 Complete transaction
  - 5.1 Take card
  - 5.2 Take cash

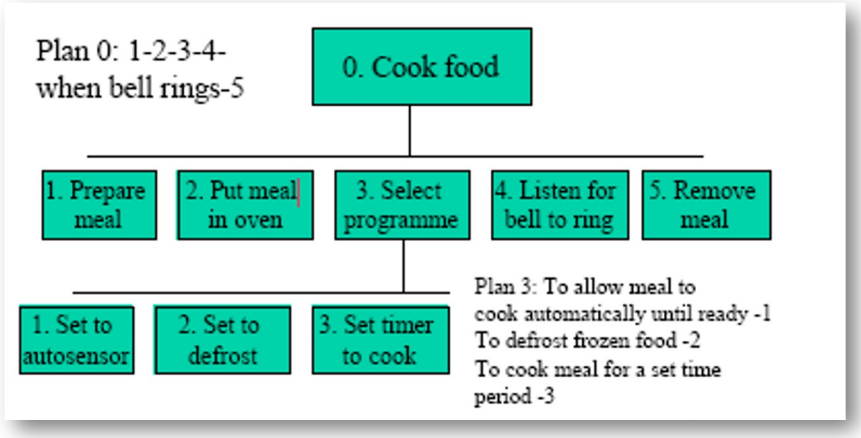
- A plan needed for each decomposed task

Plan 0: do 1; if possible do 2; repeat 3 until PIN correctly entered; do 4; do 5.
Plan 1: do 1.1, 1.2 in any order
Plan 4: do 4.1; do 4.2
Plan 5: wait until card available; do 5.1; wait until cash available; do 5.2

HTA Example 2: microwave oven

- What is the overall goal?
  - "Cook food!"
- How is this done?
  - Prepare meal
  - Put meal in oven
  - Select program
  - Listen for bell to ring
  - Remove meal
- Selecting a programme - How is this done?
  - Set to auto sensor
  - Set to defrost
  - Set timer to cook
- What are the rules that influence the order in which tasks/subtasks take place? (the plans)

HTA Example 2 Graphical Presentation



HTA Example 3: filing documents

- The manual task of filing documents in a filing cabinet…
  - You have an unsorted stack of documents
  - Your filing cabinet contains folders that are ordered somehow
  - How do you get all the documents into the correct folder?

HTA Example 3: … filing cabinet

- You probably act differently if you have a lot of documents to file rather than a few…

0. Store documents in filing cabinet

1. File lots of documents
2. File one or two documents

Plan 0: Do 1 or 2

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HTA Example 3: … filing one or two things

- Simply find the appropriate folder and put the documents in…

2. File one or two documents
   - 2.1. Open cabinet
   - 2.2. File each document
   - 2.3. Close cabinet

Plan 2: Do 2.1., (2.2. repeatedly) then 2.3.

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HTA Example 3:
… filing each document

2.2. File each document
- 2.2.1. Find appropriate file
- 2.2.2. Open file
- 2.2.3. Place document in file
- 2.2.4. Close file

Plan 2.2: Do 2.2.1, 2.2.2., 2.2.3., then 2.2.4.

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HTA Example 3:
… filing lots of document

- Sort the documents into order first
- then split the sorted documents up into 'categories' (i.e. all the documents whose author begins with 'A')
- then work through the filing cabinet, putting each category into the right file

1. File lots of documents
   - 1.1. Choose criteria on which documents are sorted
   - 1.2. Sort all documents to be filed into order
   - 1.3. Split documents up into categories
   - 1.4. Open cabinet
   - 1.5. Place each category of document into file
   - 1.6. Close cabinet

Plan 1: Do 1.1., 1.2., 1.3., 1.4., (1.5. repeatedly) then 1.6.

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HTA Example 3:
… choosing sorting criteria

1.1. Choose criteria on which documents are sorted
  1.1.1. Choose alphabetical by title of document
  1.1.2. Choose alphabetical by author of document
  1.1.3. Choose date order

Plan 1.1: Do any one of 1.1.1., 1.1.2., or 1.1.3.

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HTA Example 3:
… placing categories in files

1.5. Place each category of document into file
  1.5.1. Open file
  1.5.2. Place each document in file
  1.5.3. Close file

Plan 1.5: Do 1.5.1., (1.5.2. repeatedly) then 1.5.3.

40

40

Things to Note

- The overall task is numbered 0
- The tasks are all described by verbs
- Each task that is decomposed must have a plan to go with it
- Plan X must only refer to sub-tasks to X

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Reflecting on HTA Example 3

- Imagine that you are designing an electronic filing system…
- What does this HTA tell you about filing systems?
- Which bits of the **manual task would you keep**, and **which would you do away with** or **redesign**?
- Why?

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HTA Supports Refinement

How to check or improve the initial HTA?
Some heuristics are:

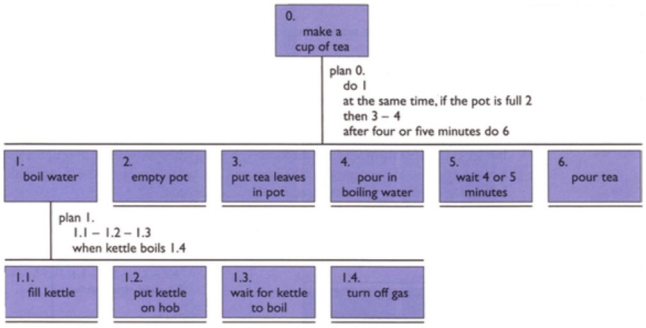
paired actions where is "turn on gas"?

restructure generate task "make pot"

balance is "pour tea" simpler than "make pot"?

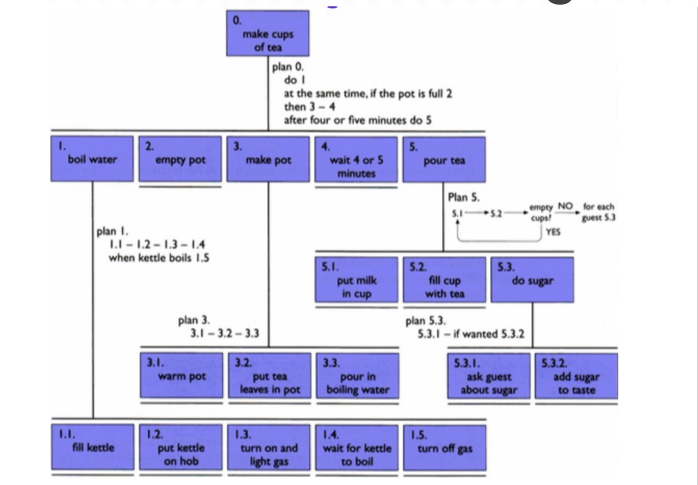
generalise make one cup . . . or more

HTA Example 4: making tea



[Dix et al, p. 515]

HTA Example 4:
… refined HTA for making tea



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HTA Example 4:
… types of plan

- **sequence** 1.1 then 1.2 then 1.3
- **optional** if the pot is full 2
- **wait** when kettle boils, do 1.4
- **cycles** do 5.1 5.2 while there are still empty cups
- **parallel** do 1; at the same time …
- **discretionary** do any of 1.3.1, 1.3.2 or 1.3.3 in any order

Most plans use several of these.

Waiting can be considered:

- a task — for "busy" waits, e.g. making tea
- part of the plan — end is the event, e.g. email reply received

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Activity

- Jill creates a stack of new catalogues in a corner of her kitchen. When she is preparing dinner, she can flip through a catalogue to see what's new or on sale or what strikes her interest. In the evening she may pick out five or six catalogues to look at while the family is watching TV. She may even take a few to bed, or to the bath.

- Generally she simply browses through pictures, reading descriptions only when the pictures look interesting. When she finds something interesting she may dog-ear the page, draw a circle round it or mark the page with a sticky note. She keeps catalogues with marked pictures around until she wants to make a purchase.

Sample Solution

0. Using catalogues
    1. Browse items
    2. Mark items

0. Using catalogues
    1. Browse items
        - 1.1. Browse by what's new
        - 1.2. Browse by sales
        - 1.3. Browse by items that interest
        - 1.4. Browse by catalogues
            - 1.4.1. Browse in kitchen
            - 1.4.2. Browse while watching TV
            - 1.4.3. Browse in bed
    2. Mark items

0. Using catalogues
    1. Browse items
    2. Mark items
        - 2.1. Circle item
        - 2.2. Mark page
            - 2.2.1. Dog-ear page
            - 2.2.2. Sticky on page

- The source doesn't say much about how the tasks are done, but…
  Plan 0: do 1 then 2 if item found
  Plan 1: do 1.1 or 1.2 or 1.3 or 1.4
  Plan 2: do 2.1 or 2.2

Reflecting the Activity

- What does this tell us?
  - Need flexible system
  - Need portable system
  - Need system that allows lots of different browsing criteria
  - Possibly need to improve system for marking selected items
- These are passed as **requirements** to the design cycle

WRITING A SCENARIO

What is a Scenario?

- A narrative description that **describes how a user is intending to use** a system
- Goal or Task-based
- Can include textual descriptions, illustrated stories, animations or even videos
- Presents the user's point of view

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Benefits

- Highlights an application within the work context
- Powerful communication tool
- Useful for testing of early prototypes
- Helps users understand the design
- Provides a contextual basis for testing
- Scenarios test the integrity of requirements and processes
- Gives an idea of current and future needs

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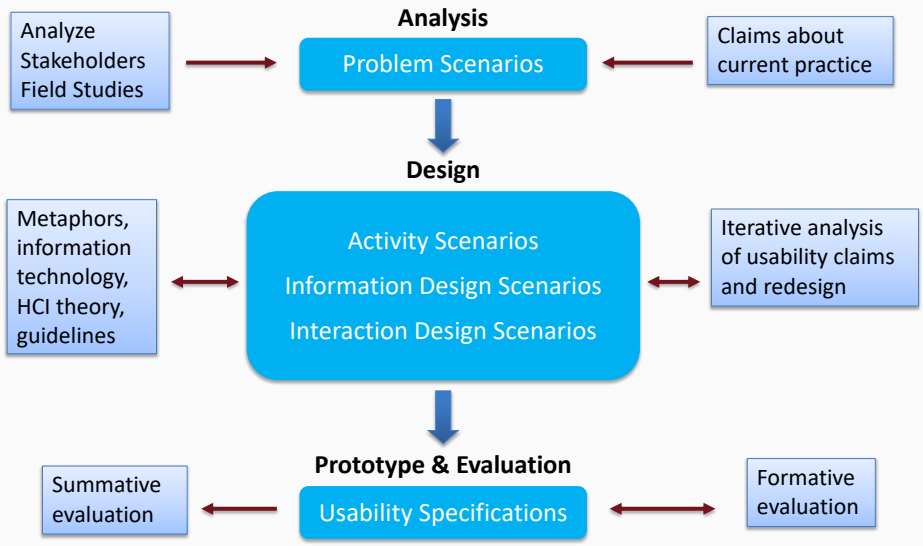
Writing Scenario Process



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How to Write a Scenario?

- Determine the target group and analyse their context of use
- Transform user goals into activities
- Determine the user tasks, and the system tasks.
- List in dot point the sequence of tasks
- Re-write task list into simple narrative language.
- Ensure you omit in your wording, any links to the technology
- Review the scenario to ensure it represents a real use context.

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How to Write a Scenario?
for using the ATM

- Persona:
  - Brad, 45, carefree office worker, travel often but does not have good time management
- User goal:
  - Withdraw money quickly from the ATM
- Tasks :
  - Identify himself
  - Key in the transaction details
  - The system responds
  - Transaction ends

55

55

Writing the Text-based Scenario

- Re-write task list into simple narrative language

It's Thursday afternoon and Brad is flying to Sydney. He doesn't have enough money for a taxi to the airport, and he's running late. He goes to the local ATM and identifies himself. He specifies that he wants $200 from his savings account. He'd like the money in $50 notes so that he can give the taxi driver the correct change. He doesn't want a printed receipt, as he doesn't bother keeping track of transactions in this account, besides he's already late. The system draws out his money, and the transaction ends.

- Ensure you omit in your wording, any links to the technology
- Review the scenario to ensure it represents a real use context

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56

Summary

- Scenarios, use cases and essential use cases can be used to articulate existing and envisioned work practices.
- Task analysis techniques such as HTA help to investigate existing systems and practices
- Task analysis gives designers a means of describing how people do their jobs
- Serves different purposes in design and development

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Topic 7: Design (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 METAPHORS AND ANALOGIES

02 INFORMATION ARCHITECTURE

2

METAPHORS AND ANALOGIES

3

3

Interface Metaphors

- Interface designed to be similar to a physical entity but also has own properties
  - e.g, desktop metaphor, web portals, search engine
- Can be based on activity, object, or a combination of both
- Exploit user's familiar knowledge, helping them to understand 'the unfamiliar'
- Conjures up the essence of the unfamiliar activity, enabling users to leverage this to understand more aspects of the unfamiliar functionality

4

4

Examples of Interface Metaphors

- Conceptualising what users are doing
  - e.g, surfing the Web
- A conceptual model instantiated at the interface
  - e.g., the desktop metaphor
- Visualising an operation
  - e.g., an icon of a shopping cart into which the user places items

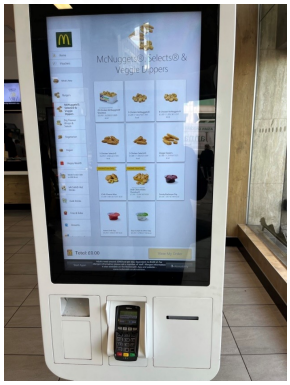
5

UTM

5

The Card Metaphor

- The card is a very popular metaphor. Why?
  - It has familiar form factor
  - It can easily be flicked through, sorted, and themed
  - Cards have strong associations, providing an intuitive way of organising content that is "card sized"
  - Structures content into meaningful chunks



A fast food display

6

UTM

6

Benefits of Interface Metaphors

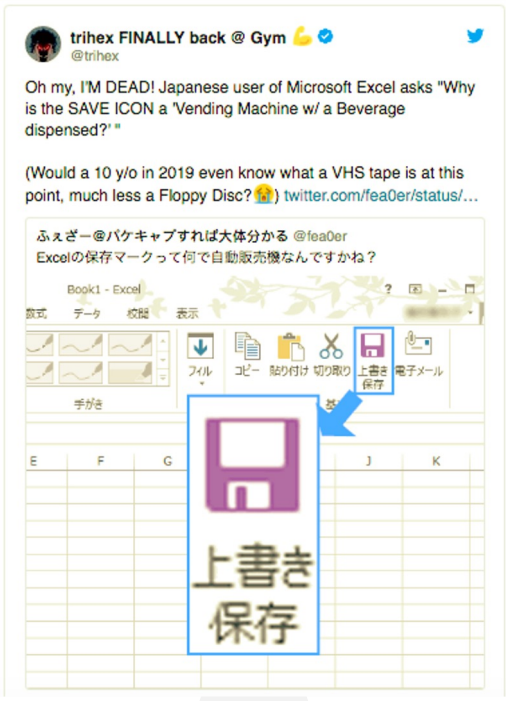
- Makes learning new systems easier
- Familiar artefact or action helps people understand the underlying conceptual model
- Can be innovative and enable apps to be made accessible to a greater diversity of people

7

Problems with Interface Metaphors

- Break conventional and cultural rules
  - e.g, recycle bin placed on desktop
- Can constrain designers in the way that they conceptualize a problem space
- Can conflict with design principles
- Designers can inadvertently use bad existing designs and transfer the bad parts over
- Limits designers' imagination in coming up with new conceptual models
- Often don't scale very well, especially when trying to describe how complex systems work (Cooper, 2020)

8



Activity

- Describe the components of the conceptual model underlying most shopping websites, for example:
  - Shopping cart
  - Proceeding to check-out
  - 1-click
  - Gift wrapping
  - Cash register

INFORMATION ARCHITECTURE

11

11

by sorting out things like this:

CLASSIFICATION and HIERARCHY

LABELS and TAGGING

NAVIGATION and WAYFINDING

SEARCH

INFORMATION ARCHITECTURE

(IA for short)

connects PEOPLE to the CONTENT they're looking for

The 'content' people are looking for could be text, images, videos, whole documents...

...or even conversations...

...and other people.

IA also helps the people who create and maintain content do their job easier...

...and links content to other content to make it more usable and easy to find.

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12

12

What is Information Architecture?

- Focuses on organising, structuring, and labeling content in an effective and sustainable way. How data is presented to users
- The goal is to help users find information and complete tasks by understanding how the pieces fit together to create the larger picture, how items relate to each other within the system
- Main components of IA:
  - Organisation Schemes and Structures: How you categorise and structure information
  - Labeling Systems: How you represent information
  - Navigation Systems: How users browse or move through information
  - Search Systems: How users look for information

13

13

How To?

- Defining taxonomies
  - bring together, like terms, so that they are sensibly classified
- To get these taxonomies right:
  - User research – structured interview to try and clarify how a user might approach a given data set
  - Card sorting technique – test how logical data sets that are already defined feel for the user to see if the users group data in the same way we (designers) did
- Sorting process include:
  - Terminology (what people call things)
  - Relationships (proximity, similarity)
  - Categories (groups and their names)

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14

14

Activity

- You are responsible for the IA of UTM Student Portal. From the recent gathered survey, most lecturers found the portal to be difficult at times, as they have to carefully look for things, and it was not so easy to understand some of the labeling
- Based on the created groups, try to sort the items for portal. You may make up your own groups names if you find them necessary.
- TWO kinds of analysis:
  - An items by groups chart shows what the groups were called and what was in each
  - An item by item chart shows which items were grouped together most often
- Online card sorting tool (example): https://www.optimalworkshop.com/

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Topic 7: Design (part 2)

SCSM2113 Human-Computer Interaction Fundamentals
Faculty of Computing
Universiti Teknologi Malaysia

1

01 DESIGN PRINCIPLES

02 VISUAL PERCEPTION

03 INDUSTRY PRACTICE

2

HCI
PRINCIPLES

3

Shneiderman's 8 Golden Rules of Interface Design

The principles	Questions to consider	Mark Complete
1. Strive for consistency	Is the style of this element maintained across your site/app? Is this content placed in the correct location according to the site hierarchy? Does this follow the conventions for your chosen platform? How can you make your designs more consistent?	☐
2. Enable frequent users to use shortcuts	Are there shortcuts available for your more experienced users? Who is this product designed for? Will there be a need to consider experienced users? How can you make it easier and quicker for experienced users?	☐
3. Offer informative feedback	Does the user know where they are at in the process? Does the user know what they have done after performing this action? How are you communicating this feedback to your user?	☐
4. Design dialogue to yield closure	Does the user have to do any guessing here? Is it clear and obvious enough for your intended audience? Are there any next steps for the user? How are you communicating the system status with the user?	☐
5. Offer simple error handling	Have you done everything imaginable to prevent this error from happening on your end? Is this error avoidable in the first place? If the user does make an error, how easy is it for them to fix it?	☐
6. Permit easy reversal of actions	How many steps does the user have to take to reverse their actions? Will the user quickly realize they need to reverse the action in the first place? How can you make your users detect the possibility of reversal?	☐
7. Support internal locus of control	Will the user feel in control at this specific touch point in your app? Will they be surprised in an unpleasant manner? Does the site feel easily navigable? Does the user feel safe and in control? How can you make the user feel more safe and in control?	☐
8. Reduce short-term memory load	Are there enough visual cues here for the user to find the functionality or item? Do they have to remember things to understand what's going on? How can you help the user recall?	☐

4

Norman's 7 Principles (1988):

1. Use both knowledge in the world and knowledge in the head.
2. Simplify the structure of tasks.
3. Make things visible.
4. Get the mappings right.
5. Exploit the power of constraints, both natural and artificial.
6. Design for error.
7. When all else fails, standardize.

5

18 Principles of Interaction Design by Bruce Tognazzini

No	Principle	Description
1	Anticipation	The system should attempt to anticipate the user's needs.
2	Autonomy	The user should feel in control of the system and its actions.
3	Color Blindness	Avoid using color as the only means of conveying information.
4	Consistency	The interface should be consistent in appearance and behavior.
5	Defaults	Reasonable default values should be provided.
6	Efficiency of the User	Design for user efficiency, not just system performance.
7	Explorable Interfaces	Users should be able to explore safely without fear of error.
8	Fitts's Law	Larger and closer targets are easier to hit.
9	Human Interface Objects	Use familiar interface objects that reflect real-world analogies.
10	Latency Reduction	Minimize delays and provide feedback to make latency feel shorter.
11	Learnability	The interface should be easy to learn and remember.
12	Metaphors	Use metaphors that are familiar and intuitive to users.
13	Protect Users' Work	Users' work should be preserved, even in the event of errors.
14	Readability	Text should be legible and easy to scan.
15	Track State	Keep users informed of the system's current state.
16	Visible Navigation	Navigation options should be easily visible and accessible.
17	User Control	Users should initiate and control actions, not the system.
18	Visible Interfaces	Important elements should be visible without unnecessary clutter.

6

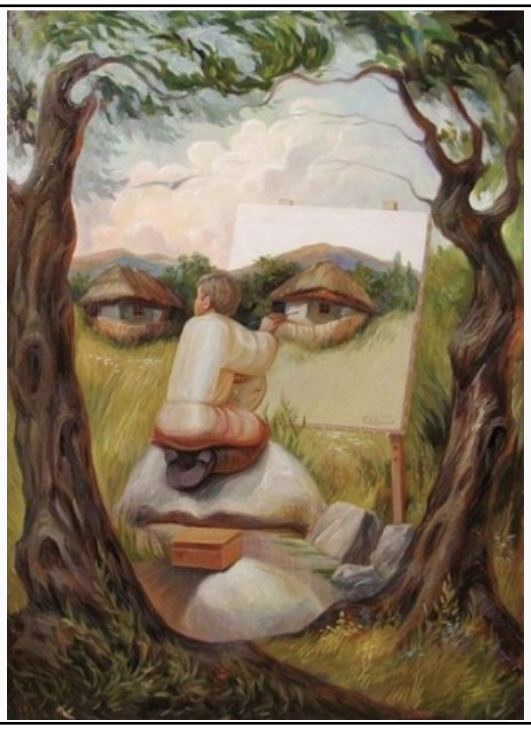
VISUAL PERCEPTION

7

7

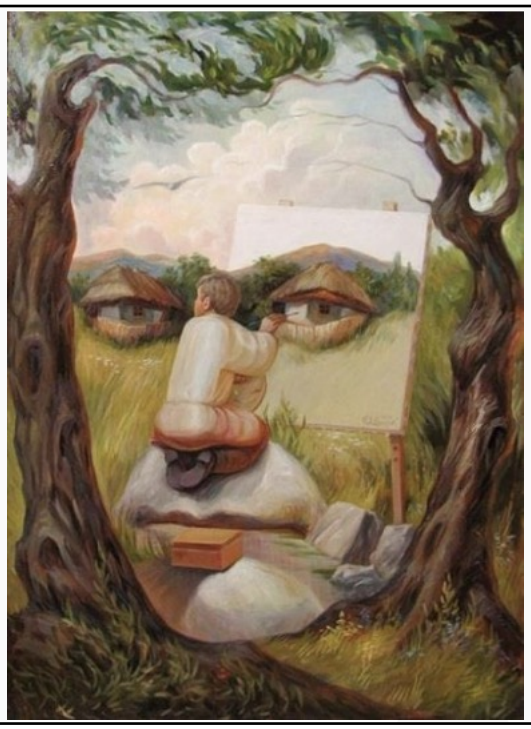
What is the ***first*** thing you see?

8



At first glance
you would see
a **human face**

9



But looking
more closely
you start to see
the details

10

The idea of Gestalt principles:

"The whole of something is more ***important*** to our understanding than the individual parts."

11

Gestalt Principles in UI Design

- Similarity
- Proximity
- Enclosure
- Continuation
- Figure & Ground

12

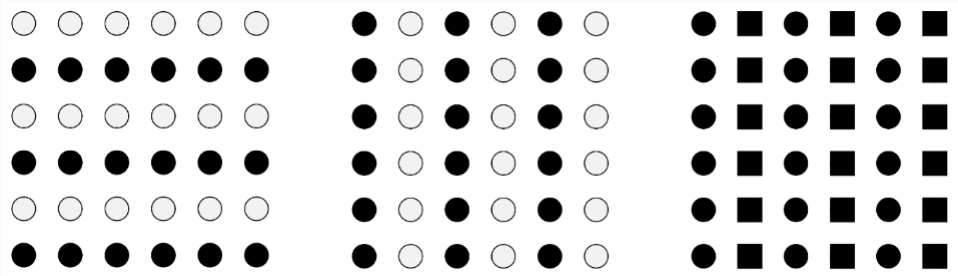
Similarity

Similar objects are seen as grouped

13

Similarity

- Similarity of objects form groupings:
  - At left, light and dark shades create similarities among horizontal groups
  - At center, light and dark shades creates similarities among vertical groups
  - At right, circle and square shapes create similarities among vertical groups



Source:
http://newmediaabington.pbworks.com/w/page/66930745/Ges

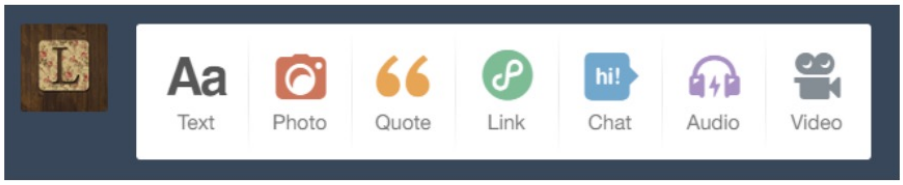
14

Similarity

- Each icon has a different image
- These icons are similar in shape and style
- They are perceived to be one group and share the same level of function

15

Similarity



- Each block has a different icon
- The blocks share similar structure: an icon with text underneath
- They are perceived to be one group and share similar function: create a new blog post
- Users know exactly what UI elements to look for

16

Proximity

When elements are *close together*, they are seen as grouped

17

Proximity

By using white space, we can separate as well as build the relationship between elements

18

Proximity

- Language & Currency fields are put close enough to be seen as one group
- The links on the right side are arranged with proper amount of white space. They are seen as 3 small groups and together form one big group

19

Proximity

- White space, however, is just one part of implementing proximity.
- A design can have a lot of white space, but if items are not **grouped correctly**, the white space will have little effect.

Two business cards

Source: https://www.webdesignerdepot.com/2010/01/the-principle-of-proximity-in-web-design/

20

Proximity

- Using white space effectively and grouping related elements are critical to giving your website a clear visual hierarchy.
- Hierarchy is conveyed by the way in which elements are grouped and sub-grouped.
- This hierarchy helps the user understand where they've been and where they want to go and, thus, helps to communicate the purpose of the website.
- A list is a good example of an element that has the potential to communicate visual hierarchy.

The list on the left has a clear visual hierarchy, showing the relationships between items (indented items are sub-categories of the primary items)

Source: https://www.webdesignerdepot.com/2010/01/the-principle-of-proximity-in-web-design/

21

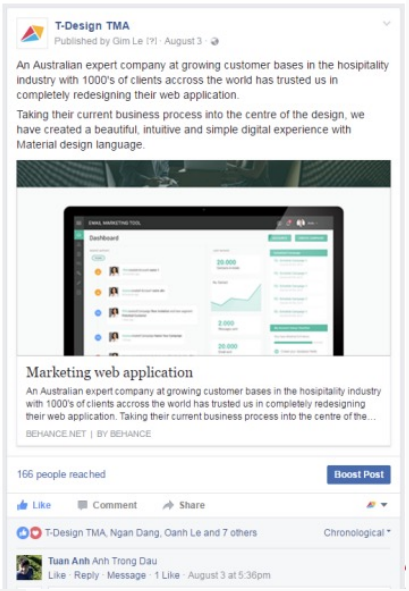
Enclosure

Things that have a *boundary* around them are seen as grouped

22

Enclosure

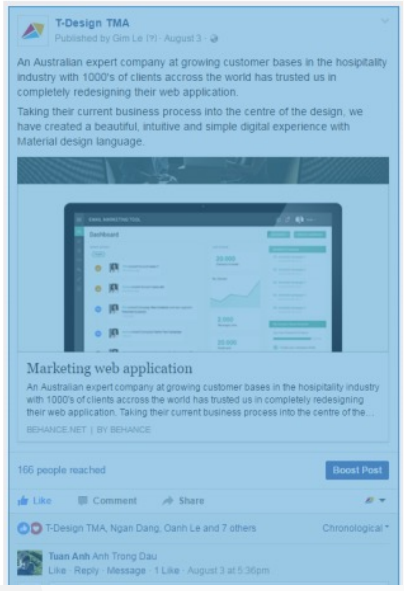
- Have you ever noticed the way Facebook elements are grouped together?



23

Enclosure

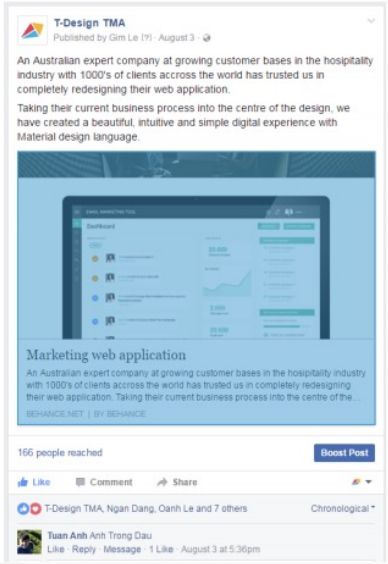
- There is a border around the whole post
- This is to separate one post to another



24

Enclosure

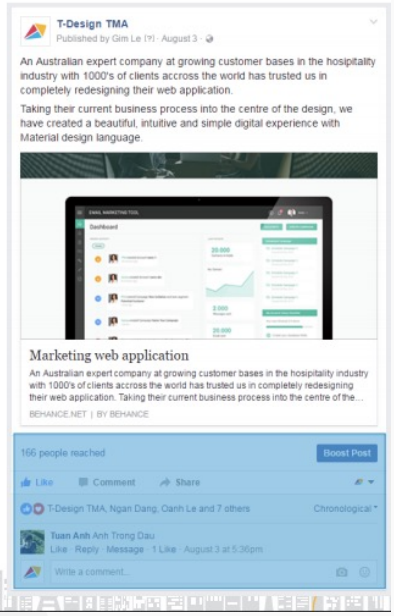
- There is another border around the link representation
- The image, the title along with a short description appear to be grouped



25

Enclosure

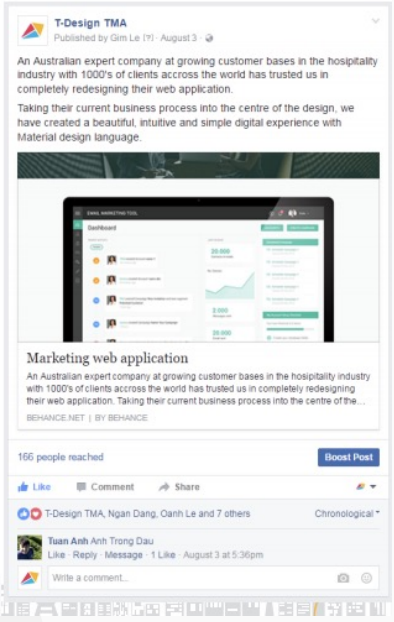
- This area at the bottom is for social interactions and is formed by several lines
- This area is also divided into different sections



26

Enclosure

- Enclosure is very helpful when you have limited space with a lot of information



27

Enclosure

To separate the link of the picture from the post

To separate one post with another posts

To separate the comment section for the related post

Source: https://www.facebook.com/univteknologimalaysia/

28

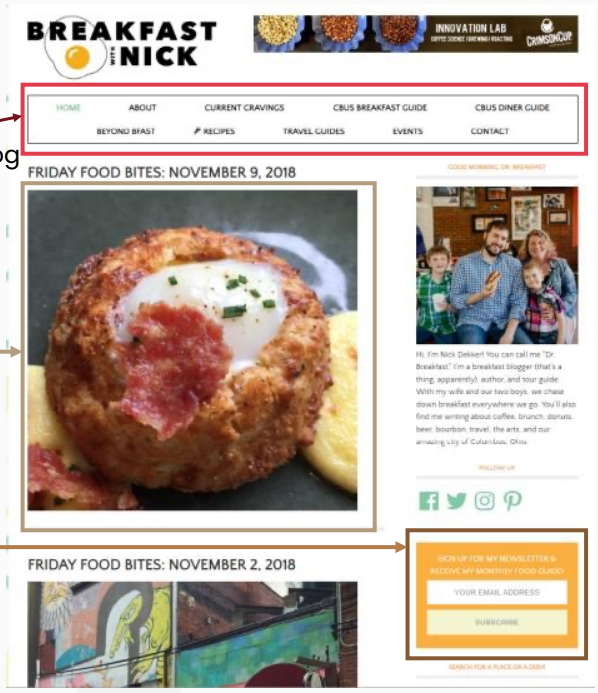
Enclosure

To separate the menu options available in the blog

To separate one post with another posts

To separate the link where users can subscribe monthly food guide by

Source: https://www.breakfastwithnick.com/



29

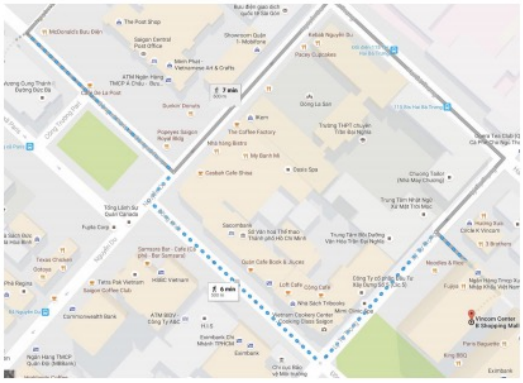
Continuation

Human eye follows *lines, curves*, or a *sequence of shapes*

30

Continuation

Human eyes tend to see a single dotted line instead of a series of blue dots



31

Continuation

- The light gray line represents the length of the track
- There is another dark gray line placed over the light gray line
- When the dark line grows, we understand the passage of time

32

Continuation

- The viewer's eye will naturally follow a line or curve.
- The smooth flowing crossbar of the "H" leads the eye directly to the maple leaf.
- Principle of good continuity is when human perceive each of two or more objects as different, singular and uninterrupted object even when they intersect.
- The alignment of the objects or forms plays a major role for this principle to take effect.
- This principle is well used by educator to teach kids on how to write letters of the alphabet and draw images.

Sources: http://graphicdesign.spokanefalls.edu/tutorials/process/gestaltprinciples/gestaltprinc.htm
https://en.wikipedia.org/wiki/Principles_of_grouping

33

Figure & Ground

Elements are seen as either "*figures*" or "*grounds*".

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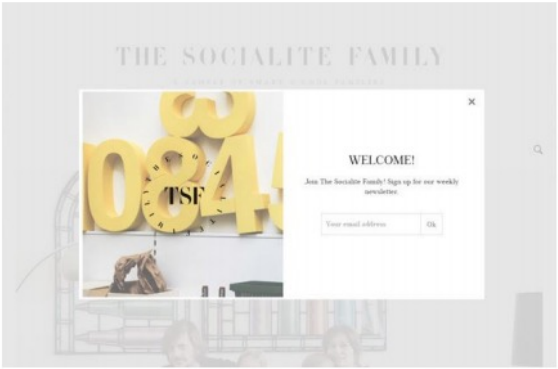
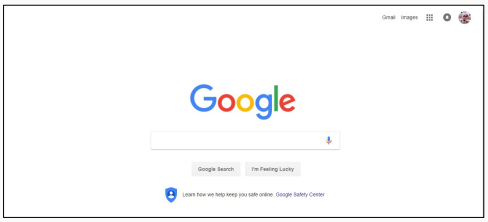
Figure & Ground

- Very common in modals
- Figure: the Welcome box
- Ground: the transparent background that softens the content in the back



35

Figure & Ground



Example of **figure** (*the Google text and search bar)* and **ground** *(the white background)*

Example of **figure** (*bird*) and **ground** (*the surrounding trees*).

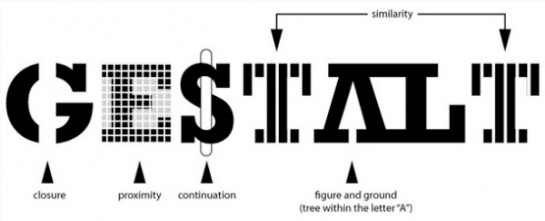
Sources:
https://wairimusensationndperception.weebly.com/figure-ground.html
https://www.google.com

36

Example of Gestalt Principles Applied to Web Design

37

Example of Gestalt Principles Applied to Web Design

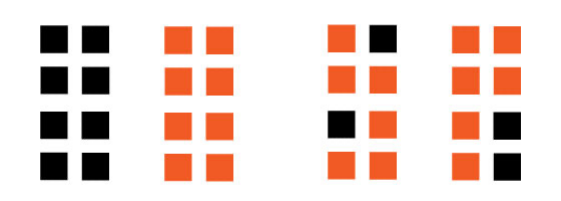


Reference: Gayle Christopher, gaylechristopher.com

38

Proximity/Grouping

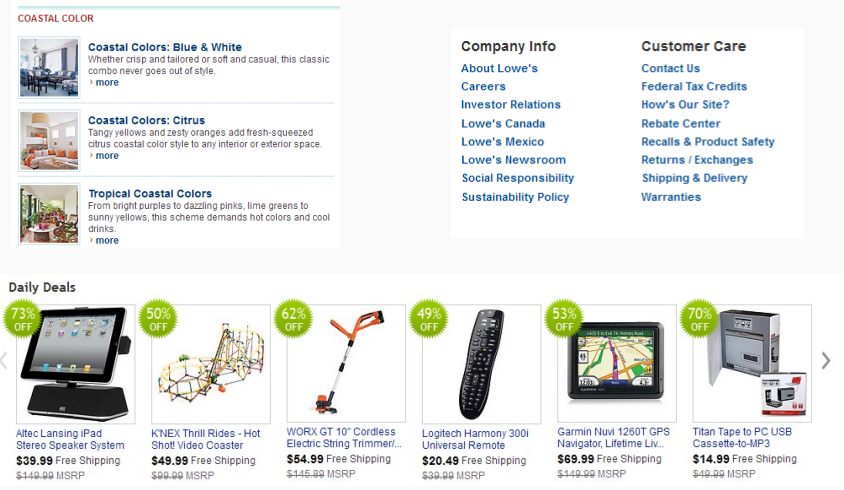
- Things that are close together are perceived to be more related than things that are spaced father apart.
- One of the first principles to impact perception
- Proximity overpowers similarity
- Only uniform connectedness overpowers proximity



Proximity overpowers similarity in color/contrast

39

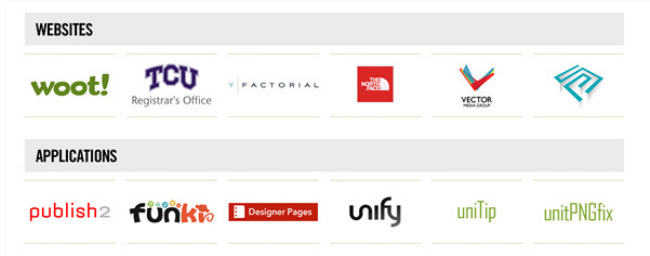
Proximity/Grouping



40

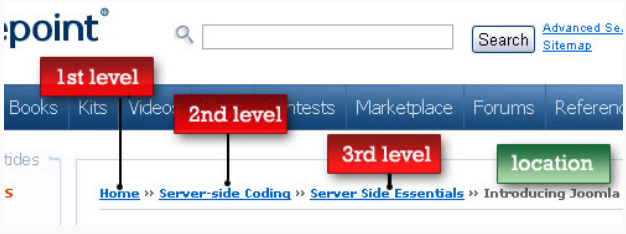
Good Continuation

- Elements arranged on a line or curve are perceived to be more related than elements not on the line or curve.
- All written language employs the principle of good continuation.
- Breadcrumbs, words in a paragraph, graphs, and linear arrangement (vertical or horizontal) communicate relatedness.



41

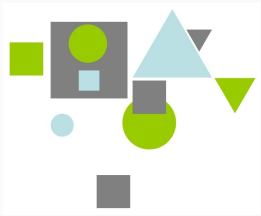
Good Continuation



42

Similarity

- Visual elements that are similar in shape, size, color and direction are perceived as part of a group.
- Different modes of similarity are not created equal.
- Color is the strongest way to suggest relationships.
- Used in links, icons, page content to suggest similar behavior, relatedness, and reinforce content hierarchies to communicate context.
- Consistency in styling and content dimension is important and implies structure.

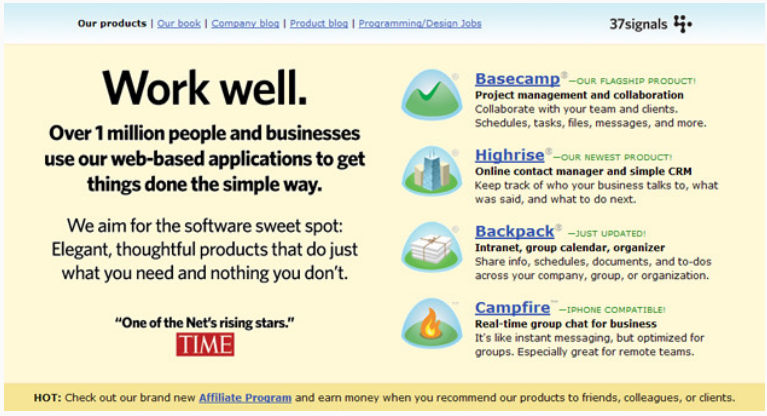


43

Similarity

Mac OS

Blue Water



44

Figure-Ground

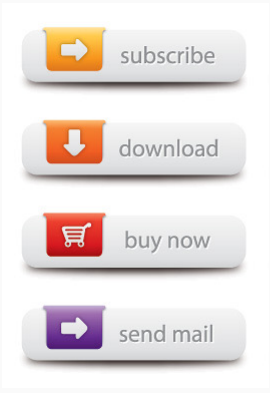
- Elements are perceived as either figures (elements of focus) or background.
- Human mind must rapidly decide which elements to focus on in a scene. This ability allows us to determine what we should pay attention to and what we can ignore even if it does provide context.
- Use color, shading, highlights to cue the viewer in on what has focus and ground depth: buttons that look like press-able buttons, links that change on rollover, overlays that use shadow.

FedEx

Hope for African Children Initiative

45

Figure-Ground



46

Law of Prägnanz

- The law of Pragnanz—or principle of simplicity—is a fundamental Gestalt psychology law concerning how humans process visual information
- People usually perceive complex things as simplified forms to easily recognise and understand what they see
- Designers cater to this tendency to see things as the sum of simple parts in digital products
- It also goes by the names "the law of good Gestalt" and "the law of simplicity."

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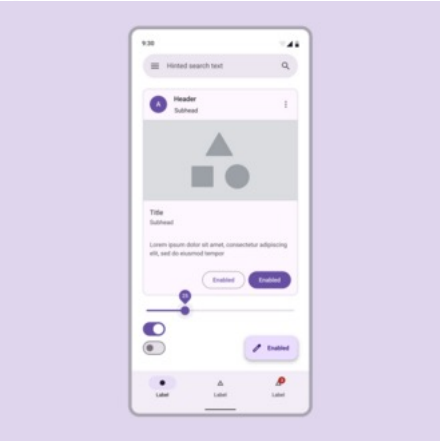
47

INDUSTRY PRACTICES

48

48

Google Material Design Principles



Principle	Description
Material is Metaphor	Uses tactile reality (shadows, surfaces, and edges) inspired by paper and ink to provide cues.
Bold, Graphic, Intentional	Emphasizes deliberate color choices, edge-to-edge imagery, large-scale typography, and intentional white space.
Motion Provides Meaning	Animations are meaningful and reinforce the user's actions, helping with orientation and continuity.
Adaptive Across Devices	Designs are responsive, scalable, and work seamlessly on all screen sizes and devices.
User Action Driven	Interfaces are focused on user input and promote direct manipulation with clear feedback.

Apple Human Interface Guideline

Principle	Description
Clarity	Ensure interfaces are easy to understand with legible text, precise icons, and straightforward navigation to eliminate ambiguity.
Deference	Minimize distractions by prioritizing content over UI elements, allowing users to focus on tasks with a clean, unobtrusive design.
Depth	Use visual hierarchy, layering, and subtle effects (e.g., shadows) to create a multi-dimensional experience that guides users naturally.
Consistency	Maintain familiar design patterns and behaviors across apps to align with Apple's ecosystem, enhancing user familiarity and comfort.
Feedback	Provide clear, immediate responses to user actions (e.g., visual cues, animations) to confirm interactions and reduce confusion.
Simplicity	Keep interfaces uncluttered, using minimal design elements and standard UI components (e.g., UIKit) to ensure intuitive functionality.

Topic 7: Design (part 3)

SCSM2113 Human-Computer Interaction Fundamentals
Faculty of Computing
Universiti Teknologi Malaysia

1

01 UI ELEMENTS IN GENERAL

02 ERROR HANDLING & PREVENTION

03 PREVENTING BOTH MISTAKES & SLIPS

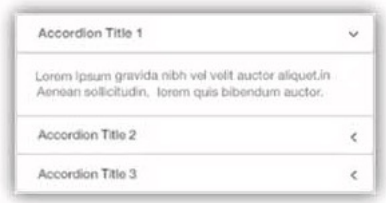
2

UI ELEMENTS IN GENERAL

3

UI Elements

- Interface elements include but are not limited to:
  - Input Controls: checkboxes, radio buttons, dropdown lists, list boxes, buttons, toggles, text fields, date field
  - Navigational Components: breadcrumb, slider, search field, pagination, slider, tags, icons, image carousel
  - Informational Components: tooltips, icons, progress bar, notifications, message boxes, modal windows
  - Containers: accordion



4

ERROR HANDLING & PREVENTION

SLIPS

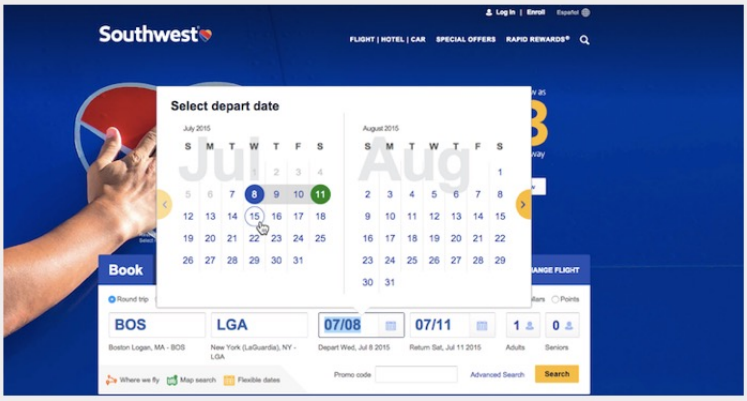
5

Slips

- Slips occur when users intend to perform one action, but end up doing another (often similar) action.
- For example, typing an “i” instead of an “o” counts as a slip; accidentally putting liquid hand soap on one’s toothbrush instead of toothpaste is also a slip.
- Slips are typically made when users are on autopilot, and when they do not fully devote their attention resources to the task at hand

6

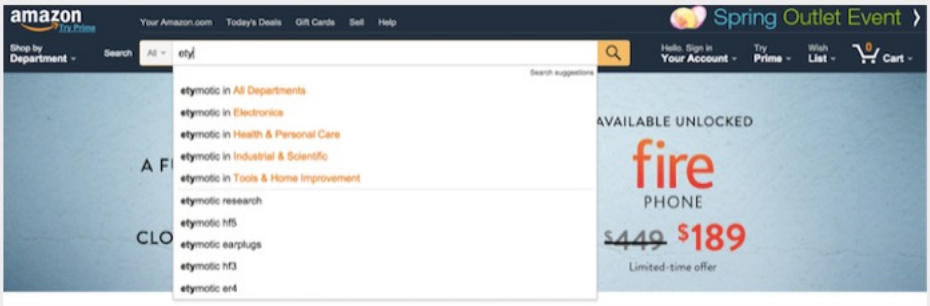
Include Helpful Constraints



Southwest's calendar widget for picking flight dates uses helpful constraints to prevent users from accidentally setting a nonsensical date range. Even if users attempt to set the return date before the departure date, this widget forces them to pick a departure date first. In addition, it subtly uses color to provide context about which date is about to be changed (in this case, blue for departure), which helps users see which field they are selecting (instead of having to keep that information in their working memory).

7

Offer Suggestions

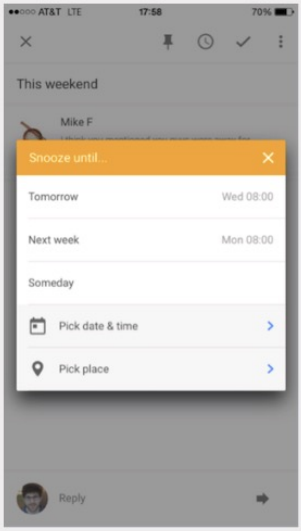


Remembering how to spell Etymotic Research *is difficult for users searching for high-quality headphones, and typing is likely to be low accuracy as well. Amazon's clickable search suggestions enable users to type less, thereby making fewer slips or mistakes that would produce no results.*

8

8

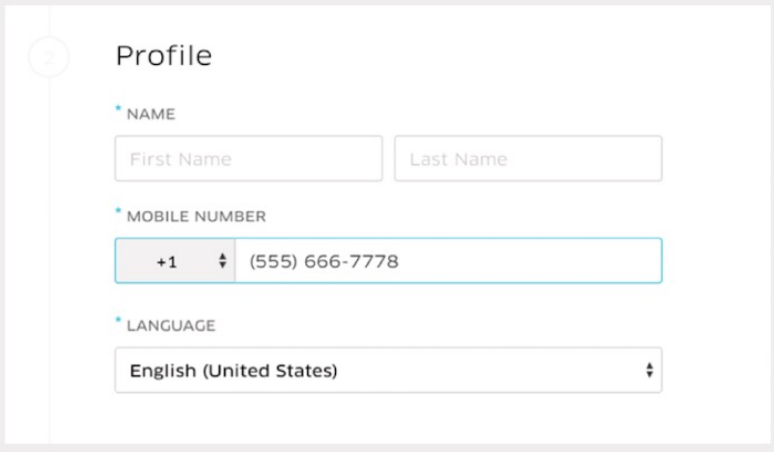
Choose Good Defaults



Google's Inbox app for iOS allows you to "snooze" an email until a later time. The default options are sensible and prevent typing errors for common choices.

9

Use Forgiving Formatting



Uber.com automatically displays the phone number in the desired format as users type, so that they can more easily scan their work to confirm that it's correct.

10

10

ERROR HANDLING & PREVENTION

MISTAKES

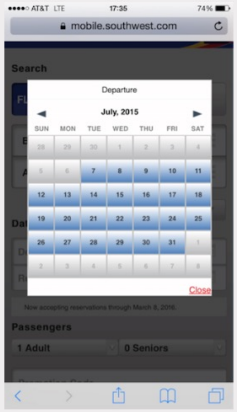
11

Mistakes

- Mistakes are made when users have goals that are inappropriate for the current problem or task; even if they take the right steps to complete their goals, the steps will result in an error.
- For example, if I misunderstood the meaning of the oil-pressure warning light in my car, and thought it was the tire-pressure monitor, no matter how carefully I added air to my tires, it would not fix the issue with my oil pressure.
- This would be a mistake, since the goal that I was attempting to accomplish was inappropriate for the situation, even though I made no errors in executing my plan.
- Mistakes are conscious errors, and often (though not exclusively) arise when a user has incomplete or incorrect information about the task, and develops a mental model that doesn't match how the interface actually works.

12

Gather User Data (follow Design Conventions)



Southwest's mobile site uses the convention of greying out dates in the past, letting you know you can't select them when booking a flight. Unfortunately, it also uses the same design for the next month's dates, implying unavailability.

13

13

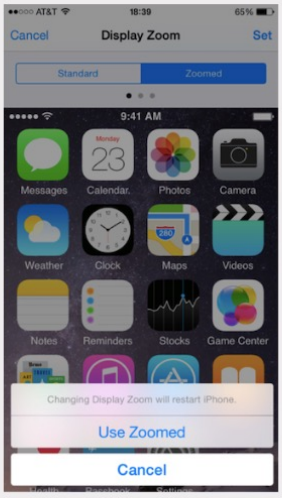
Communicate Affordance

Polarr is a popular photo editor on iOS. The right-hand editing controls (Temp, Tint, etc.) require that you tap on the box, and then swipe left or right to change the numeric value of that parameter. However, the controls don't indicate how users should interact with them, so novices are likely to accidentally set those controls to the wrong values several times before learning how to interact with them properly. An additional interaction difficulty is present: since the controls are on the far right side of the display, you can easily decrease the parameters by swiping left, but you quickly run out of horizontal space to swipe right and increase the value.

14

14

Preview Results



*In iOS 8, there is an accessibility option that allows users with low vision to zoom the display so that icons and text are larger. Applying zoom requires restarting the phone, which is a **heavyweight** action that will take a while, so iOS shows a preview of what things will look like before you commit to applying this change. This helpful preview allows you to evaluate whether or not your goal really is to zoom the display.*

15

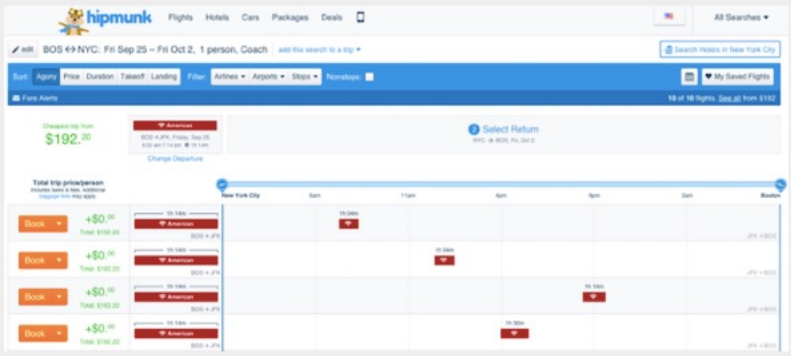
ERROR HANDLING & PREVENTION

MISTAKES

16

16

Remove Memory Burdens

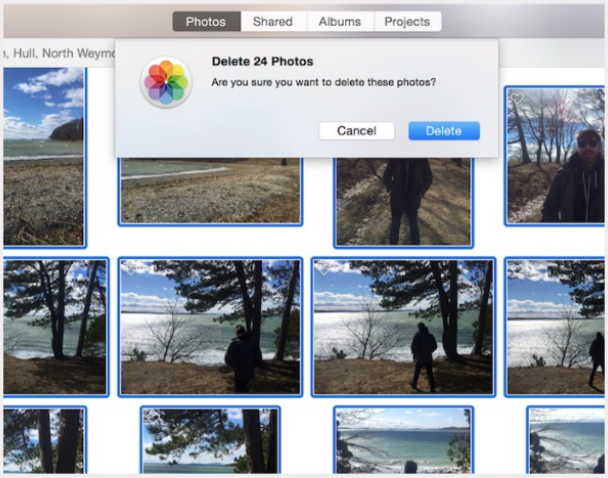


Hipmunk provides a quick glance at the contextual information needed to resume the process of choosing a flight, even after an interruption. At this second step in the booking process, it clearly shows the dates of travel, the airports in question, the fact that the lowest-priced fare for departure was chosen, and that the user is required to pick a return flight. Even users who had been distracted for quite some time could easily resume this process without accidentally deviating from their original requirements and plan for this flight, or attempting to repeat the steps already completed.

17

17

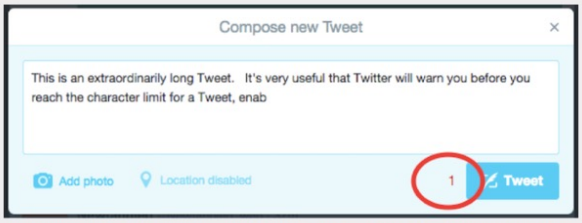
Confirm before Destructive Actions



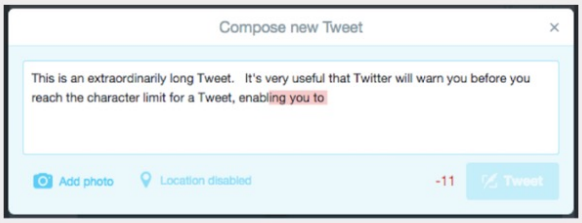
Apple's new Photos app uses a conventional dialog box to make sure that the user really intended to delete these photos of a recent hiking trip, and indicates that the scope of the action is the highlighted 24 photos. Even better, the button to confirm the delete action is clearly labeled Delete, *rather than the generic* Confirm.

18

Warn before Errors are Made



Twitter famously has a strict character limit for Tweets, and warns users before they exceed that limit with a remaining character count.



Once a Tweet is longer than the limit, it shows a negative counter, highlights the excess characters, and deactivates the Tweet button, which lets users know exactly what they need to do to fix their mistake.

19

Topic 8: Prototyping and Construction

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 PROTOTYPING

02 CONCEPTUAL DESIGN

03 CONCRETE DESIGN

04 GENERATING PROTOTYPES

05 CONSTRUCTION

2

Designing *with* or *for* People?

- Is stakeholder engagement one-way?
- Co-design
  - Emphasises creativity and mutual learning
  - Often multidisciplinary
- Participatory design (Scandinavian version)
  - Stakeholders are design partners
  - Co-operative prototyping is important
- Community-based design
  - Scale in terms of numbers and diversity

3

UTM

3

PROTOTYPING

4

4

Prototyping

- What is a prototype?
- Why prototype?
- Different kinds of prototyping
  - Low fidelity
  - High fidelity
- Compromises in prototyping
  - Vertical
  - Horizontal
- Final product needs to be engineered

5

What is a Prototype?

- One manifestation of a design that allows stakeholders to interact with it

- In other design fields, a prototype is a small-scale model:
  - A miniature car
  - A miniature building or town

Source: PalmPilot wooden model © Mark Richards

6

3D Printing Examples

(a) (b) (c)

Examples of 3D printing: (a) model jet engine, (b) Synapse Dress by Anouk Wipprecht: embedded with sensors, the wearer can control the dress's lighting pattern, and (c) custom-made climbing shoes based on a scan of the wearer's feet
Source: (a) Catiav5ftw / MakerBot Industries, LLC / CC BY-NC 4.0, www.thingiverse.com/thing:392115. Licensed under CC-BY-3.0, (b) ANOUK WIPPRECHTSYNAPSE DRESS created for Intel in 2014, www.niccolocasas.com/SYNAPSE-DRESS, and (c) Photo Credits: ATHOS

7

What is a Prototype in Interaction Design (ID)?

- Among other things an interaction design prototype can be:
  - A series of screen sketches
  - A storyboard, for example, a cartoon-like series of scenes
  - A PowerPoint slide show
  - A video simulating the use of a system
  - A lump of wood (for instance, the PalmPilot)
  - A loosely connected set of electronic elements
  - An animation of product use
  - A piece of software with limited functionality written in the target language or in another language

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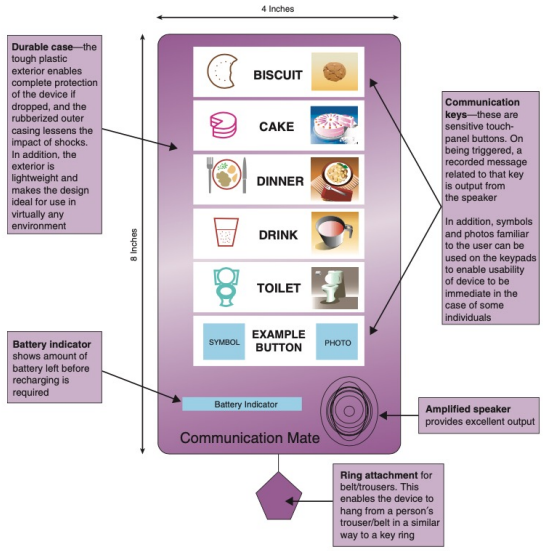
Example ID Prototype



A paper-based prototype of a handheld device to support an autistic child
Used courtesy of Sigil Khwaja

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What to Prototype?

- Technical issues
- Work flow, task design
- Screen layouts and information display
- Difficult, controversial, critical areas

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Why Prototype?

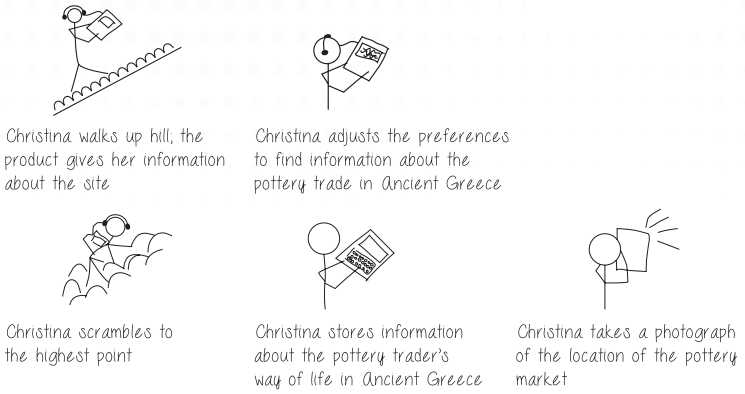
- Evaluation and feedback are central to interaction design
- Stakeholders can see, hold, and interact with a prototype more easily than a document or a drawing
- Team members can communicate effectively
- Ideas can be tested out
- Prototyping encourages reflection: an important aspect of design
- Prototypes answer questions and support designers in choosing between alternatives

Low-fidelity Prototyping

- Uses a medium which is unlike the final medium, for example, paper or cardboard
- Is simple, quick, cheap & easily changed
- Examples:
  - Sketches of screens and task sequences
  - Index cards or sticky notes
  - Storyboards
  - Wizard-of-Oz

Example 1: Storyboards

- A series of sketches showing how someone might progress through a task using the product
- Often used with scenarios, bringing in more detail and a chance to role play



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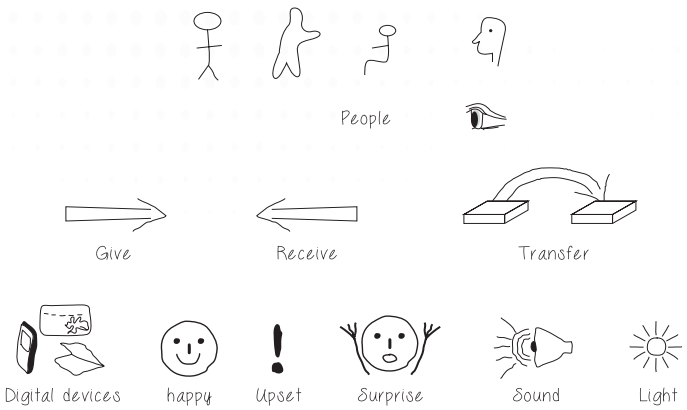
Example 2: Sketching

- Low-fidelity prototyping often relies on sketching
- Don't be inhibited about drawing ability – Practice simple symbols



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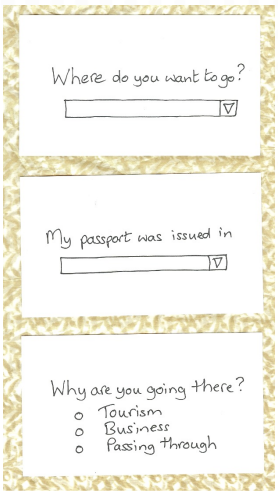
Example 3: Index Cards

- Index cards (3 x 5 inches)
- Each card represents one element of interaction
- In evaluation, can step through the cards



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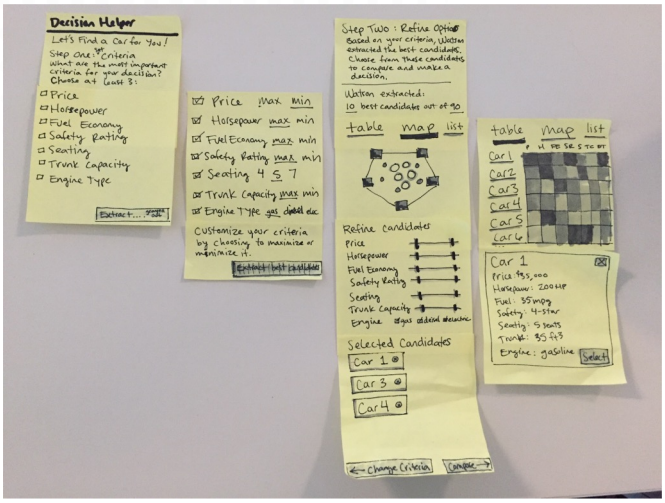
Example 4: Post-it Notes

- Low-fidelity paper prototypes don't look like the final product

Source:
https://www.oreilly.com/ideas/prototyping-physical-digital-products



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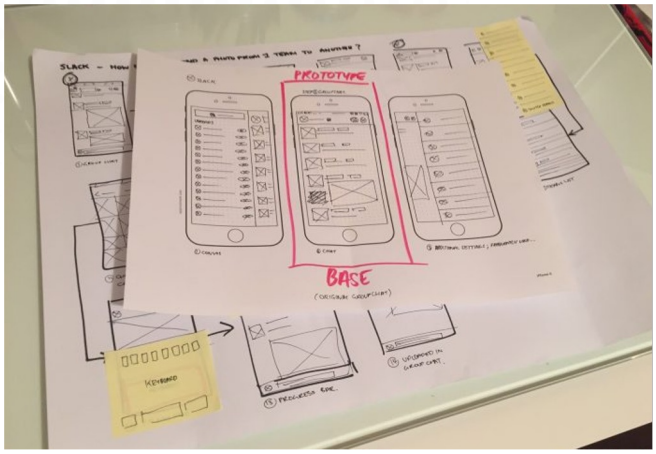
Example 5: Wireframe/Paper

- Paper prototypes can assist in documentation. Notes and revisions will support designers and developers when they will create an actual product

Source: https://theblog.adobe.com/prototyping-difference-low-fidelity-high-fidelity-prototypes-use



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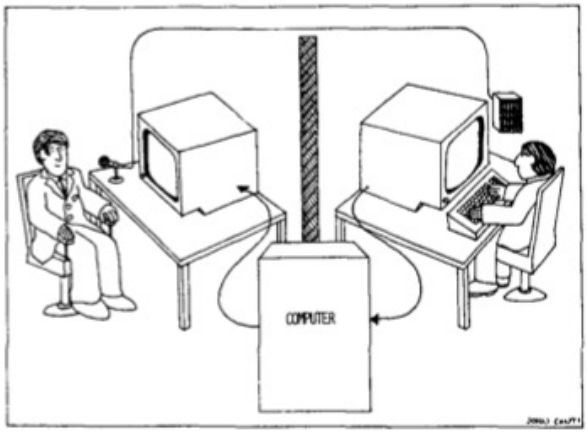
Example 6: Wizard-of-Oz

- The participant thinks they are interacting with a computer, but a human is responding to output rather than the system
- Usually done early in design to understand peoples' expectations



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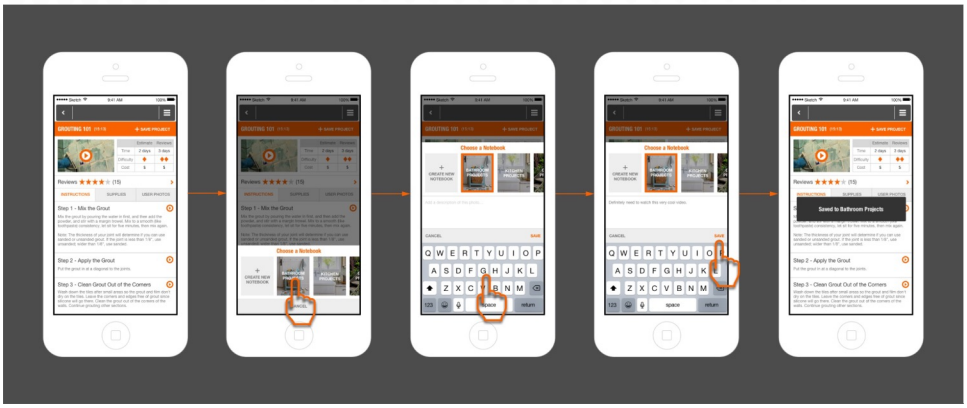
High-fidelity Prototyping

- Uses materials that would be in the final product
- Prototype looks more like the final system than a low-fidelity version
- High-fidelity prototypes can be developed by integrating existing hardware and software components
- Danger that people think they have a complete system…see compromises

Example: Mobile App



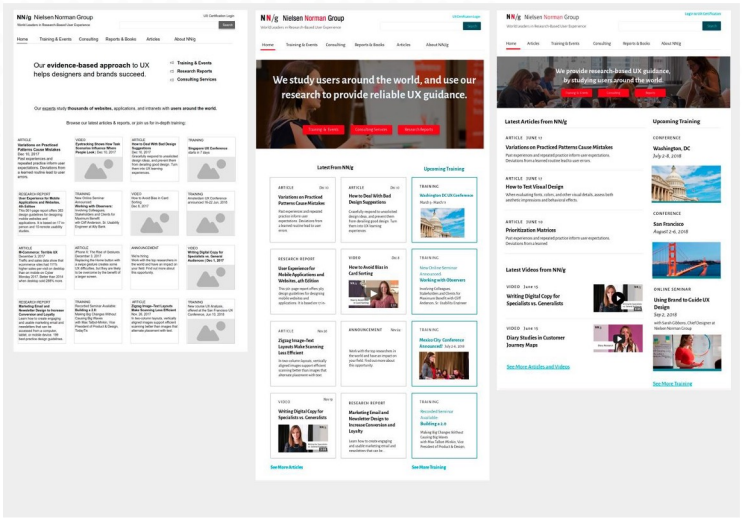
In the design process of high-fidelity prototyping, you need to work closely with the interaction designers to design a product as simple and easy to use as possible

Source: https://www.mockplus.com/blog/post/different-stages-of-prototyping

Evolution of Prototypes

- Iterative design process evolved from black–and–white wireframes for content planning to multiple versions of high-fidelity visual mockups created with cloud-based prototyping tools

Sourcehttps://www.nngroup.com/articles/case-study-iterative-design-prototyping/



Advantages and Disadvantages

Type	Advantages	Disadvantages
Low-fidelity prototype	Lower development cost Evaluates multiple design concepts Useful communication device Addresses screen layout issues Useful for identifying market requirements Proof of concept	Limited error checking Poor detailed specification to code to Facilitator-driven Limited utility after requirements established Limited usefulness for usability tests Navigational and flow limitations
High-fidelity prototype	Complete functionality Fully interactive User-driven Clearly defines navigational scheme Use for exploration and test Look and feel of final product Serves as a living specification Marketing and sales tool	More resource-intensive to develop Time-consuming to create Inefficient for proof-of-concept designs Not effective for requirements gathering

Table 11.3 Advantages and disadvantages of low- and high-fidelity prototypes

Compromises in Prototyping

- Prototyping involve compromises
- For software-based prototyping, maybe there is a slow response? sketchy icons? limited functionality?
- In-the-wild prototypes are operational but not necessarily robust
- Two common types of compromise:
  - **Horizontal:** wide range of functions, with little detail
  - **Vertical:** a lot of detail for only a few functions
- Other common compromise: robust vs changeable
- Beware of compromises in prototypes as final product needs to be engineered

Read more:
https://www.youtube.com/watch?v=HJmK_wuZWp4

CONCEPTUAL DESIGN

Conceptual Design

- Transforms user requirements/needs into a conceptual model
- A conceptual model is an outline of what people can do with a product and what concepts are needed to understand and interact with it
- Understand problem space and current requirements; empathize with users
- Creativity and brainstorming techniques
- Mood board may capture desired feel
- Consider alternatives: scenarios and prototyping helps

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Building Scenarios

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Emphathising with Users – The Third Age Suit

Source: Ford Motor Company

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Choosing an Interface Metaphor

- Interface metaphors combine familiar knowledge with new knowledge in a way that will help the user understand the product.
- Three steps:
  - understand functionality → identify potential problem areas → generate metaphors
- Evaluate metaphors:
  - How much structure does it provide?
  - How much is relevant to the problem?
  - Is it easy to represent?
  - Will the audience understand it?
  - How extensible is it?

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Considering Interaction and Interface Types

- Which interaction type?
  - How the user invokes actions
  - Instructing, conversing, manipulating, exploring, or responding
- Do different interface types provide insight?
  - Shareable, tangible, augmented reality, and so forth

Expanding the Initial Conceptual Model

- What functions will the product perform?
  - What will the product do and what will the user do?
- How are the functions related to each other?
  - Sequential or parallel?
  - Categorisations, for example, all actions related to privacy on a smartphone
- What information is needed?
  - What data is needed to perform the task?
  - How is this data to be transformed by the system?

CONCRETE DESIGN

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Concrete Design

- Difference between conceptual and concrete is emphasis
- Many aspects to concrete design
  - Color, icons, buttons, interaction devices, and so on
- User characteristics and context
  - Inclusiveness, input, and output modes
- Accessibility
  - Web Content Accessibility Guidelines
- Localisation and internationalisation
  - Language, navigation, icons, and metaphor
  - Indigenous knowledge and perspectives

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GENERATING PROTOTYPES

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Generating Prototypes

- Generate a storyboard from a scenario
  - Break down scenario into steps
  - Create a scene for each step
- Sketching out a storyboard prompts designers to think about design issues
- Generate a card-based prototype from a storyboard or from a use case
  - Consider each step in the use case – what interaction element is needed
  - Draw a card that captures it

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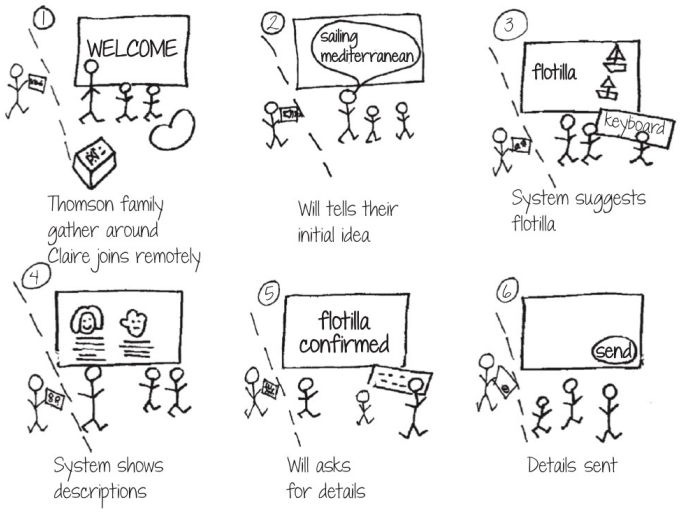
Generating Storyboard



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Generating Card-based Prototype

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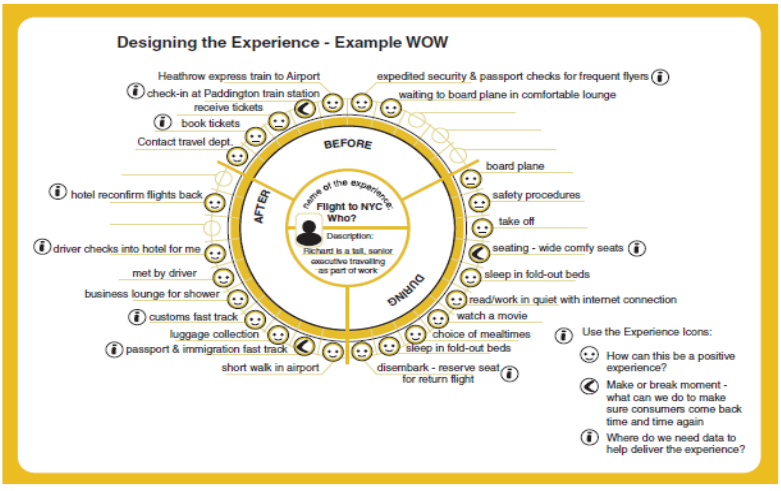
Mapping the Overall Experiences

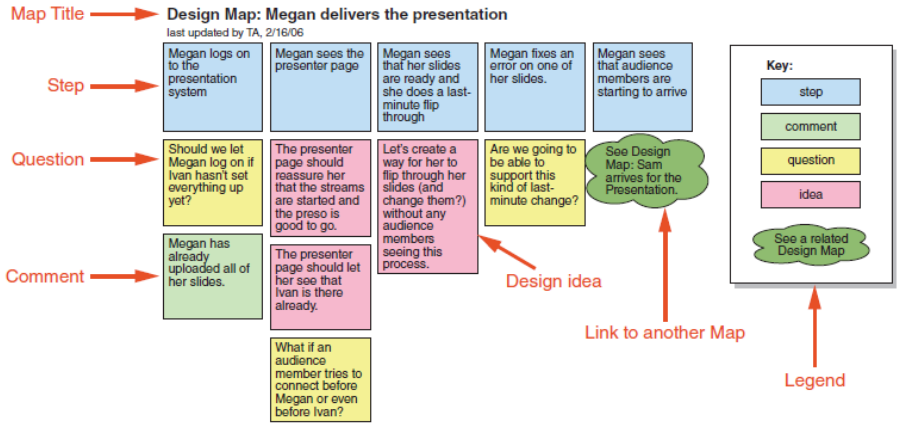
- Combination of personas, prototypes, or stickies to model the overall experience
- Visual representation called:
  - Design map
  - Customer journey map (see part 2)
  - Experience map
- Two common representations
  - Wheel
  - Timeline
- User flows focus on screen content and design, particularly used for mobile devices

An experience map drawn as a Wheel



Source: LEGO

An experience map drawn as a Timeline



Source: Adlin and Pruitt (2010), p134. Used courtesy of Morgan Kaufmann.

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CONSTRUCTION

40

Construction: Physical Computing

- Build and code prototypes using electronics
- Toolkits available include
  - Arduino
  - LilyPad (for fabrics)
  - Raspberry Pi
  - BBC micro:bit
  - MaKey MaKey
- Designed for use by a wide range of people

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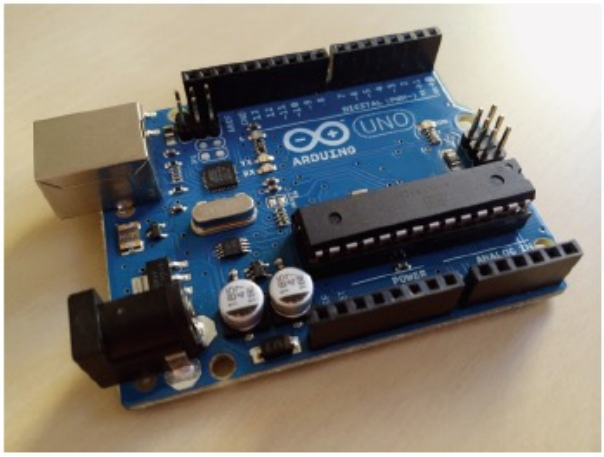
Physical Computing Kits: Arduino Board



Source: Used courtesy of Dr. Nicolai Marquardt

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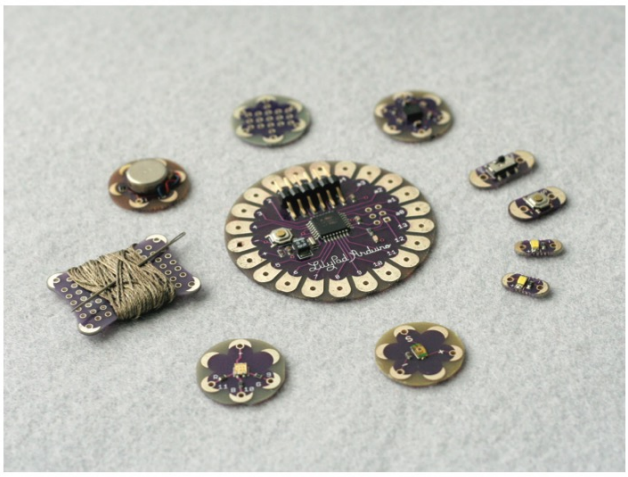
Physical Computing Kits: Lilypad Arduino



Source: Courtesy of Leah Beuchley

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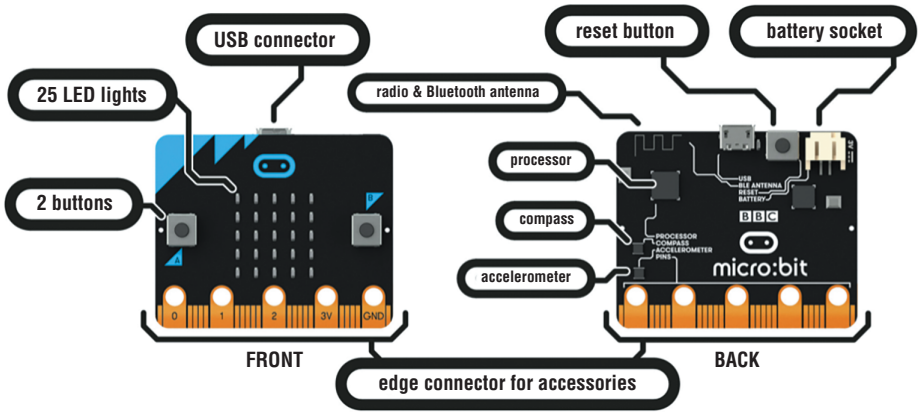
Physical Computing Kits: Micro:bit



Source: micro:bit. Used courtesy of Micro:bit Foundation

44

Physical Computing Kits: MaKey MaKey

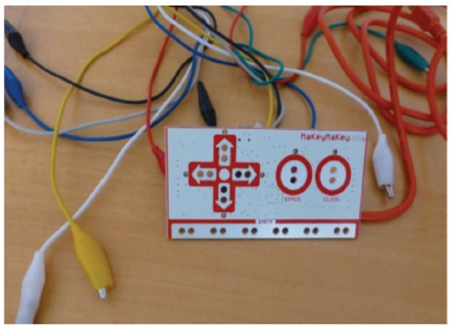


Figure 11.24 The MaKey MaKey toolkit

Figure 11.25 A group of retired friends playing with a MaKey MaKey toolkit

Construction: SDKs

- Software Development Kits
  - Programming tools and components to develop for a specific platform, for example, iOS
- Includes: IDE, documentation, drivers, sample code, and application programming interfaces (APIs)
- Makes development much easier
- Examples:
  - Amazon's Alexa Skills Kit for voice-based services
  - Apple's ARKit for augmented reality

Summary

- Prototyping may be low fidelity (such as paper-based) or high fidelity (such as software-based)
- Ready-made software and hardware helps create prototypes
- Two aspects to design: conceptual and concrete
- Conceptual design develops an outline of what people can do and what concepts are needed to understand the product.
- Concrete design specifies design details, for example, layout or navigation
- Three approaches to develop an initial conceptual model: interface metaphors, interaction styles, and interface styles.
- Expand an initial conceptual model by considering whether product or user performs each function, how those functions are related, and what information is required to support them
- Scenarios and prototypes can be used to explore design ideas
- Physical computing kits and software development kits facilitate the transition from design to construction

Topic 9: Introducing Evaluation (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 OVERVIEW

02 TYPES OF EVALUATION

03 EVALUATION CASE STUDIES

04 OTHER ISSUES TO CONSIDER

2

OVERVIEW

3

Why, What, Where, and When to evaluate

Iterative design and evaluation is a continuous process:

- **Why:** To check that all aspects of the user experience are taken into account in design
- **What:** A conceptual model, early and subsequent prototypes of a new system, more complete prototypes, and to compare a design with competitors' products
- **Where:** In natural, in-the-wild, in living lab and lab settings
- **When:** Throughout design; finished products can be evaluated to collect information to inform new products and for comparisons with other products

4

TYPES OF EVALUATION

5

Evaluation Types

- **Controlled settings** that directly involve users (for example, usability and research labs)
- **Natural settings** involving users (for instance, online communities and products that are used in public places)
- Often there is little or no control over what users do, especially in in-the-wild settings
- Any **setting that doesn't directly involve users** (for example, consultants and researchers critique the prototypes, and may predict and model how successful they will be when used by users

- Remote evaluation can be done with studies in all three categories above.
- Remote evaluation was practiced since the 1990s but it became important recently for coping with the impact of the Covid-19 pandemic
- Different methods my also be used together to provide different perspectives. For example, testing in a lab maybe used to compare different parts of a design, such as two different menu structures. Then the design may be tested in a natural setting to see whether people like using it in their homes or in other natural settings

6

Living Labs

- People's use of technology in their everyday lives can be evaluated in living labs. For example, evaluators can see how people use technology over a period of time, and how it fits in/or not with their routines
- Such evaluations are too difficult to do in a usability lab
- An early example was the Aware Home that was embedded with a complex network of sensors and audio/video recording devices (Abowd et al., 2000)

7

Living Labs (cont.)

- More recent examples include whole blocks and cities that house hundreds of people, for example, Verma et al., research in Switzerland (2017), and the huge PEARL facility in the UK
- Many citizen science projects can also be thought of as living labs, for instance, iNaturalist.org
- These examples illustrate how the concept of a lab is changing to include other spaces where people's use of technology can be studied in realistic environments

8

EVALUATION
CASE STUDIES

9

Constrasting Case Studies

- A study to test how well an automated vehicle supports drivers' behavior as they transition control to and frow between the automated vehicle and the drive.
- An ethnographic study of visitors at the the Royal Highland show in which participants are directed and tracked using a mobile phone app
- Crowdsourcing in which the opinions and reactions of volunteers (for example, from the crowd) inform technology evaluation

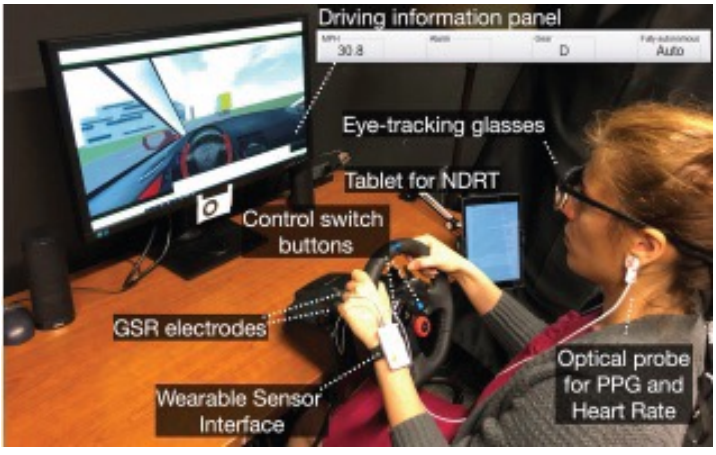
10

DeepTake and Automated Vehicles

- Large amounts of data were collected about drivers' driving and non-driving behavior and analyzed using machine learning techniques
- A user study was conducted to examine driver takeover behavior – ie, when a driver took control of the vehicle and when they left control to the automated system
- 11 females and 9 males aged 18-30 years performed tasks of varying difficulty
- Each task took about an hour and was followed up by a questionnaire. Eye tracking was used to track where the driver was looking and the time and processes to complete each task was recorded

11

The user study set-up

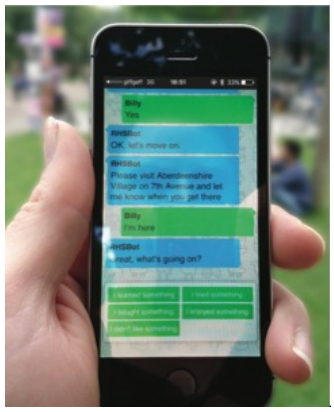


Source: Pakdamanian et al., 2021. CHI'2021 https://dl.acm.org/doi/pdf/10.1145/3411764.3445563

12

Ethnobot: Gathering Ethnographic data at the Royal Highland Show

- The Ethnobat directed Billy to a particular place (Aberdeenshire Village)
- Next, Ethnobot asks "…what's going on?"
- The screen shows five of the experience buttons from which Billy needs to select a response



Source: Tallyn et al. (2018) Reproduced with permission of ACM Publications.

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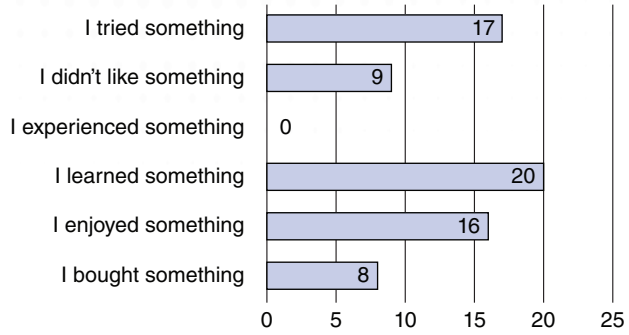
Experience responses to questions submitted in Ethnobot

Number of prewritten experience responses submitted by participants to the pre-established questions that Ethnobot asked them about their experiences



Source: Tallyn et al. (2018) Reproduced with permission of ACM Publications.

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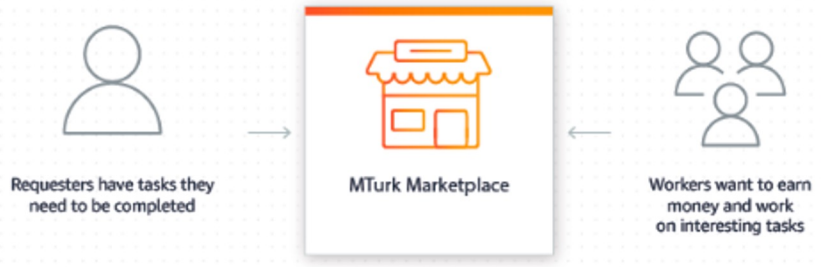
Mechanical Turk Test - Amazon



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What did we learn from the case studies?

- How to observe users in a lab simulation and in natural settings
- How evaluators use different levels of control in the lab and in natural settings and in crowdsourcing evaluation studies
- Use of different evaluation methods
- How to develop different data collection and analysis techniques to evaluate user experience goals such as engagement
- The ability to run experiments on the Internet that are quick and inexpensive using crowdsourcing
- How a large number of participants can be involved in crowdsourcing using systems such as Mechanical Turk

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Evaluation Methods

Method	Controlled settings	Natural settings	Without users
Observing	X	X	
Asking users	X	X	
Asking experts		X	X
Testing	X		
Modeling			X

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17

The Language of Evaluation

- Analytics
- Analytical evaluation
- Biases
- Controlled experiment
- Crowdsourcing
- Distributed evaluation
- Ecological validity
- Expert review
- Field study
- Formative evaluation
- Heuristic evaluation
- Informed consent
- In the wild evaluation
- Living lab
- Pain points
- Predictive evaluation
- Reliability
- Research in the wild studies
- Scope
- Summative evaluation
- Usability lab
- User studies
- Usability testing
- Users or participants
- Validity

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18

OTHER ISSUES TO CONSIDER

19

Participants' Rights and Getting Consent

- Participants need to be told why the evaluation is being done, what they will be asked to do and informed about their rights
- Informed consent forms provide this information and act as a contract between participants and researchers
- The design of the informed consent form, the evaluation process, data analysis, and data storage methods are typically approved by a higher authority.

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Things to Consider when Interpreting Data

- **Reliability:** Does the method produce the same results on separate occasions?
- **Validity:** Does the method measure what it is intended to measure?
- Ecological validity: Does the environment of the evaluation distort the results?
- **Biases:** Are there biases that distort the results?
- **Scope:** How generalizable are the results?

Summary

- Evaluation and design are very closely integrated
- Some of the same data gathering methods are used in evaluation as for establishing requirements and identifying users' needs, for example, observation, interviews, and questionnaires
- Evaluations can be done in controlled settings such as lab, less controlled field settings, and remotely or without users being involved
- Usability testing and experiments enable the evaluator to have a high level of control over what gets tested, whereas evaluators typically impose little or no control on participants in field studies, particularly in in the wild studies
- Different methods can be combined to get different perspectives
- Participants need to be made aware of their rights
- It is important not to over-generalize findings from an evaluation

Topic 9: From Controlled to Natural Settings (part 2)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 USABILITY TESTING

02 CONDUCTING EXPERIMENTS

03 IN-THE-WILD STUDIES

2

USABILITY TESTING

3

What is Usability Testing?

- Involves testing how people perform on products in controlled settings
- The people being tested are observed and the time it takes them to complete a task and the number and kinds of errors they make are recorded
- Data is recorded on video and key presses are logged
- The data is used to calculate performance times and to identify and explain errors
- User satisfaction is evaluated using questionnaires and interviews
- Observations about how the product is used in more natural contexts, including in-the wild may be included

4

Quantitative Measures

- Number of participants successfully completing the task
- Time to complete the task
- Time to complete the task after time away from it
- Number and type of errors per task
- Number of errors per unit of time
- Number of times people navigate to an item such as online help
- Number of people making the same or similar errors

Source: Wixon and Wilson, 1997

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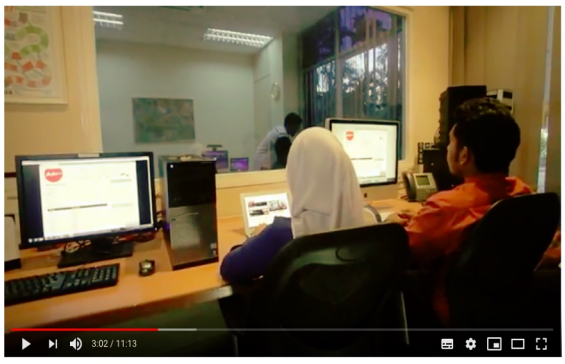
Usability lab with designers watching a user and assistant through a one-way mirror and recording



MIMOS UX Lab

6

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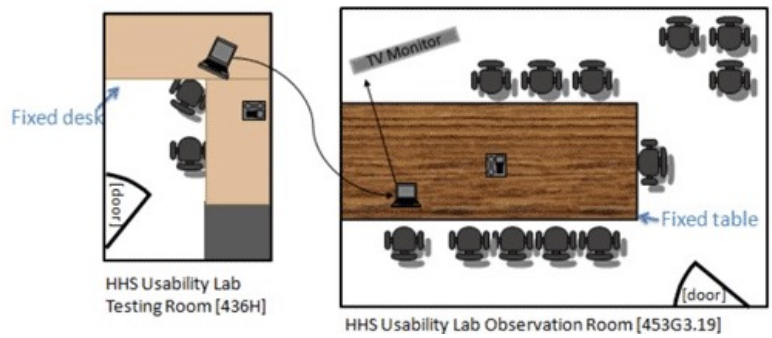
Layout of a usability lab used by US Health & Human Services Dept.



source: https://www.usability.gov/how-to-and-tools/guidance/hhs-usability-lab.html

7

7

Tobii Glasses Mobile Eye-Tracking System

Source: https://www.tobiipro.com/news-events/on-demand-webinars/Wearable-eye-tracking-for-research/

Source: Dalton et al., 2015, p.3891. Reproduced with permission of ACM Publications.

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8

Equipment is Getting Smaller

- As more testing is done remotely and in-the- wild, equipment has got smaller
- New lightweight models of eye-tracking glasses that look like ordinary glasses
- Video and audio recording is often done using mobile phones which produce increasingly good quality with each new model of phone
- Replace bulky recording devices and tripods, also more discrete

Remote Testing

- Participants can carry out tasks in their own environment without an evaluator being present
- Zoom, Teams and other digital communications are often used
- Testing can be synchronous or asynchronous
- Advantage of remote testing, especially asynchronous testing, is that several participants can be tested at the same time in their own environments
- Covid-19 pandemic encouraged creative ways of conducting remote testing that was safe
- Remote testing may make it easier for people with disabilities to participate
- Downside: no one present to help with equipment problems and less personal than co-present testing

Case 1: iPad usability
(Budiu & Nielsen, 2010)

- Now a classic study – innovative in its day!
- Study conducted in two cities: Fremont, CA and Chicago, IL
- Tests had to be done quickly, as information was needed by third-party app developers
- Also needed to be done secretly so that the competition was not aware of the study before the iPad was launched
- Seven participants with over three months experience with iPhones

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iPad Usability Testing Procedure

- Signed an informed consent form explaining:
  - What the participant would be asked to do
  - The length of time needed for the study
  - The compensation that would be offered for participating
  - Participants' right to withdraw from the study at any time
  - A promise that the person's identity would not be disclosed
  - An agreement that the data collected would be confidential and available only to the evaluators
- Participants were asked to explore the iPad
- Next, they were asked to perform randomly-assigned specified tasks

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Examples of the Tasks used in the iPad Evaluation

App or website	Task
iBook	Download a free copy of *Alice's Adventures in Wonderland* and read through the first few pages.
Craigslist	Find some free mulch for your garden.
eBay	You want to buy a new iPad on eBay. Find one that you could buy from a reputable seller.
Time Magazine	Browse through the magazine and find the best pictures of the week.
Epicurious	You want to make an apple pie for tonight. Find a recipe and see what you need to buy in order to prepare it.
Kayak	You are planning a trip to Death Valley in May this year. Find a hotel located in the park or close to the park.

Adapted from Budiu and Nielsen, 2010
Source: iPad App and Website Usability Study. Used courtesy of the Neilsen Norman Group.

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Problems and Actions

- Examples of problems detected:
  - Accessing the Web was difficult
  - Lack of affordance and feedback
  - Getting lost in an application
  - Knowing where to tap
- Actions by evaluators:
  - Reported to developers
  - Made available to public on Neilsen Norman Group
- Accessibility for all users is important
- Study did not address how iPad would be used in people's everyday lives
- Another study was done a year later to examine this and other issues that there was insufficient time to address in this study

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Case 2: Remote testing with extended VR (XR) (Sanni Siltanen et al., 2021)

- Study at Krone Corp. (Finland) to examine new immersive experiences in VR
- Needed to find ways to test experts and company employees remotely because of Covid-19 pandemic
- Also needed a sophisticated setup to test the collaborative XR platforms across multiple locations
- Testing was conducted remotely with experts in 8 countries: Finland, India, China, Germany, Indonesia, Malaysia, USA, United Arab Emirates
- Some tests in participants premises where special disinfecting routines had to be carried out, other tests were done remotely
- Little advice available about how to do remote testing during a pandemic

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A Participant in a Test Session

Source: Siltanen et al. (2021) MDPI / CC BY 4.0

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Screenshots from the DesignSpace VR Environments

Source: Siltanen et al. (2021) MDPI / CC BY 4.0

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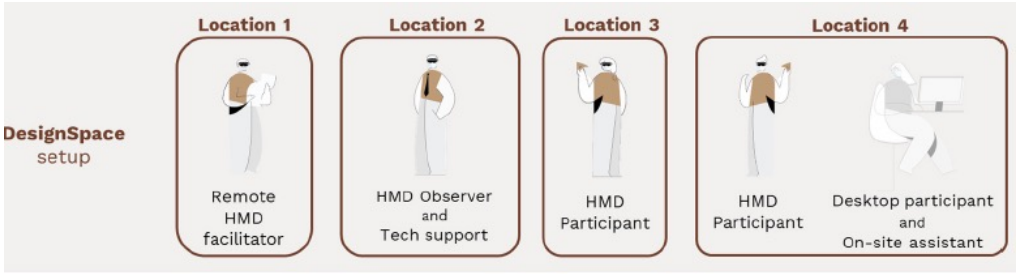
Testing setup used with DesignSpace environments



Source: Siltanen et al. (2021) MDPI / CC BY 4.0

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CONDUCTING EXPERIMENTS

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Experiments

- Test hypotheses
- Predict the relationship between two or more variables
- Independent variable is manipulated by the researcher
- Dependent variable influenced by the independent variable
- Typical experimental designs have one or two independent variables
- Validated statistically and replicable

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Experimental Designs

- Different participants (between participants):
  - Single group of participants is allocated randomly to the experimental conditions
- Same participants (within participants):
  - All participants participate in both conditions
- Matched participants (pairwise design):
  - Participants are matched in pairs, for example, based on expertise, gender, and so on

Different, Same, Matched Participant Design

Design	Advantages	Disadvantages
Different	No order effects	Many subjects and individual differences a problem
Same	Few individuals, no individual differences	Counter-balancing needed because of ordering effects
Matched	Same as different participants, but individual differences reduced	Cannot be sure of perfect matching on all differences

IN-THE-WILD STUDIES

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In-the-Wild Studies

- Done in natural settings
- "In-the-wild" is a term for prototypes being used freely in natural settings – broader and typically less controlled than traditional field studies
- Seek to understand what users do naturally and how technology impacts them
- In-the-wild studies are used in product design to:
  - Identify opportunities for new technology
  - Determine design requirements
  - Decide how best to introduce new technology
  - Evaluate technology in use

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An In-the-wild Study of a Pain-monitoring Device

- Monitoring patients' pain is a known challenge for physicians
- Goal of the study was to evaluate the use of a pain-monitoring device for use after ambulatory surgery
- Painpad is a keypad device
- It was usability tested extensively in the lab before being brought into two hospitals
- Goal was to understand how Painpad was used in the natural environment and as part of routines in two UK hospitals.
- How pain-monitoring changed with Painpad

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Painpad

A tangible device for inpatient self-logging of pain

Source: Price et al., 2018. Reproduced with permission of ACM Publications.

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Data Collection and Participants

- Two studies in two hospitals involving 54 people
- 13 males, 41 females
- Privacy was an important concern
- Hospital stay ranged from 1-7 days, mean and median age 64.6, 64.5 years
- Patients given Painpad after surgery and prompted to report pain levels every two hours
- Nurses also collected scores
- All data entered into charts
- Patients in one hospital were given a user-satisfaction survey when they left
- Also rated Painpad on a 1-5 Likert scale

Data Analysis and Presentation

- Three types of data were collected:
  - Satisfaction with Painpad was based on questionnaire responses
  - Patients' compliance with the two-hour routine
  - How data collected from Painpad compared with data collected by nurses

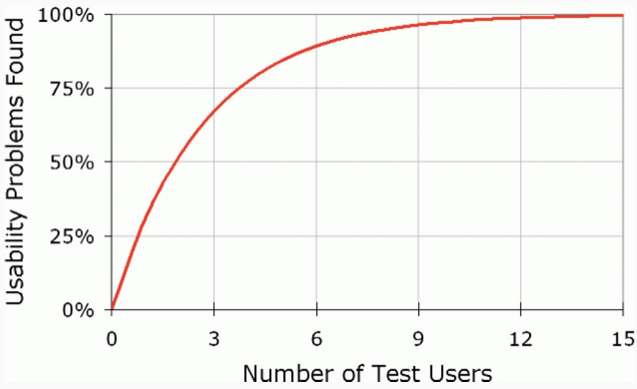
- Data showed:
  - Satisfaction with Painpad 4.63 on Likert scale
  - Patients' compliance was mixed: some liked it while others disliked or didn't notice the prompts
  - Patients recorded more scores with Painpad than through the nurses

How many participants is enough for user testing?

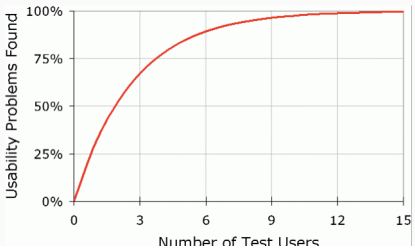
- The number is a practical issue
- Depends on:
  - Schedule for testing
  - Availability of participants
  - Cost of running tests
- Typically 5-10 participants if possible
- Some experts argue that testing should continue until no new insights are gained
- Others suggest that 5 participants is enough to identify any serious problems

Usability testing: Why Need to Test with 5 Users?

The typical value of L is 31%, averaged across a large number of projects studied. Plotting the curve for L =31% gives the following result:



Usability testing: Why Need to Test with 5 Users?



Justify Why 5 users is enough ?

1. As soon as you collect data from a single test user, your insights shoot up and you have already learned almost a third of all there is to know about the usability of the design.

2. When you test the second user, you will discover that this person does some of the same things as the first user, so there is some overlap in what you learn. People are definitely different, so there will also be something new that the second user does that you did not observe with the first user. So the second user adds some amount of new insight, but not nearly as much as the first user did.

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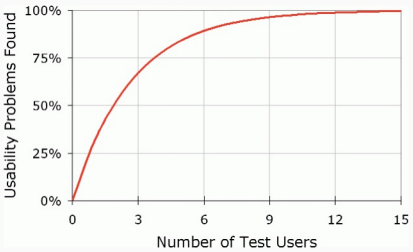
Usability testing: Why Need to Test with 5 Users?



Justify Why 5 users is enough ?

3. The third user will do many things that you already observed with the first user or with the second user and even some things that you have already seen twice. Plus, of course, the third user will generate a small amount of new data, even if not as much as the first and the second user did.

4. As you add more and more users, you learn less and less because you will keep seeing the same things again and again. There is no real need to keep observing the same thing multiple times, and you will be very motivated to go back to the drawing board and redesign the site to eliminate the usability problems.

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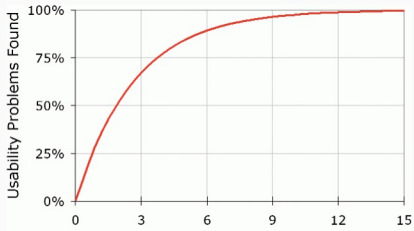
Usability testing: Why Need to Test with 5 Users?



Justify Why 5 users is enough ?

5. After the fifth user, you are wasting your time by observing the same findings repeatedly but not learning much new.

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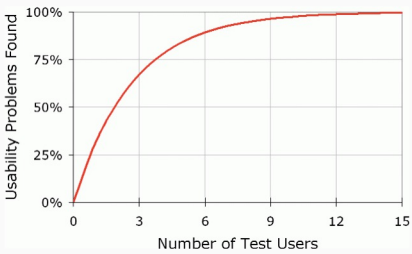
Usability testing: Why Need to Test with 5 Users?



When you need more users for usability testing ?

When To Test More Users?

You need to test additional users when a website has several **highly distinct groups of users**. The formula only holds for comparable users who will be using the site in fairly similar ways.

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How many participants is enough for other types of evaluation?

- Experiments test hypotheses to discover new knowledge by investigating the relationship between two or more variables
- The number of participants needed depends on the type of experiment being conducted
- It is advisable to consult a statistician before deciding
- In-the-wild studies also vary from a few people in a home to a software team to a whole community

Usability Testing vs Research

Usability Testing	Experiments for Research
• Improve products	• Discover knowledge
• Few participants	• Often many participants
• Results inform design	• Results validated statistically
• Usually not completely replicable	• Must be replicable
• Conditions controlled as much as possible	• Strongly controlled conditions
• Procedure planned	• Experimental design
• Results reported to developers	• Scientific report to scientific community

Summary

- Usability testing takes place in controlled spaces - usability labs, temporary labs
- Usability testing focuses on performance measures - how long and how many errors are made when completing a set of predefined tasks
- Indirect observation (video and keystroke logging), user satisfaction questionnaires, and interviews are also collected
- Remote testing has been conducted since the early 1990s but it became important during the Covid19 pandemic
- Remote testing uses portable equipment - video and audio recording using smart phones, mobile eye-tracking and automated keystroke logging

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Summary (cont.)

- Long-term studies of several weeks are used to evaluate complex products that participants need time to learn and use in their own work
- Experiments test a hypothesis by manipulating certain variables while keeping others constant
- The experimenter controls independent variable(s) in order to measure changes in the dependent variable(s)
- In-the-wild studies are carried out in natural settings to discover how people interact with technology in the real world
- In-the-wild studies involve deployment of prototypes or technologies in natural settings
- Sometimes the findings of in-the-wild studies are unexpected, especially for studies that explore how novel technologies are used by participants in their own homes, places of work, or outside

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Topic 9: Inspections, Analytics & Models (part 3)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 INSPECTION: HEURISTIC EVALUATION

02 INSPECTION: WALK-THROUGHS

03 ANALYTICS AND A/B TESTING

04 PREDICTIVE MODELS

2

No Direct Involvement of Users

- The evaluation methods described in previous chapters involved users
- The methods described in this chapter do not directly involve users
- These methods are based on an understanding of users' behaviour but users are not directly involved

- Three types of methods are discussed:
  - Knowledge codified in heuristics
  - Data collected remotely
  - Models that predict people's performance

3

3

INSPECTION: HEURISTIC EVALUATION

4

4

Inspections: Heuristic Evaluation & Walkthroughs

- Several kinds
- Experts or researchers use their knowledge of people and technology to review software usability
- Critiques can be formal or informal
- Heuristic evaluation is a review guided by a set of heuristics
- Walkthroughs involve stepping through a pre-planned scenario noting potential problems

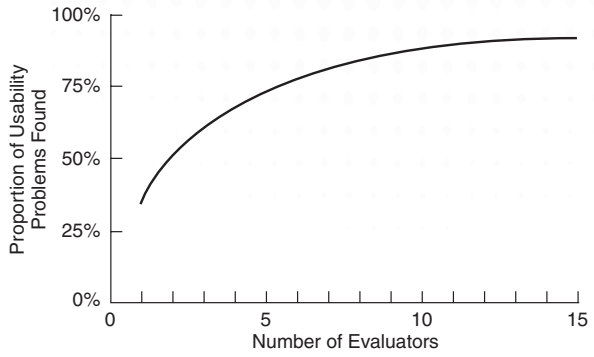
Heuristic Evaluation

- Developed by Jacob Nielsen in the early 1990s
- Based on heuristics distilled from an empirical analysis of 249 usability problems
- These heuristics have been revised for current technology by Nielsen and others for:
  - Mobile devices
  - Wearables
  - Virtual worlds
  - Social media
  - Other types of systems
- Design guidelines often form a basis for developing heuristics

Revised Version (2014) of Nielsen's Original Heuristics

- Visibility of system status
- Match between system and real world
- User control and freedom
- Consistency and standards
- Error prevention
- Recognition rather than recall
- Flexibility and efficiency
- Aesthetic and minimalist design
- Support error recovery
- Provide help and documentation

No of Evaluators and Problems



Curve showing the proportion of usability problems in an interface found by heuristic evaluation using different numbers of evaluators

Source: Nielsen and Mack, 1994. Courtesy of Wiley.

No of Evaluators

- Nielsen suggests that on average 5 evaluators identify 75-80 % of usability problems
- This often depends on the context or type of application (Cockton & Woolrych, 2001)
- Also depends on the skills and experience of the evaluators

Heuristics for Websites Focus on Key Criteria

- Ensure website is designed for people who will use it
- Clarity - avoid jargon, be concise & clear
- Minimise unnecessary complexity & reduce cognitive load
- Provide users with context – clear name & statement of purpose
- Promote positive & pleasurable experience

Heuristics for UI Design based on Gestalt Principles

- Simplicity
- Proximity
- Enclosure
- Continuation
- Figure and Ground

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Heuristics for Ambient Displays

Figure 15.2 Two ambient devices: (a) bus indicator, (b) lightness and darkness indicator

Source: J. Mankoff, A. K. Dey, G. Hsich, J. Kientz, Lederer and A. Morgan (2003) Heuristic evaluation of ambient devices. In *Proceedings of CHI 2003*, ACM Fig.1, p. 170. ©2003 Association for Computing Machinery, Inc. Reprinted by permission.

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Doing Heuristic Evaluation

1. Briefing session to tell evaluators what to do

2. Evaluation period of 1-2 hours in which:
   - Each evaluator works separately
   - Take one pass to get a feel for the product
   - Take a second pass to focus on specific features

3. Debriefing session in which evaluators work together to prioritize problems

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Severity Ratings in Heuristic Evaluation

- Severity ratings can be used to allocate the most resources to fix the most serious problems and can also provide a rough estimate of the need for additional usability efforts.
- If the severity ratings indicate that several disastrous usability problems remain in an interface, it will probably be inadvisable to release it.
- But one might decide to go ahead with the release of a system with several usability problems if they are all judged as being cosmetic in nature.

https://www.nngroup.com/articles/how-to-rate-the-severity-of-usability-problems/

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Contributing factors to a problem

The severity of a usability problem is a combination of three factors:

1) The **frequency** with which the problem occurs: Is it common or rare?
2) The **impact** of the problem if it occurs: Will it be easy or difficult for the users to overcome?
3) The **persistence** of the problem: Is it a one-time problem that users can overcome once they know about it or will users repeatedly be bothered by the problem?

If a problem has most of these factors, it should be rated more severe

https://www.nngroup.com/articles/how-to-rate-the-severity-of-usability-problems/

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Severity Rating Scale

Nielsen (1994) proposed the following 0 to 4 rating scale to rate the severity of usability problems:

0 = I don't agree that this is a usability problem at all

1 = Cosmetic problem only: need not be fixed unless extra time is available on project

2 = Minor usability problem: fixing this should be given low priority

3 = Major usability problem: important to fix, so should be given high priority

4 = Usability catastrophe: imperative to fix this before product can be released

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Severity Rating Scale (cont..)

Barnum, C. (2011) proposed a 3-point severity rating scale:

1 = Catastrophe—user cannot complete task; or user can complete the process but expresses extreme irritation at the process; or user needs assistance (business goal is to make the process independent of user assistance).

2 = Serious problem—user is frustrated but gets through it; suggests that others may be less inclined to put up with the inconvenience or that frustration level will be high.

3 = Cosmetic problem—user may hesitate or pick the wrong option, but user corrects it without incident; or user expresses minor irritation or annoyance, but it doesn't affect ability to complete tasks.

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Problem prioritisation

- Besides the rating scales shown in previous slides, there are other examples of severity rating scales that has been proposed over the years.
- To effectively understand the impact of each usability problems found, some researchers estimate its severity using dual ratings, for example in terms of usability principles and the ease to solve the problem. This can further help in prioritising the problem areas

https://irp-cdn.multiscreensite.com/b18f9b96/files/uploaded/HE_NWApp_v01.pdf

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Prioritisation Steps (dual rating example)

1. Identify the severity of an issue based on its ranking

Severity Ranking	
Rating	**Definition**
0	This is not a usability problem at all
1	Cosmetic problem only: need not be fixed unless extra time is available on project
2	Minor usability problem: fixing this should be given low priority
3	Major usability problem: important to fix, so should be given high priority
4	Usability catastrophe: imperative to fix this before product can be released

2. For each issue identified, rank its ease of fixing

Ease of Fixing Ranking	
Rating	**Definition**
0	Problem would be extremely easy to fix.
1	Problem would be easy to fix. Involves specific interface elements and solution is clear.
2	Problem would require some effort to fix. Involves multiple aspects of the interface or would require team of developers to implement changes before next release or solution is not clear.
3	Problem would be difficult to fix. Requires much development effort to finish before next release, involves multiple aspects of interface. Solution may not be immediately obvious or may be disputed.

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Understanding the Impact

(1) Overview: Tabulate the identified issue with how frequent the issue occurs, its severity ranking and its ease of fixing

Usability / UI Issue	Heuristic #	Frequency	Severity ranking (0 – 4)	Ease of fixing ranking (0 – 3)

(2) Identify which heuristic is the most problematic

Heuristic	Frequency
h1	
h2	
h3	
h4	
h5	
h6	
h7	
h8	
h9	
h10	
	(total)

(3) Identify issues by their severity & ease of fixing

Severity	Freq	Ease of Fixing	Freq
0	2	**0**	6
1	0	**1**	9
2	5	**2**	5
3	4	**3**	1
4	10		
	total = 21		total = 21

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Advantages and Problems

- Few ethical and practical issues to consider because users are not involved
- Can be difficult and expensive to find expert evaluators with knowledge of application domain and the people using it
- Biggest problems:
  - Important problems may get missed
  - Many trivial problems are often identified, such as false alarms
  - Evaluators have biases

Turning Design Guidelines and Golden Rules into Heuristics

- Ask questions like the following:

"Does the application include a visible title page, section or site? Does the user always know where they are in the application? Does the user always know what the system or application is doing? Are the links clearly defined? Can all actions be visualized directly (i.e., no other actions are required)?"

Granollers, 2018, p. 62

Evaluating Accessibility using WCAG Guidelines

- Web Content Accessibility Guidelines (WCAG) (Lazar et al., 2015)
- Guidelines can be used as heuristics for evaluating websites
- Governments and large corporations have to make their websites accessible by law
- Four key concepts (POUR):
  - Perceivable – eg, provide text alternatives for non-text content
  - Operable – eg, make functionality available via a keyboard
  - Understandable – eg, make content appear and be readable in predictable ways
  - Robust – eg ensure compatibility with other tools

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Source: WCAG 2.1 at a Glance.

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INSPECTION: COGNITIVE WALKTHROUGH

24

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Cognitive Walkthroughs

- Focus on ease of learning
- Designer presents an aspect of the design and usage scenarios
- Evaluator is told the assumptions about people using the system, context of use, task details
- One or more evaluators walk through the design prototype with the scenario
- Evaluators are guided by three questions

The THREE (3) Questions

1. Will the correct action be sufficiently evident to the people using the system?
2. Will the they notice that the correct action is available?
3. Will the they associate and interpret the response from the action correctly?

As the evaluators work through the scenario, they note problems

Pluralistic Walkthroughs

- Variation on the cognitive walkthrough theme
- Performed by a carefully managed team
- A panel of evaluators begins by working separately
- This is followed by a managed discussion that leads to agreed decisions
- The approach lends itself well to participatory design
- There are also other adaptations of basic cognitive walkthroughs

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ANALYTICS
& A/B TESTING

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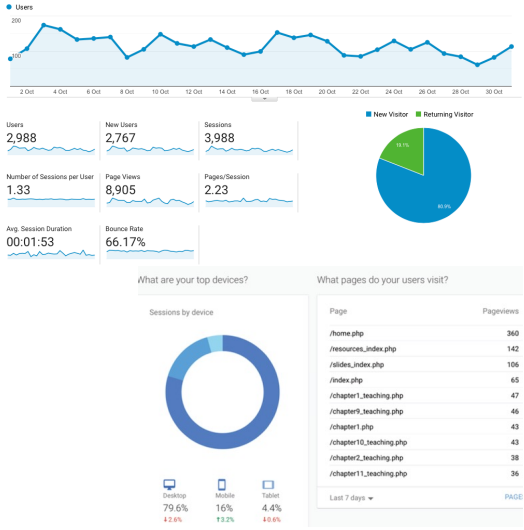
Web Analytics

- A form of interaction logging that analyses users' activities on websites
- Web designers use the analysis to improve their designs
- When designs don't meet peoples' needs, they will not return to the site
- They become one-time users
- Web analytics enable designers to track the activities of people on their site
- They can see how many people come to the site, how long they stay, and where they go next
- Web analytics offer designers a "big picture" about how their site performs based on users' activity
- Google provides is one of the most well-known analytics

Segment of Google Analytics for Interaction Design 5e website, December 2022



	Language	Acquisition			Behaviour		
		Users	New Users	Sessions	Bounce Rate	Pages/Session	Avg. Session Duration
		699 % of Total: 100.00% (699)	617 % of Total: 100.16% (616)	846 % of Total: 100.00% (846)	70.69% Avg for View: 70.69% (0.00%)	1.99 Avg for View: 1.99 (0.00%)	00:01:41 Avg for View: 00:01:41 (0.00%)
1.	en-us	379 (54.14%)	323 (52.35%)	466 (55.08%)	68.67%	2.12	00:01:45
2.	en-gb	89 (12.71%)	79 (12.80%)	111 (13.12%)	71.17%	1.77	00:02:22
3.	zh-cn	69 (9.86%)	67 (10.86%)	79 (9.34%)	79.75%	1.44	00:00:33
4.	sv-se	31 (4.43%)	28 (4.54%)	37 (4.37%)	62.16%	2.11	00:02:12
5.	en	13 (1.86%)	12 (1.94%)	15 (1.77%)	73.33%	1.33	00:00:27
6.	ko-kr	12 (1.71%)	12 (1.94%)	14 (1.65%)	78.57%	1.43	00:00:15
7.	en-ph	10 (1.43%)	10 (1.62%)	13 (1.54%)	69.23%	1.85	00:03:28
8.	id-id	8 (1.14%)	5 (0.81%)	11 (1.30%)	81.82%	1.27	00:02:25
9.	en-ca	7 (1.00%)	6 (0.97%)	7 (0.83%)	85.71%	1.14	00:00:02
10.	ar	5 (0.71%)	4 (0.65%)	6 (0.71%)	83.33%	3.50	00:03:32

A/B Testing

- A type of large-scale experiment
- Offers another way to evaluate a website, application or app running on a mobile device
- Often used for evaluating changes in design of social media applications
- Compares how two groups of users perform on two versions of a design
- Can involve thousands of people
- May create ethical dilemmas if users don't know they are part of the test
- Care is needed to ensure that other issues are not affecting users' behavior

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PREDICTIVE MODELS

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Predictive Models

- Provide a way of evaluating products or designs without directly involving users
- Less expensive than testing with users and some kinds of inspections
- Usefulness limited to systems with predictable tasks, for example, voicemail systems, smartphones, and dedicated mobile devices
- Based on knowledge of expert error-free behavior

Fitts' Law (1954)

- Fitts' Law predicts that the time to point at an object using a device is a function of the distance from the target object and the object's size
- The further away and the smaller the object, the longer the time to locate it and point to it
- It is particularly useful for determining where on a screen to position an object
- Fitts' Law is useful for evaluating systems for which the time to locate an object is important, for example, smartphones, handhelds, and mobile devices

Fitts' Law: A Model of Human Motor Performance

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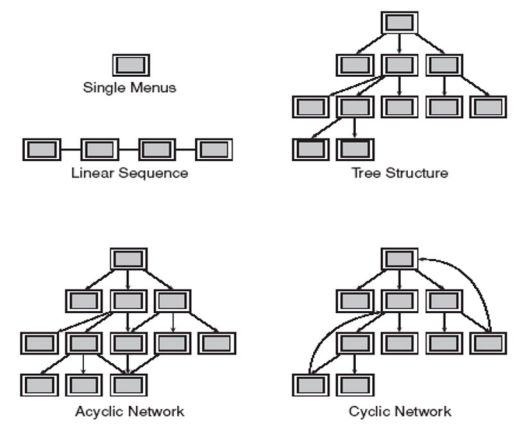
Task-Related Organisation

"The primary goal for menu, form-fillin, and dialog-box designers is to create a sensible, comprehensible, memorable, and convenient organization relevant to the user's task."



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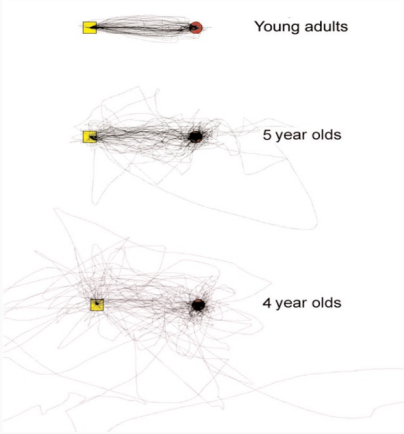
Fitts' Law

- Index of difficulty = $\log_2$ (D / W+1)
  - D = distance between buttons
  - W = target size
- Time to point = C1 + C2 (index of difficulty)
- C1 and C2 and constants that depend on the device
  - C1 = start/stop time in seconds
  - C2 = speed of the device
- Index of difficulty is $\log_2$ (2*8/1) = $\log_2$ (16) = 4 bits



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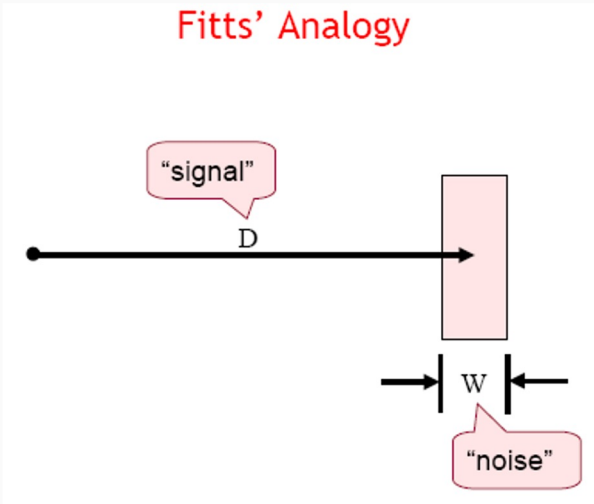
Fitts' Analogy



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Task Difficulty

- Fitts noted that the difficulty of a task is related to...
  - D - the distance[a] to move, and
  - W - the width of the target to select
- The larger the D, the harder the task
- The larger the W, the easier the task

[a] "distance" is often referred to as "amplitude" (A)

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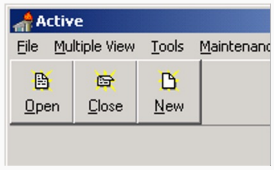
Fitts in Practice

Microsoft Toolbars allow you to either keep or remove the labels under Toolbar buttons

According to Fitts' Law, which is more efficient?



Source: http://www.asktog.com/columns/022DesignedToGiveFitts.html

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Fitts in Practice

You have a toolbar with 16 icons, each with dimensions of 16x16

Without moving the array from the left edge of the screen, or changing the size of the icons, how can you make this more efficient?

Adapted from Hearst,

41

Fitts in Practice

Answer: Line up all 16 icons on the left hand edge of the screen

Make sure that each button can be activated up the last pixel on the left hand edge

Why? Because you cannot move your mouse off of the screen, the effective width s is infinite

Adapted from Hearst,

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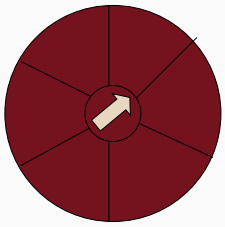
Pie menus

- A circular pop-up menu
  - no bounds on selection area
    - basically only angle counts
    - do want a "dead area" at center
  - What are Fitts' law properties?



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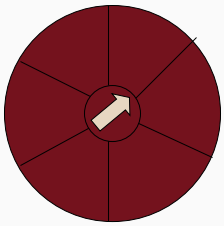
Pie menus

- A circular pop-up menu
  - no bounds on selection area
    - basically only angle counts
    - do want a "dead area" at center
  - Fitts' law properties:
    - minimum distance to travel
    - minimum required accuracy
    - very fast



- Why don't we see these much?
  - Just not known
  - Harder to implement
    - particularly drawing labels
    - but there are variations that are easier
  - Don't scale past a few items
    - No hierarchy

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GOMS

- **Goals** - the state the user wants to achieve e.g., find a website.
- **Operators** - the cognitive processes & physical actions needed to attain the goals, e.g., decide which search engine to use.
- **Methods** - the procedures for accomplishing the goals, e.g., drag mouse over field, type in keywords, press the go button.
- **Selection rules** - decide which method to select when there is more than one.

Keystroke Level Model (KLM)

- GOMS has also been developed to provide a quantitative model - the keystroke level model.
- The keystroke model allows predictions to be made about how long it takes an expert user to perform a task.

Response times for keystroke level operators (Card et al., 1983)

Operator	Description	Time (sec)
K	Pressing a single key or button Average skilled typist (55 wpm) Average non-skilled typist (40 wpm) Pressing shift or control key Typist unfamiliar with the keyboard	 0.22 0.28 0.08 1.20
P P1	Pointing with a mouse or other device on a display to select an object. This value is derived from Fitts' Law which is discussed below. Clicking the mouse or similar device	0.40 0.20
H	Bring 'home' hands on the keyboard or other device	0.40
M	Mentally prepare/respond	1.35
R(t)	The response time is counted only if it causes the user to wait.	t

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Summary

- Inspections can be used to evaluate requirements, mockups, functional prototypes, or systems
- User testing and heuristic evaluation may reveal different usability problems
- Design guidelines can be used to develop heuristics
- Web Content Accessibly Guidelines (WCAG) provide guidance for evaluating the accessibility of websites
- Walkthroughs are a fine-grained focused method for evaluating small parts of a prototype or product

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Summary

- Analytics involves collecting data about people's activity on a website or product to see which parts are used
- A/B testing is a form of large-scale experiment
- Fitts' Law can be used to predict expert, error-free performance for clearly defined tasks with limited key presses, for example, to evaluate keypress sequences for handheld devices and the position of objects on a small screen

Topic 10: Data Analysis, Interpretation and Presentation

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

Goals

- Discuss the difference between qualitative and quantitative data and analysis
- Enable you to analyse data gathered from:
  - Questionnaires
  - Interviews
  - Observation studies
- Make you aware of software packages that are available to help your analysis
- Identify common pitfalls in data analysis, interpretation, and presentation
- Enable you to interpret and present your findings in a meaningful and appropriate manner

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01 QUANTITATIVE AND QUALITATIVE
02 BASIC QUANTITATIVE ANALYSIS
03 BASIC QUALITATIVE ANALYSIS
04 ANALYTICAL FRAMEWORKS
05 TOOLS TO SUPPORT DATA ANALYSIS
06 INTERPRETING AND PRESENTING FINDINGS

3

UTM

3

QUANTITATIVE AND QUALITATIVE

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Quantitative and Qualitative

- **Quantitative data:** Expressed as numbers
- **Qualitative data**: In the form of words and images. Can be expressed as numbers, but not always meaningful to do so
- **Quantitative analysis:** Numerical methods to ascertain size, magnitude, or amount
- **Qualitative analysis:** Expresses the nature of elements and is represented as themes, patterns, or stories

Be careful how you manipulate data and numbers!

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BASIC QUANTITATIVE ANALYSIS

6

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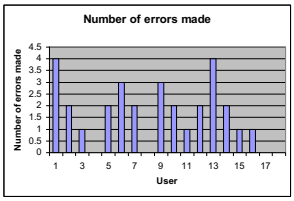
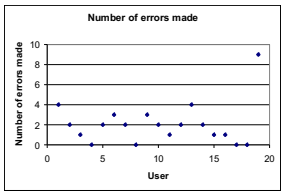
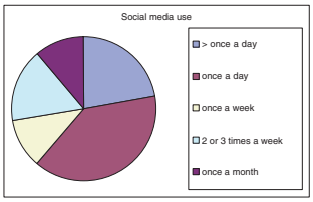
Basic Quantitative Analysis

- Averages:
  - Mean: Add up values and divide by number of data points
  - Median: Middle value of data when ranked
  - Mode: Value that appears most often in the data
- Percentages
- Be careful not to mislead with numbers!
- Graphical representations give overview of data







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Visualising Log Data

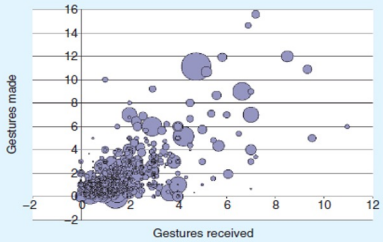

Figure 8.6 Interaction profiles of players in the cantina

Source: N. Ducheneaut and R.J. Morris (2004): "The social side of gaming: a study of interaction patterns in a massively multiplayer online game" in *Proceedings of CSCW 04.* ©2004 Association for Computing Machinery, Inc. Reprinted by permission.

Interaction profiles of players in online game

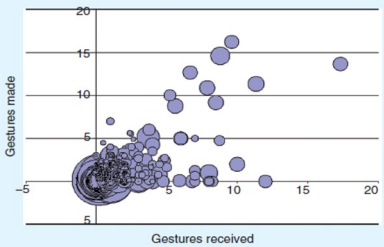

Figure 8.7 Interaction profiles of players in the starport

Source: N. Ducheneaut and R.J. Morris (2004): "The social side of gaming: a study of interaction patterns in a massively multiplayer online game" in *Proceedings of CSCW 04.* ©2004 Association for Computing Machinery, Inc. Reprinted by permission.

Log of web page activity

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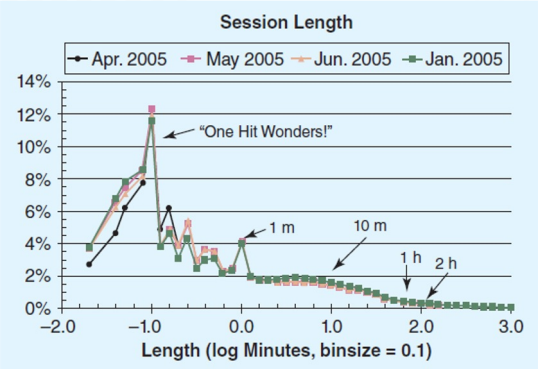
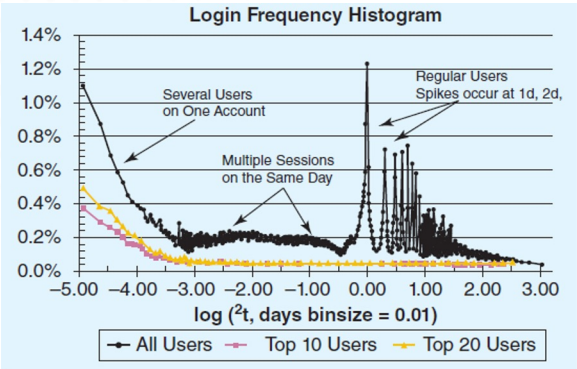
Web Analytics





session length data of four different months from Teachers' Domain (NSDL)

Source: Khoo, M., Pagano, J., Washington, A. L., Recker, M., Palmer, B., and Donahue, R. A. (2008) Using web metrics to analyze digital libraries. *Proceedings of Joint Conference on Digital Libraries*, Pittsburgh, June 16–20. ©2008 Association for Computing Machinery, Inc. Reprinted by permission.

Question Design Affects Data Analysis

- Open question: Each answer analysed separately
- Closed question: Can be analysed quantitatively
- Fixed alternative answers restrict what can be said in findings

BASIC QUANTITATIVE ANALYSIS

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Basic Qualitative Analysis

- Coding is central to qualitative analysis
  - Inductive (bottom-up) and Deductive (top-down)
  - Meaningful codes that don't overlap
  - What level of granularity is appropriate?
- Identifying themes
  - Emergent from data, dependent on observation framework if used
  - Inductive analysis
- Categorising data
  - Categorisation scheme pre-specified
  - Deductive analysis
- Combination of inductive and deductive is common

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QUALITATIVE ANALYSIS
ANALYTICAL FRAMEWORKS

Conversation Analysis

- Examines the semantics of a conversation in fine detail

```
01  SUS  i'd like to play beat the intro in a minute
02  LIA  [ oh no:: ]
03  SUS  [ alexa   ][ (1.1)  ] beat the in[tro
04  CAR             [ °yeah° ]
05  LIA                                   [°no:::...°
06  CAR  (0.6) it's mother's day? (0.4)
07  SUS  it's (    ) yep (.) listen (.) you need to keep
08       on eating your orange stuff (.) liam
09       (0.7)
10  CAR  and your green stuff
11  SUS  alexa (1.3) alexa (0.5)=
12  CAR                          =°and your brown stuff°
13  SUS  play beat the intro
```

An extract of the conversation between a family and Alexa

Discourse Analysis

- Focuses on dialogue; that is, the meaning of what is said and how words convey meaning
- Assumption that there is no objective scientific "truth"
- Language is viewed as a constructive tool
- Discourse analysis is useful when trying to identify subtle meaning
- Time-consuming but tools can help as long as context isn't lost

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Content Analysis

- Involves classifying data into themes or categories and studying their frequencies
- Can be used for any "text": video, newspapers, advertisements, images, and sounds
- Often used in conjunction with other techniques

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Figure 8.8 Building the affinity diagram of Indian ATM usage

Source: Figure 1, A. DeAngeli, U. Athavamker, A. Joshi, L. Coventry and G.I. Johnson (2004) "Introducing ATMs in India: a contextual inquiry", *Interacting with Computers* 16(1), 29–44. Reproduced with permission.

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Interaction Analysis

- A way to investigate and understand interactions between people and between people and artefacts
- Based on empirical observations such as videos
- Inductive process in teams, collaboratively
- Contents of the material is logged
- Materials are extracted, classified, or removed
- Instances of a salient event are assembled and played one after the other
- The team of researchers studies the assemblage together

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Grounded Theory

- Seeks to develop theory from systematic analysis of empirical data
- Three levels of coding
  - Open: Identify categories
  - Axial: Flesh out and link to subcategories
  - Selective: Form theoretical scheme
- Researchers are encouraged to draw on their own theoretical backgrounds to inform analysis
- Analytic tools to help stimulate thinking:
  - Question the data
  - Analyse words, phrases or sentence
  - Comparisons between objects or abstract categories

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Illustration of Open Coding

Game Feature	Observations
Game name	*AdVenture Capitalist* [G38]
Play description	You start CLICKING on a lemonade stand and collect money. Spend money to make upgrades, INCREASE PRODUCTION PER CLICK. Start hiring workers and INCREASE PRODUCTION PER SECOND. When you have enough money, you can buy new businesses, automate all your businesses to INCREMENT more money, and leave the game progress.
Game mechanics	Click to gain money, AUTOMATE production, make upgrades to DAMAGE/SEC.
Rewards	ONE CURRENCY, which is money, is rewarded in return.
Interface	GRAPHICAL
Interactivity level	7
Progress rate	9
Overview	This is a SINGLE-PLAYER game, which requires LONG CYCLES OK CLICKING at the start, and making a number of upgrades. Production rate reaches $390/sec in less that 10 minutes and you gain 1M in cash making the game progress faster.

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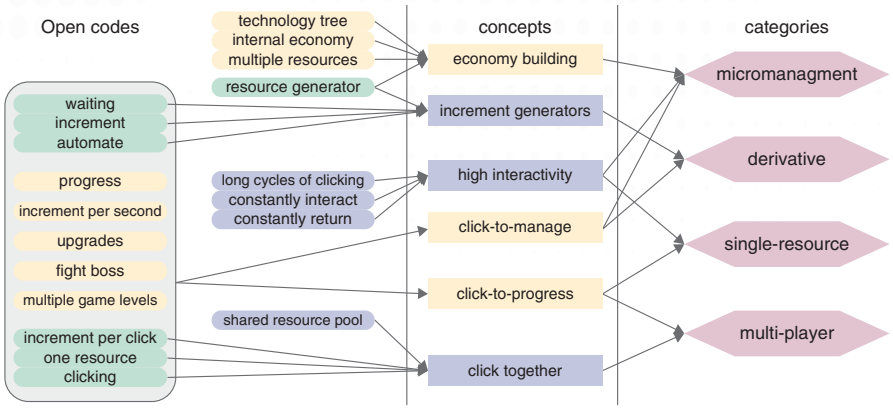
Development of Open Coding

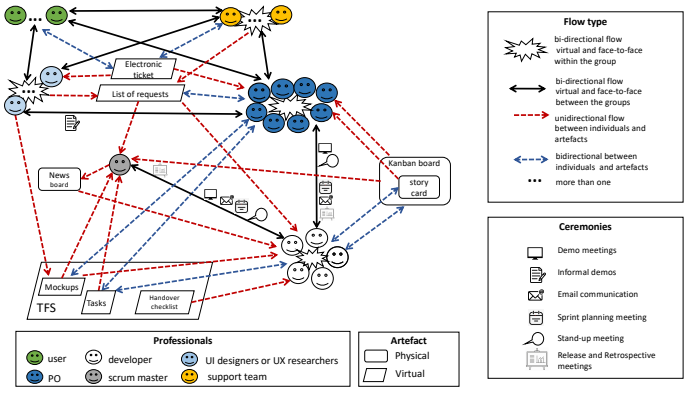


The analysis process that developed the incremental games super-category (each category above is part of incremental games). The process started with open coding of observations on idle games: multiple codes are created. Concepts are discovered through analyzing the open codes and identifying common features. This is an iterative process, where new codes are added, combined, or deleted. Each code is connected to one or more games and can be combined to form new concepts. Concepts are analyzed to find common relationships, and, thus, categories emerge. In the diagram, coloration is only to aid in reading. The left grouping is to show that all contained codes are part of click-to-manage and click-to-progress.

Source: Alharti et al (2018)

System-based Frameworks

- Understanding a whole socio-technical system requires different analytical framework
- Socio-technical Systems Theory
- Distributed Cognition of Teamwork



Source: Zaina et al (2021)

Which Analytical Framework?

Framework	Data	Focus	Expected outcomes	Level of granularity
Conversation analysis	Recordings of spoken conversations	How conversations are conducted	Insights into how conversations are managed and how they progress	Word-level, or finer, for instance, pauses and inflection
Discourse analysis	Recordings of speech or writing from individuals or several participants	How words are used to convey meaning	Implicit or hidden meanings in texts	Word, phrase, or sentence-level
Content analysis	Any form of "text" including written pieces, video and audio recordings, or photographs	How often something is featured or is spoken about	Frequency of items appearing in a text	A wide range of levels from words, to feelings or attitudes, to artifacts or people
Interaction analysis	Video recordings of a naturally-occurring activity	Verbal and non-verbal interactions between people and artifacts	Insights about how knowledge and action are used within an activity	At the level of artifact, dialogue, and gesture
Grounded theory	Empirical data of any kind	Constructing a theory around the phenomenon of interest	A theory grounded in empirical data	Varying levels, depending on the phenomenon of interest
Systems-based frameworks	Large-scale and heterogeneous data	Large-scale involving people and technology, such as a hospital or airport	Insights about organizational effectiveness and efficiency	Macro-level, organizational level

Proceedings of CHIuXiD 2019 – The 5th ACM In Cooperation International Conference in HCI and UX

An Observational Study for Developing Digital Technology in the Museum

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ABSTRACT
This study aims to understand the behaviour of visitors of the Negeri Sembilan State Museum, Malaysia based on visitor demographic information and activity patterns. 30 volunteers were observed during their visit to investigate how they interact with tangible objects that are exhibited in the museum space. All participants were never exposed to any digital technologies in the museum before. A qualitative approach was adopted where an evaluation toolkit for museum practitioners and observation framework was used for data collection. Our findings indicated that dioramas received much more attention from the participants than the artefacts and labels display. This pointed to the inadequacy of the exhibition style where it did not capture nor attract the participants to stop, interact or obtain information from the artefacts. The gap may be filled in by the usage of digital technology in museum exhibitions as it has the potential of providing interactive features that would attract visitors and create engagement with the artefacts in the museum.

Author keywords
HCI; Digital technology; Museum; Observation.

ACM Classification Keywords
H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

INTRODUCTION
Digital technologies in museums have seen massive growth in the last few years given the big effort in producing a more comprehensible system to give visitors the best museum experience.

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https://doi.org/10.1145/3328243.3328264

To improve engagement with visitors, many museums nowadays take advantage in the use of a variety of applications and technology to make the visiting experience more impactful [2].

The use of digital technologies allows the museum to create a new experience and to look for ways to create their exhibition more exciting in order to meet visitors' expectations, especially in this digital age [3, 4]. With the evolution of technologies combined with the innovative abilities of many converging media forms such as games, mobile application, virtual and mix-reality environments [5], the museum has changed from a static exhibition towards a non-linear, interactive and immersive environment [6].

This evolution occurred to fulfil the needs to attract visitors, especially the young generation. Nowadays, museum managers realised the importance of giving attention to accommodate visitor needs and expectation while touring the museums. This approach is also a part of a method to make visitors feel engaged with the exhibition and have good experience during the visit. In this technological era as well, visiting museums can also happen through a website, social media and cyberspace, especially for visitors who are not able to attend physically [7, 8]. Museum managers need to plan on technology application to interact and communicate with visitors and potential visitors. By adopting digital technology in the context of a museum exhibition physically or virtually, visitors can easily get access to the exhibition presented by the museum.

The rapid advancement in the digital world has contributed to changes in the modes of visiting the museums from something bored to something thrilling. This development of technology has given impact to visitors' expectation to explore and engage with the museum. Visitors can enhance and transform their experience beyond the physical objects exhibited [3, 9]. At the moment, there are many digital applications developed and are available in the museum in order to enhance visitors' experience and engagement with the museums collections [10]. For example, iMuse, which is a mobile type of digital technology that contains a virtual presenter which can expand visitors' experience by assisting

Method used

The observation framework is integrated with nine elements suggested by Robson which are space, actors, activities, objects, acts, events, time, goals and lastly, feelings, as in Table 1. By using this framework, Robson is encouraging the observer to focus on the context of the activity [15].

Space	What is the physical space like and how is it laid out?
Actors	What are the names and relevant details of the people involved?
Activities	What is the actor doing and why?
Objects	What are physical objects are present, such as furniture?
Acts	What are specifics individual actions?
Events	Is what you observe part of a special event?
Time	What is the sequence of the event?
Goals	What is the actor trying to accomplish?
Feelings	What is the mood of the group and individuals?

Table 1. The Observation Framework [15].

TOOLS TO SUPPORT DATA ANALYSIS

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Tools Support

- Spreadsheet — Simple to use, basic graphs
- Statistical packages, for example, R and SPSS

- Qualitative data analysis tools
  - Categorisation and theme-based analysis
  - Quantitative analysis of text-based data
- Nvivo and Dedoose support qualitative data analysis
- Computer Assisted Qualitative Data Analysis (CAQDAS) Networking Project, based at the University of Surrey

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INTERPRETING AND PRESENTING FINDINGS

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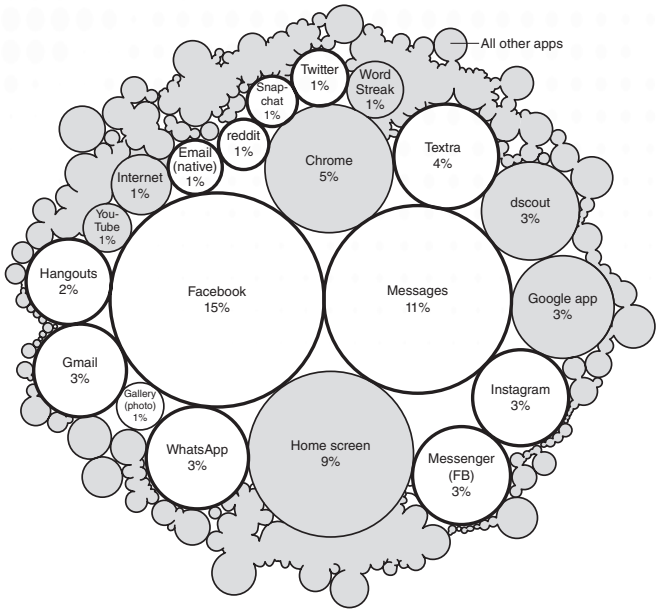

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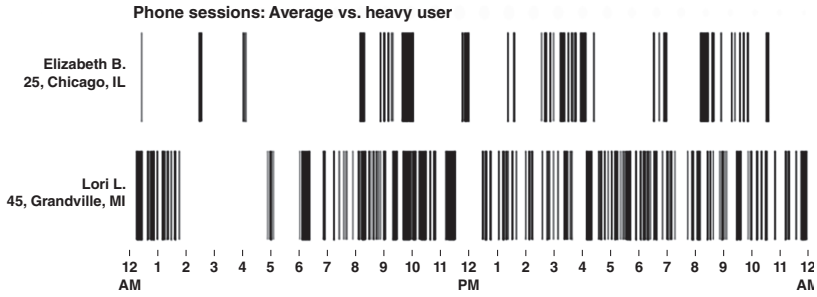

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Presenting Findings

- Structured notations have clear syntax and semantics to present particular viewpoint
- Using stories is an easy and intuitive approach to communicate ideas
- Summarise findings using a range of notations

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Key Points

- The data analysis that can be done depends on the data gathering technique(s) used
- Qualitative and quantitative data may be gathered from any of the three main data gathering approaches
- Percentages and averages are commonly used in Interaction Design
- Graphical representations help to identify patterns and outliers
- Mean, median, and mode are different kinds of 'average' and can have very different answers for the same set of data
- Analysis of qualitative data analysis may be inductive (extracted from the data), or deductive (pre-existing concepts)
- Several analytical frameworks exist that focus on different levels of granularity with different purposes